1510.06699: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE's — a formula has none of its own. A line's confidence is not a formula's.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00001	1	0.755	`{ }^{1,2, \mathrm{a}}`	${ }^{1,2, \mathrm{a}}$	A.N. Lasenby[1, 2, a)] and M.P. Hobson[1, b)]
1510.06699_F00002	1	0.755	`{ }^{1, \mathrm{~b}}`	${ }^{1, \mathrm{~b}}$	A.N. Lasenby[1, 2, a)] and M.P. Hobson[1, b)]
1510.06699_F00003	1	0.969	`{ }^{1)}`	${ }^{1)}$	[1)] *Astrophysics Group, Cavendish Laboratory, JJ Thomson Avenue, Cambridge CB3 0HE, UK*
1510.06699_F00004	1	0.979	`{ }^{2)}`	${ }^{2)}$	[2)] *Kavli Institute for Cosmology, Madingley Road, Cambridge CB3 0HA, UK*
1510.06699_F00005	1	0.947	`{ }^{\underline{1}}`	${ }^{\underline{1}}$	Following an initial attempt by Utiyama[1], the idea
1510.06699_F00006	1	0.998	`\varphi`	φ	(or fields) φ with energy-momentum and, in general,
1510.06699_F00007	1	0.798	`{ }^{\text {a) }}`	${ }^{\text {a) }}$	[a)] Electronic mail: a.n.lasenby@mrao.cam.ac.uk
1510.06699_F00008	1	0.997	`{ }^{\text {b) }}`	${ }^{\text {b) }}$	[b)] Electronic mail: mph@mrao.cam.ac.uk
1510.06699_F00009	1	1.000	`S_{\mathrm{M}}=`	$S_{\mathrm{M}}=$	field dynamics are described by a matter action $S_{\mathrm{M}} =$
1510.06699_F00010	1	1.000	`\int L_{\mathrm{M}}\left(\varphi, \partial_{\mu} \varphi \right) d^{4} x`	$\int L_{\mathrm{M}}\left(\varphi, \partial_{\mu} \varphi \right) d^{4} x$	$\int L_{\mathrm{M}}(\varphi, \partial_\mu\varphi)\, d^4x$ that is invariant under global Poincaré
1510.06699_F00011	1	1.000	`h_{a}{ }^{\mu}`	$h_{a}{ }^{\mu}$	new field variables $h_a{}^\mu$ and $A^{ab}{}_\mu$, which are interpreted
1510.06699_F00012	1	1.000	`A^{a b}{ }_{\mu}`	$A^{a b}{ }_{\mu}$	new field variables $h_a{}^\mu$ and $A^{ab}{}_\mu$, which are interpreted
1510.06699_F00013	1	0.990	`\mathcal{D}_{a} \varphi`	$\mathcal{D}_{a} \varphi$	a covariant derivative $\mathcal{D}_a\varphi$, and the matter action in the
1510.06699_F00014	1	1.000	`S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}\left(\varphi, \mathcal{D}_{a} \varphi\right) d^{4} x`	$S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}\left(\varphi, \mathcal{D}_{a} \varphi\right) d^{4} x$	to obtain $S_{\mathrm{M}} = \int h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a\varphi)\, d^4x$ (where the factor
1510.06699_F00015	1	1.000	`h \equiv \operatorname{det}\left(h_{a}{ }^{\mu}\right)`	$h \equiv \operatorname{det}\left(h_{a}{ }^{\mu}\right)$	containing $h \equiv \det(h_a{}^\mu)$ is required to make the inte-
1510.06699_F00016	2	1.000	`\left[\mathcal{D}_{a}, \mathcal{D}_{b}\right] \varphi`	$\left[\mathcal{D}_{a}, \mathcal{D}_{b}\right] \varphi$	ering $[\mathcal{D}_a, \mathcal{D}_b]\varphi$. This procedure yields two field strength
1510.06699_F00017	2	1.000	`\mathcal{R}^{a b}{ }_{c d}(h, A, \partial A)`	$\mathcal{R}^{a b}{ }_{c d}(h, A, \partial A)$	tensors: $\mathcal{R}^{ab}{}_{cd}(h, A, \partial A)$ corresponding to the gauge field
1510.06699_F00018	2	0.999	`\mathcal{T}^{a}{ }_{b c}(h, \partial h, A)`	$\mathcal{T}^{a}{ }_{b c}(h, \partial h, A)$	Poincaré group; and $\mathcal{T}^a{}_{bc}(h, \partial h, A)$ corresponding to
1510.06699_F00019	2	1.000	`S_{\mathrm{G}}=\int h^{-1} L_{\mathrm{G}}\left( \mathcal{R}^{a b}{ }_{c d}, \mathcal{T}^{a}{ }_{b c} \right) d^{4} x`	$S_{\mathrm{G}}=\int h^{-1} L_{\mathrm{G}}\left(\mathcal{R}^{a b}{ }_{c d}, \mathcal{T}^{a}{ }_{b c} \right) d^{4} x$	$S_{\mathrm{G}} = \int h^{-1} L_{\mathrm{G}}(\mathcal{R}^{ab}{}_{cd}, \mathcal{T}^a{}_{bc})\, d^4x$, where the Lagrangian
1510.06699_F00020	2	1.000	`L_{\mathrm{G}}`	L_{G}	L_{G} is a scalar depending on the two field strengths.
1510.06699_F00021	2	1.000	`\tau^{k}{ }_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_{a}{ }^{\mu}`	$\tau^{k}{ }_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_{a}{ }^{\mu}$	energy-momentum $\tau^k{}_\mu \equiv \delta\mathcal{L}_{\mathrm{M}}/\delta h_a{}^\mu$ and spin-angular-
1510.06699_F00022	2	1.000	`\sigma_{a b}{ }^{\mu} \equiv \delta \mathcal{L}_{ \mathrm{M}} / \delta A^{a b}{ }_{\mu}`	$\sigma_{a b}{ }^{\mu} \equiv \delta \mathcal{L}_{ \mathrm{M}} / \delta A^{a b}{ }_{\mu}$	momentum $\sigma_{ab}{}^\mu \equiv \delta\mathcal{L}_{\mathrm{M}}/\delta A^{ab}{}_\mu$ of the matter field act
1510.06699_F00023	2	1.000	`\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}} \left(\varphi, \mathcal{D}_{a} \varphi\right)`	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}} \left(\varphi, \mathcal{D}_{a} \varphi\right)$	as sources, where $\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}}(\varphi, \mathcal{D}_a\varphi)$. In contrast to
1510.06699_F00024	2	1.000	`\mathcal{R} \equiv \mathcal{R}^{a b}{ }_{a b}`	$\mathcal{R} \equiv \mathcal{R}^{a b}{ }_{a b}$	derivatives of the gauge fields, namely $\mathcal{R} \equiv \mathcal{R}^{ab}{}_{ab}$. In-
1510.06699_F00025	2	1.000	`L_{\mathrm{G}} \propto \mathcal{R}`	$L_{\mathrm{G}} \propto \mathcal{R}$	deed, Kibble chose simply to set $L_{\mathrm{G}} \propto \mathcal{R}$, which defines
1510.06699_F00026	2	0.733	`{ }^{\underline{5}}`	${ }^{\underline{5}}$	transformations has been criticised by Hehl et al.[5], who
1510.06699_F00027	2	1.000	`\delta \varphi \equiv \varphi^{\prime}\left(x^{\prime} \right)-\varphi(x)`	$\delta \varphi \equiv \varphi^{\prime}\left(x^{\prime} \right)-\varphi(x)$	ble considered the 'total' variation $\delta\varphi \equiv \varphi'(x') - \varphi(x)$ of
1510.06699_F00028	2	1.000	`\delta_{0} \varphi \equiv \varphi^{\prime}(x)-\varphi(x)`	$\delta_{0} \varphi \equiv \varphi^{\prime}(x)-\varphi(x)$	ation $\delta_0\varphi \equiv \varphi'(x) - \varphi(x)$. The latter approach is closer
1510.06699_F00029	2	1.000	`\delta_{0} \varphi`	$\delta_{0} \varphi$	matter field transformation $\delta_0\varphi$, the partial derivative in
1510.06699_F00030	2	0.988	`{ }^{6,9}`	${ }^{6,9}$	theory in Minkowski spacetime[6,9], in the same way as the
1510.06699_F00031	2	0.991	`{ }^{\underline{10}}`	${ }^{\underline{10}}$	see Blagojevic[10], in which many of the issues above are
1510.06699_F00032	2	1.000	`\mathcal{R}^{a b}{ }_{c d}`	$\mathcal{R}^{a b}{ }_{c d}$	sors $\mathcal{R}^{ab}{}_{cd}$ and $\mathcal{T}^a{}_{bc}$, and their inclusion in L_{G} cannot
1510.06699_F00033	2	1.000	`\mathcal{T}^{a}{ }_{b c}`	$\mathcal{T}^{a}{ }_{b c}$	sors $\mathcal{R}^{ab}{}_{cd}$ and $\mathcal{T}^a{}_{bc}$, and their inclusion in L_{G} cannot
1510.06699_F00034	2	1.000	`{ }^{11,12}`	${ }^{11,12}$	adding the linear Hilbert-Einstein term[11,12]. Hence, one

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1510.06699_F00035	2	1.000	`\kappa=8 \pi G / c^{4}`	$\kappa=8 \pi G / c^{4}$	where $\kappa = 8\pi G/c^4$ is Einstein's gravitational constant,
1510.06699_F00036	2	1.000	`\Lambda`	Λ	Λ is a cosmological constant, a is a dimensionless free
1510.06699_F00037	2	1.000	`a`	a	Λ is a cosmological constant, a is a dimensionless free
1510.06699_F00038	3	1.000	`\alpha_{i}`	α_{i}	in which the α_i and β_i are also dimensionless free parame-
1510.06699_F00039	3	1.000	`\beta_{i}`	β_{i}	in which the α_i and β_i are also dimensionless free parame-
1510.06699_F00040	3	0.362	`{ }^{\underline{12,14}-\underline{16}}`	${ }^{\underline{12,14}-\underline{16}}$	investigated previously[12,14–16] and may, in principle, be
1510.06699_F00041	3	1.000	`\alpha_{1}, \alpha_{3}`	α_{1}, α_{3}	alised Gauss–Bonnet identity to set one of α_1, α_3 or α_6
1510.06699_F00042	3	1.000	`\alpha_{6}`	α_{6}	alised Gauss–Bonnet identity to set one of α_1, α_3 or α_6
1510.06699_F00043	3	0.999	`c=\hbar=1`	$c=\hbar=1$	in the form (1) so that, in natural units ($c = \hbar = 1$,
1510.06699_F00044	3	0.944	`{ }^{-4}`	${ }^{-4}$	$(\text{length})^{-4}$, and hence the corresponding action S_{G} is di-
1510.06699_F00045	3	0.944	`S_{\mathrm{G}}`	S_{G}	$(\text{length})^{-4}$, and hence the corresponding action S_{G} is di-
1510.06699_F00046	3	1.000	`\kappa`	κ	mensionless; note that in natural units κ has dimensions
1510.06699_F00047	3	0.986	`{ }^{-2}`	${ }^{-2}$	of $(\text{length})^2$ and Λ has dimensions of $(\text{length})^{-2}$.
1510.06699_F00048	3	0.585	`{ }^{19,20}`	${ }^{19,20}$	suffer from Ostrogradsky's instability[19,20], wherein the
1510.06699_F00049	3	1.000	`b^{a}{ }_{\mu}`	$b^{a}{ }_{\mu}$	$b^a{}_\mu$ of the inverse h-field) and $A^{ab}{}_\mu$, the 'curvature' and
1510.06699_F00050	3	1.000	`h`	h	$b^a{}_\mu$ of the inverse h-field) and $A^{ab}{}_\mu$, the 'curvature' and
1510.06699_F00051	3	1.000	`b^{a}{ }_{0}`	$b^{a}{ }_{0}$	derivatives of $b^a{}_0$ and $A^{ab}{}_0$; thus 10 of the field equa-
1510.06699_F00052	3	1.000	`A^{a b}{ }_{0}`	$A^{a b}{ }_{0}$	derivatives of $b^a{}_0$ and $A^{ab}{}_0$; thus 10 of the field equa-
1510.06699_F00053	3	1.000	`b^{a}{ }_{\alpha}`	$b^{a}{ }_{\alpha}$	remaining 30 variables $b^a{}_\alpha$ and $A^{ab}{}_\alpha$ $(\alpha = 1,2,3)$, one
1510.06699_F00054	3	1.000	`A^{a b}{ }_{\alpha}(\alpha=1,2,3)`	$A^{a b}{ }_{\alpha}(\alpha=1,2,3)$	remaining 30 variables $b^a{}_\alpha$ and $A^{ab}{}_\alpha$ $(\alpha = 1,2,3)$, one
1510.06699_F00055	3	1.000	`\mathcal{R}+\mathcal{T}^{2}`	$\mathcal{R}+\mathcal{T}^{2}$	so-called $\mathcal{R}+\mathcal{T}^2$ theory, for which $L_{\mathrm{G}} = \kappa^{-1}(a\mathcal{R}+L_{\mathcal{T}^2})$.
1510.06699_F00056	3	1.000	`L_{\mathrm{G}}=\kappa^{-1}\left(a \mathcal{R}+L_{ \mathcal{T}^{2}}\right)`	$L_{\mathrm{G}}=\kappa^{-1}\left(a \mathcal{R}+L_{\mathcal{T}^{2}}\right)$	so-called $\mathcal{R}+\mathcal{T}^2$ theory, for which $L_{\mathrm{G}} = \kappa^{-1}(a\mathcal{R}+L_{\mathcal{T}^2})$.
1510.06699_F00057	3	1.000	`A`	A	namics imposes an algebraic relation between the A-field
1510.06699_F00058	3	1.000	`L_{\mathcal{R}^{2}}`	$L_{\mathcal{R}^{2}}$	in $L_{\mathcal{R}^2}$, in which case the 'torsion' becomes a propagating
1510.06699_F00059	3	1.000	`\mathcal{R}+\mathcal{R}^{2}`	$\mathcal{R}+\mathcal{R}^{2}$	field. Consequently, a number of $\mathcal{R}+\mathcal{R}^2$-type theories
1510.06699_F00060	3	0.970	`\mathcal{R}^{2}+\mathcal{T}^{2}`	$\mathcal{R}^{2}+\mathcal{T}^{2}$	have been investigated, as have $\mathcal{R}^2+\mathcal{T}^2$ theories[21]. It
1510.06699_F00061	3	0.986	`\mathcal{R}+\mathcal{R}^{2}+\mathcal{T}^{2}`	$\mathcal{R}+\mathcal{R}^{2}+\mathcal{T}^{2}$	$\mathcal{R}+\mathcal{R}^2+\mathcal{T}^2$ theories is given by Obukhov et al.[12]. The
1510.06699_F00062	3	1.000	`\mathcal{R}`	$\mathcal{R}$	if L_{G} consists of $\mathcal{R}$ and $\mathcal{T}^2$ terms, then it contains dimen-
1510.06699_F00063	3	1.000	`\mathcal{T}^{2}`	$\mathcal{T}^{2}$	if L_{G} consists of $\mathcal{R}$ and $\mathcal{T}^2$ terms, then it contains dimen-
1510.06699_F00064	3	1.000	`\mathcal{R}^{2}`	$\mathcal{R}^{2}$	quantisation. On the other hand, the $\mathcal{R}^2$ terms contain
1510.06699_F00065	3	1.000	`{ }^{\underline{25}}`	${ }^{\underline{25}}$	can eliminate such higher derivatives from the theory[25],
1510.06699_F00066	3	0.881	`{ }^{\underline{26}}`	${ }^{\underline{26}}$	well-defined initial value problem[26]. Nonetheless, the
1510.06699_F00067	3	0.995	`a, \alpha_{i}`	a, α_{i}	that the parameters a, α_i and β_i in the Lagrangian L_{G}
1510.06699_F00068	3	0.924	`{ }^{\underline{27}}`	${ }^{\underline{27}}$	obey certain conditions[27].
1510.06699_F00069	4	1.000	`h_{a}^{\mu}`	h_{a}^{μ}	**Scale-invariant gauge theories of grav**
1510.06699_F00070	4	1.000	`L_{\mathrm{G}}^{(2)}`	$L_{\mathrm{G}}^{(2)}$	2) *Kavli Institute for Cosmology, Madingley Road, C*
1510.06699_F00071	4	1.000	`k`	k	Poincaré invariance to include invariance under
1510.06699_F00072	4	1.000	`J^{P}`	J^{P}	in Minkowski spacetime. Our approach therefo
1510.06699_F00073	4	1.000	`k^{a}`	k^{a}	scale-invariant Poincaré gauge theory (PGT) ar
1510.06699_F00074	4	0.998	`J^{P}=2^{+}`	$J^{P}=2^{+}$	general invariance properties than those impose
1510.06699_F00075	4	1.000	`J^{P}=2^{ \pm}, 1^{ \pm}, 0^{ \pm}`	$J^{P}=2^{ \pm}, 1^{ \pm}, 0^{ \pm}$	dilations, which includes its 'normal' transforma
1510.06699_F00076	4	1.000	`2(5+3+1)=18`	$2(5+3+1)=18$	Weyl gauge theory (eWGT) has a number of int
1510.06699_F00077	4	1.000	`S`	S	show that they are special cases of WGT and e
1510.06699_F00078	4	1.000	`k^{2}`	k^{2}	fundamental interactions, the gauge principle has proved
1510.06699_F00079	4	1.000	`-R /\left(k^{2}-m^{2}\right)`	$-R /\left(k^{2}-m^{2}\right)$	it is natural to consider the gauge group description of
1510.06699_F00080	4	1.000	`R>0`	$R>0$	it is natural to consider the gauge group description of
1510.06699_F00081	4	1.000	`m^{2} \geq 0`	$m^{2} \geq 0$	gravitation. In the gravitational case, however, in place

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1510.06699_F00082	4	1.000	`{ }^{18,33}`	^{18,33}	Following an initial attempt by Utiyama, the idea of gauging the Poincaré group was fully developed by
1510.06699_F00083	4	1.000	`J^{P}=1^{-}`	$J^{P}=1^{-}$	Kibble (and also considered by Sciama). The physical model envisaged in Kibble's approach is an un-
1510.06699_F00084	4	1.000	`\sim 1 / k^{4}`	$\sim 1 / k^{4}$	as $\sim 1/k^4$, whereas those for the tordions should be
1510.06699_F00085	4	1.000	`\sim 1 / k^{2}`	$\sim 1 / k^{2}$	$\sim 1/k^2$, since tordion vertices are better behaved than
1510.06699_F00086	4	1.000	`{ }^{29,31,35}`	^{29,31,35}	those of gravitons[29,31,35]. Pure $1/k^4$ poles can occur in
1510.06699_F00087	4	1.000	`1 / k^{4}`	$1 / k^{4}$	those of gravitons[29,31,35]. Pure $1/k^4$ poles can occur in
1510.06699_F00088	4	1.000	`\left(k^{2}-m_{1}^{2}\right)^{-1}\left(k^{2}-m_{2}^{2} \right)^{-1}=`	$\left(k^{2}-m_{1}^{2}\right)^{-1}\left(k^{2}-m_{2}^{2}\right)^{-1}=$	pole structures of the form $(k^2-m_1^2)^{-1}(k^2-m_2^2)^{-1} =$
1510.06699_F00089	4	1.000	`\left(m_{2}^{2}-m_{1}^{2}\right)^{-1}\left[ \left(k^{2}-m_{2}^{2}\right)^{-1}-\left(k^{2}-m_{1}^{2} \right)^{-1}\right]`	$\left(m_{2}^{2}-m_{1}^{2}\right)^{-1}\left[\left(k^{2}-m_{2}^{2}\right)^{-1}-\left(k^{2}-m_{1}^{2}\right)^{-1}\right]$	$(m_2^2-m_1^2)^{-1}[(k^2-m_2^2)^{-1}-(k^2-m_1^2)^{-1}]$. These can
1510.06699_F00090	4	1.000	`{ }^{6,39}`	^{6,39}	have been investigated[6,39]. More innovative approaches
1510.06699_F00091	4	0.945	`{ }^{40,41}`	^{40,41}	to quantisation of PGT have been pursued[40,41], but are
1510.06699_F00092	4	0.991	`{ }^{\underline{31,43}}`	$^{\underline{31,43}}$	tary and renormalizable theory[31,43], without having to
1510.06699_F00093	4	0.566	`{ }^{44-46}`	$^{44-46}$	resort to non-local theories[44–46]. In particular, perhaps
1510.06699_F00094	4	1.000	`m^{2} \neq 0`	$m^{2} \neq 0$	either continuous (if $m^2 \neq 0$) or all the masses vanish.
1510.06699_F00095	4	0.838	`\mathrm{C}(1,3)^{\underline{48}-50}`	$\mathrm{C}(1,3)^{\underline{48}-50}$	of the full conformal group C(1,3)[48–50]. This may be for-
1510.06699_F00096	5	1.000	`S_{\mathrm{M}}=\int L_{\mathrm{M}}\left(\varphi, \partial_{\mu} \varphi\right) d^{4} x`	$S_{\mathrm{M}}=\int L_{\mathrm{M}}\left(\varphi, \partial_{\mu} \varphi\right) d^{4} x$	ter action $S_{\mathrm{M}} = \int L_{\mathrm{M}}(\varphi,\partial_\mu\varphi)\, d^4x$ that is invariant under
1510.06699_F00097	5	1.000	`B_{\mu}`	B_{μ}	a new gravitational gauge field B_μ corresponding to the
1510.06699_F00098	5	0.870	`{ }^{51-53}`	$^{51-53}$	dilation part of the Weyl group[51–53]. Similar to PGT, the
1510.06699_F00099	5	1.000	`h_{a}{ }^{\mu}, A^{a b}{ }_{\mu}`	$h_{a}{ }^{\mu}, A^{a b}{ }_{\mu}$	gravitational gauge fields $h_a{}^\mu$, $A^{ab}{}_\mu$ and B_μ are used to
1510.06699_F00100	5	0.991	`\mathcal{D}_{a}^{*} \varphi`	$\mathcal{D}_{a}^{*} \varphi$	assemble a covariant derivative $\mathcal{D}^*_a\varphi$, and the matter ac-
1510.06699_F00101	5	1.000	`S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}\left(\varphi, \mathcal{D}_{a}^{*} \varphi\right) d^{4} x`	$S_{\mathrm{M}}=\int h^{-1} L_{\mathrm{M}}\left(\varphi, \mathcal{D}_{a}^{*} \varphi\right) d^{4} x$	coupling procedure, so that $S_{\mathrm{M}} = \int h^{-1}L_{\mathrm{M}}(\varphi,\mathcal{D}^*_a\varphi)\,d^4x$.
1510.06699_F00102	5	1.000	`L_{\mathrm{M}}`	L_{M}	cise form of L_{M}. In particular, since h^{-1} has a Weyl
1510.06699_F00103	5	1.000	`h^{-1}`	h^{-1}	cise form of L_{M}. In particular, since h^{-1} has a Weyl
1510.06699_F00104	5	0.788	`w\left(h^{-1}\right)=4`	$w\left(h^{-1}\right)=4$	(or conformal) weight $w(h^{-1})=4$ (see Section II B), the
1510.06699_F00105	5	1.000	`w\left(L_{\mathrm{G}}\right)=-4`	$w\left(L_{\mathrm{G}}\right)=-4$	Lagrangian L_{M} must have a weight $w(L_{\mathrm{G}})=-4$.
1510.06699_F00106	5	1.000	`\left[\mathcal{D}_{a}^{*}, \mathcal{D}_{b}^{*}\right] \varphi`	$\left[\mathcal{D}_{a}^{*}, \mathcal{D}_{b}^{*}\right] \varphi$	by considering $[\mathcal{D}^*_a,\mathcal{D}^*_b]\varphi$. One obtains $\mathcal{R}^{ab}{}_{cd}(h,A,\partial A)$
1510.06699_F00107	5	1.000	`\mathcal{T}^{* a}{ }_{b c}(h, \partial h, A, B)`	$\mathcal{T}^{* a}{ }_{b c}(h, \partial h, A, B)$	and $\mathcal{T}^{*a}{}_{bc}(h,\partial h,A,B)$ as before (but where the form
1510.06699_F00108	5	1.000	`\mathcal{H}_{a b}(h, \partial B)`	$\mathcal{H}_{a b}(h, \partial B)$	tional field strength $\mathcal{H}_{ab}(h,\partial B)$ corresponding to the new
1510.06699_F00109	5	1.000	`S_{\mathrm{G}}=`	$S_{\mathrm{G}}=$	free gravitational action then has the general form $S_{\mathrm{G}} =$
1510.06699_F00110	5	0.999	`\int h^{-1} L_{\mathrm{G}}\left(\mathcal{R}^{a b}{ }_{c d}, \mathcal{T}^{* a}{ }_{b c}, \mathcal{H}_{a b}\right) d^{4} x`	$\int h^{-1} L_{\mathrm{G}}\left(\mathcal{R}^{a b}{ }_{c d}, \mathcal{T}^{* a}{ }_{b c}, \mathcal{H}_{a b}\right) d^{4} x$	$\int h^{-1}L_{\mathrm{G}}(\mathcal{R}^{ab}{}_{cd},\mathcal{T}^{*a}{}_{bc},\mathcal{H}_{ab})\,d^4x$. As usual, the total ac-
1510.06699_F00111	5	1.000	`\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu}`	$\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu}$	$\sigma_{ab}{}^\mu \equiv \delta\mathcal{L}_{\mathrm{M}}/\delta A^{ab}{}_\mu$ and dilation current $\zeta^\mu \equiv \delta\mathcal{L}_{\mathrm{M}}/\delta B_\mu$
1510.06699_F00112	5	1.000	`w\left(\mathcal{R}^{a b}{ }_{c d}\right)=w\left( \mathcal{H}_{a b}\right)=-2`	$w\left(\mathcal{R}^{a b}{ }_{c d}\right)=w\left(\mathcal{H}_{a b}\right)=-2$	$w(\mathcal{R}^{ab}{}_{cd}) = w(\mathcal{H}_{ab}) = -2$ and $w(\mathcal{T}^{*a}{}_{bc}) = -1$, which
1510.06699_F00113	5	1.000	`w\left(\mathcal{T}^{* a}{ }_{b c}\right)=-1`	$w\left(\mathcal{T}^{* a}{ }_{b c}\right)=-1$	$w(\mathcal{R}^{ab}{}_{cd}) = w(\mathcal{H}_{ab}) = -2$ and $w(\mathcal{T}^{*a}{}_{bc}) = -1$, which
1510.06699_F00114	5	1.000	`\mathcal{H}_{a b}`	$\mathcal{H}_{a b}$	means that L_{G} can be quadratic in $\mathcal{R}^{ab}{}_{cd}$ and $\mathcal{H}_{ab}$, while
1510.06699_F00115	5	1.000	`\mathcal{T}^{* a}{ }_{b c}`	$\mathcal{T}^{* a}{ }_{b c}$	terms linear in $\mathcal{R}\equiv\mathcal{R}^{ab}{}_{ab}$ or quadratic in $\mathcal{T}^{*a}{}_{bc}$ are
1510.06699_F00116	5	1.000	`\mathcal{R}^{2}+\mathcal{H}^{2}`	$\mathcal{R}^{2}+\mathcal{H}^{2}$	in the first derivatives of gauge fields, is of $\mathcal{R}^2+\mathcal{H}^2$ type:
1510.06699_F00117	5	1.000	`\xi`	ξ	where the α_i and ξ are dimensionless free parameters.
1510.06699_F00118	5	1.000	`\mathcal{T}^{* 2}`	$\mathcal{T}^{* 2}$	in PGT; essentially the $\mathcal{R}$ and $\mathcal{T}^{*2}$ terms, which possess
1510.06699_F00119	5	1.000	`w=-4`	$w=-4$	tal Lagrangian must have a Weyl weight $w=-4$ clearly
1510.06699_F00120	5	0.999	`\phi`	ϕ	ϕ with Weyl weight $w(\phi) = -1$, often termed the

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1510.06699_F00121	5	0.999	`w(\phi)=-1`	$w(\phi)=-1$	ϕ with Weyl weight $w(\phi) = -1$, often termed the
1510.06699_F00122	5	0.785	`(\mathrm{s}){ }^{\underline{10}}`	$(\mathrm{s}){ }^{\underline{10}}$	compensator(s)[10]. This opens up possibilities for the in-
1510.06699_F00123	5	0.803	`\phi^{2} \mathcal{R}`	$\phi^{2} \mathcal{R}$	ample, terms proportional to $\phi^2\mathcal{R}$ or $\phi^2 L_{\mathcal{T}^{*2}}$ are Weyl-
1510.06699_F00124	5	0.803	`\phi^{2} L_{\mathcal{T}^{* 2}}`	$\phi^{2} L_{\mathcal{T}^{* 2}}$	ample, terms proportional to $\phi^2\mathcal{R}$ or $\phi^2 L_{\mathcal{T}^{*2}}$ are Weyl-
1510.06699_F00125	5	1.000	`{ }^{54,55}`	${ }^{54,55}$	been widely investigated[54,55], whereas the phenomeno-
1510.06699_F00126	5	0.999	`\phi^{2} L_{\mathcal{T} * 2}`	$\phi^{2} L_{\mathcal{T} * 2}$	logical value of the $\phi^2 L_{\mathcal{T}^{*2}}$ terms has received relatively
1510.06699_F00127	5	0.969	`{ }^{56,57}`	${ }^{56,57}$	little attention[56,57]. In any case, the resulting total La-
1510.06699_F00128	5	1.000	`\psi`	ψ	tion for a free Dirac field ψ, which has Weyl weight
1510.06699_F00129	5	1.000	`w(\psi)=w(\bar{\psi})=-\frac{3}{2}`	$w(\psi)=w(\bar{\psi})=-\frac{3}{2}$	$w(\psi) = w(\bar{\psi}) = -\frac{3}{2}$. This action is not scale-invariant
1510.06699_F00130	5	1.000	`m \bar{\psi} \psi`	$m \bar{\psi} \psi$	owing to the mass term $m\bar{\psi}\psi$, so it appears that one re-
1510.06699_F00131	5	0.983	`{ }^{10} \zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu}`	${ }^{10} \zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu}$	vergence of the dilation current[10] $\zeta^\mu \equiv \delta\mathcal{L}_{\mathrm{M}}/\delta B_\mu$. In
1510.06699_F00132	6	1.000	`m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi`	$m \bar{\psi} \psi \rightarrow \mu \phi \bar{\psi} \psi$	replacement $m\bar{\psi}\psi \rightarrow \mu\phi\bar{\psi}\psi$ in the Dirac action, where μ
1510.06699_F00133	6	1.000	`\mu`	μ	replacement $m\bar{\psi}\psi \rightarrow \mu\phi\bar{\psi}\psi$ in the Dirac action, where μ
1510.06699_F00134	6	1.000	`\mu \phi`	$\mu \phi$	is a dimensionless parameter but $\mu\phi$ has the dimensions
1510.06699_F00135	6	1.000	`G`	G	'constant' G to vary with cosmic epoch. This idea can
1510.06699_F00136	6	1.000	`\phi=\phi_{0}`	$\phi=\phi_{0}$	stein gauge. Setting $\phi = \phi_0$ in the equation of motion
1510.06699_F00137	6	1.000	`m=\mu \phi_{0}`	$m=\mu \phi_{0}$	tation as a massive field with $m = \mu\phi_0$. It is usually
1510.06699_F00138	6	1.000	`w`	w	$h_a{}^\mu$ and $A^{ab}{}_\mu$ transform covariantly with weights of w,
1510.06699_F00139	6	1.000	`\mathcal{R}_{a b c d}`	$\mathcal{R}_{a b c d}$	strength (or 'curvature') $\mathcal{R}_{abcd}$ and translational gauge
1510.06699_F00140	6	1.000	`\mathcal{T}_{a b c}`	$\mathcal{T}_{a b c}$	field strength (or 'torsion') $\mathcal{T}_{abc}$ are treated in a more
1510.06699_F00141	6	0.617	`{ }^{59-62,89}`	${ }^{59-62,89}$	so is the total action)[59–62,89].
1510.06699_F00142	6	1.000	`Y_{4}`	Y_{4}	Weyl–Cartan Y_4 spacetime (although we do give a brief
1510.06699_F00143	7	1.000	`\widehat{Y}_{4}`	$\widehat{Y}_{4}$	a new spacetime geometry (that we term $\widehat{Y}_4$), which is
1510.06699_F00144	7	1.000	`\mathcal{M}`	$\mathcal{M}$	We begin with a Minkowski spacetime $\mathcal{M}$, labelled
1510.06699_F00145	7	1.000	`x^{\mu}`	x^{μ}	using Cartesian inertial coordinates x^μ, on which the
1510.06699_F00146	7	1.000	`\varphi(x)`	$\varphi(x)$	gravitational) fields $\varphi(x)$ is described by a matter action
1510.06699_F00147	7	0.959	`\boldsymbol{e}_{\mu}`	$\boldsymbol{e}_{\mu}$	$\boldsymbol{e}_\mu$, which are tangents to the coordinate curves and for
1510.06699_F00148	7	0.997	`\boldsymbol{e}_{\mu} \cdot \boldsymbol{e}_{\nu}=\eta_{\mu \nu}`	$\boldsymbol{e}_{\mu} \cdot \boldsymbol{e}_{\nu}=\eta_{\mu \nu}$	Cartesian inertial coordinates satisfy $\boldsymbol{e}_\mu \cdot \boldsymbol{e}_\nu = \eta_{\mu\nu}$, where
1510.06699_F00149	7	1.000	`\eta_{\mu \nu}=\operatorname{diag}(1,-1,-1,-1)`	$\eta_{\mu \nu}=\operatorname{diag}(1,-1,-1,-1)$	$\eta_{\mu\nu} = \mathrm{diag}(1,-1,-1,-1)$. At each point of $\mathcal{M}$, we can
1510.06699_F00150	7	1.000	`\hat{\boldsymbol{e}}_{a}(x)`	$\hat{\boldsymbol{e}}_{a}(x)$	mal tetrad $\hat{\boldsymbol{e}}_a(x)$, such that $\hat{\boldsymbol{e}}_a \cdot \hat{\boldsymbol{e}}_b = \eta_{ab}$. In global
1510.06699_F00151	7	1.000	`\hat{\boldsymbol{e}}_{a} \cdot \hat{\boldsymbol{e}}_{b}=\eta_{a b}`	$\hat{\boldsymbol{e}}_{a} \cdot \hat{\boldsymbol{e}}_{b}=\eta_{a b}$	mal tetrad $\hat{\boldsymbol{e}}_a(x)$, such that $\hat{\boldsymbol{e}}_a \cdot \hat{\boldsymbol{e}}_b = \eta_{ab}$. In global
1510.06699_F00152	7	1.000	`\boldsymbol{e}_{\mu}(x)`	$\boldsymbol{e}_{\mu}(x)$	$\boldsymbol{e}_\mu(x)$, i.e. $\hat{\boldsymbol{e}}_a = \delta_a^\mu \boldsymbol{e}_\mu$. We will adopt the common con-
1510.06699_F00153	7	1.000	`\hat{\boldsymbol{e}}_{a}=\delta_{a}^{\mu} \boldsymbol{e}_{\mu}`	$\hat{\boldsymbol{e}}_{a}=\delta_{a}^{\mu} \boldsymbol{e}_{\mu}$	$\boldsymbol{e}_\mu(x)$, i.e. $\hat{\boldsymbol{e}}_a = \delta_a^\mu \boldsymbol{e}_\mu$. We will adopt the common con-
1510.06699_F00154	7	1.000	`\operatorname{SO}(3,1)`	$\operatorname{SO}(3,1)$	$\mathrm{SO}(3,1)$ by $\mathrm{SL}(2,C)$, so that spinor fields can be
1510.06699_F00155	7	1.000	`\operatorname{SL}(2, C)`	$\operatorname{SL}(2, C)$	$\mathrm{SO}(3,1)$ by $\mathrm{SL}(2,C)$, so that spinor fields can be
1510.06699_F00156	7	1.000	`\left\{\varphi^{a}(x)\right\}`	$\left\{\varphi^{a}(x)\right\}$	ter field components $\{\varphi^a(x)\}$ simply carry a generic Ro-
1510.06699_F00157	8	0.999	`\mathrm{S}(\Lambda)`	$\mathrm{S}(\Lambda)$	where $\mathrm{S}(\Lambda)$ is the matrix corresponding to the element Λ
1510.06699_F00158	8	1.000	`\Lambda^{\mu}{ }_{\nu}=\delta_{\nu}^{\mu}`	$\Lambda^{\mu}{ }_{\nu}=\delta_{\nu}^{\mu}$	formation, i.e. with $\Lambda^\mu{}_\nu = \delta^\mu_\nu$ and $a^\mu = 0$ in (6) and
1510.06699_F00159	8	1.000	`a^{\mu}=0`	$a^{\mu}=0$	formation, i.e. with $\Lambda^\mu{}_\nu = \delta^\mu_\nu$ and $a^\mu = 0$ in (6) and
1510.06699_F00160	8	1.000	`e^{\rho}`	e^{ρ}	tor e^ρ. Adopting global inertial coordinates, such that
1510.06699_F00161	8	0.999	`d s^{2}=\eta_{\mu \nu} d x^{\mu} d x^{\nu}`	$d s^{2}=\eta_{\mu \nu} d x^{\mu} d x^{\nu}$	$ds^2 = \eta_{\mu\nu}\,dx^\mu\,dx^\nu$, the metric must therefore transform
1510.06699_F00162	8	1.000	`\eta_{\mu \nu} \rightarrow \gamma_{\mu \nu}=e^{2 \rho} \eta_{\mu \nu}`	$\eta_{\mu \nu} \rightarrow \gamma_{\mu \nu}=e^{2 \rho} \eta_{\mu \nu}$	according to $\eta_{\mu\nu} \rightarrow \gamma_{\mu\nu} = e^{2\rho}\eta_{\mu\nu}$. Hence, under this
1510.06699_F00163	8	1.000	`w=2`	$w=2$	active dilation the metric has a Weyl weight of $w = 2$.
1510.06699_F00164	8	1.000	`x^{\prime \mu}=e^{\rho} x^{\mu}`	$x^{\prime \mu}=e^{\rho} x^{\mu}$	ordinate transformation $x'^\mu = e^\rho x^\mu$, which 'shrinks' the

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1510.06699_F00165	8	1.000	`\gamma_{\mu \nu}^{\prime}=e^{-2 \rho} \gamma_{\mu \nu}= \eta_{\mu \nu}`	$\gamma_{\mu \nu}^{\prime}=e^{-2 \rho} \gamma_{\mu \nu}= \eta_{\mu \nu}$	coordinate transformation, $\gamma'_{\mu\nu} = e^{-2\rho}\gamma_{\mu\nu} = \eta_{\mu\nu}$ and so,
1510.06699_F00166	8	1.000	`e^{4 \rho}`	$e^{4 \rho}$	transformation (6) (for which the Jacobian equals $e^{4\rho}$), a
1510.06699_F00167	8	1.000	`m`	m	m, one has the Klein–Gordon Lagrangian
1510.06699_F00168	8	1.000	`m=0`	$m=0$	dilation invariant only for $m = 0$ and provided the Weyl
1510.06699_F00169	8	1.000	`\bar{\psi} \psi`	$\bar{\psi} \psi$	pretation of $\bar{\psi}\psi$ as a number density. As mentioned in the
1510.06699_F00170	8	1.000	`A_{\mu}`	A_{μ}	Finally, we consider the electromagnetic field A_μ
1510.06699_F00171	8	1.000	`F_{\mu \nu} \equiv \partial_{\mu} A_{\nu}-\partial_{\nu} A_{\mu}`	$F_{\mu \nu} \equiv \partial_{\mu} A_{\nu}-\partial_{\nu} A_{\mu}$	where $F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$ and J^μ is some source cur-
1510.06699_F00172	8	1.000	`J^{\mu}`	J^{μ}	where $F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$ and J^μ is some source cur-
1510.06699_F00173	8	1.000	`w_{J}`	w_{J}	where w and w_J are the Weyl weights of A_μ and J_μ
1510.06699_F00174	8	1.000	`J_{\mu}`	J_{μ}	where w and w_J are the Weyl weights of A_μ and J_μ
1510.06699_F00175	8	1.000	`w\left(A_{\mu}\right)=-1`	$w\left(A_{\mu}\right)=-1$	(9), provided that $w(A_\mu) = -1$ and $w(J_\mu) = -3$. It is
1510.06699_F00176	8	1.000	`w\left(J_{\mu}\right)=-3`	$w\left(J_{\mu}\right)=-3$	(9), provided that $w(A_\mu) = -1$ and $w(J_\mu) = -3$. It is
1510.06699_F00177	8	1.000	`J^{\mu} \equiv \bar{\psi} \gamma^{\mu} \psi`	$J^{\mu} \equiv \bar{\psi} \gamma^{\mu} \psi$	worth noting that the Dirac current $J^\mu \equiv \bar{\psi}\gamma^\mu\psi$ satisfies
1510.06699_F00178	9	1.000	`a^{\mu}`	a^{μ}	the four translation parameters a^μ and the dilation pa-
1510.06699_F00179	9	1.000	`\rho`	ρ	rameter ρ into eleven independent arbitrary functions of
1510.06699_F00180	9	1.000	`x`	x	x. This leads to a complete decoupling of the translation
1510.06699_F00181	9	1.000	`x^{\prime}`	$x^{\prime}$	lation parts. Thus, it is customary to view x' as arising
1510.06699_F00182	9	1.000	`\Lambda(x)`	$\Lambda(x)$	local Lorentz rotation $\Lambda(x)$ of the orthonormal tetrad
1510.06699_F00183	9	0.997	`\rho(x)`	$\rho(x)$	$\hat{e}_a(x)$, or local (physical) dilation $\rho(x)$, at each point.
1510.06699_F00184	9	1.000	`\Lambda^{a}{ }_{b}(x)`	$\Lambda^{a}{ }_{b}(x)$	Lorentz rotations $\Lambda^a{}_b(x)$. This means that any spinor,
1510.06699_F00185	9	1.000	`\partial_{\mu} \varphi`	$\partial_{\mu} \varphi$	field. Consequently, the partial derivative, $\partial_\mu\varphi$, is co-
1510.06699_F00186	9	1.000	`\eta_{a b}=\eta^{a b}=\operatorname{diag}(1,-1,-1,-1)`	$\eta_{a b}=\eta^{a b}=\operatorname{diag}(1,-1,-1,-1)$	ric $\eta_{ab} = \eta^{ab} = \operatorname{diag}(1, -1, -1, -1)$, whereas Greek in-
1510.06699_F00187	9	1.000	`\gamma_{\mu \nu}=\boldsymbol{e}_{\mu} \cdot \boldsymbol{e}_{\nu}`	$\gamma_{\mu \nu}=\boldsymbol{e}_{\mu} \cdot \boldsymbol{e}_{\nu}$	ric $\gamma_{\mu\nu} = \boldsymbol{e}_\mu \cdot \boldsymbol{e}_\nu$ and its inverse $\gamma^{\mu\nu}$ corresponding to
1510.06699_F00188	9	1.000	`\gamma^{\mu \nu}`	$\gamma^{\mu \nu}$	ric $\gamma_{\mu\nu} = \boldsymbol{e}_\mu \cdot \boldsymbol{e}_\nu$ and its inverse $\gamma^{\mu\nu}$ corresponding to
1510.06699_F00189	9	1.000	`\hat{\boldsymbol{e}}^{a}=\eta^{a b} \hat{ \boldsymbol{e}}_{b}`	$\hat{\boldsymbol{e}}^{a}=\eta^{a b} \hat{ \boldsymbol{e}}_{b}$	by $\hat{\boldsymbol{e}}^a = \eta^{ab}\hat{\boldsymbol{e}}_b$ and $\boldsymbol{e}^\mu = \gamma^{\mu\nu}\boldsymbol{e}_\nu$, respectively. Moreover,
1510.06699_F00190	9	1.000	`\boldsymbol{e}^{\mu}=\gamma^{\mu \nu} \boldsymbol{e}_{ \nu}`	$\boldsymbol{e}^{\mu}=\gamma^{\mu \nu} \boldsymbol{e}_{ \nu}$	by $\hat{\boldsymbol{e}}^a = \eta^{ab}\hat{\boldsymbol{e}}_b$ and $\boldsymbol{e}^\mu = \gamma^{\mu\nu}\boldsymbol{e}_\nu$, respectively. Moreover,
1510.06699_F00191	9	1.000	`\gamma_{\mu \nu}`	$\gamma_{\mu \nu}$	metric $\gamma_{\mu\nu}$ of the Minkowski spacetime obeys the trans-
1510.06699_F00192	9	1.000	`(\Lambda, \rho)`	(Λ, ρ)	two steps. Firstly, one defines the (Λ, ρ)-covariant deriva-
1510.06699_F00193	9	1.000	`D_{\mu} \varphi(x)`	$D_{\mu} \varphi(x)$	where $D_\mu\varphi(x)$ is the Λ-covariant derivative introduced
1510.06699_F00194	9	1.000	`A^{a b}{ }_{\mu}(x)`	$A^{a b}{ }_{\mu}(x)$	of complex conjugation. Most importantly, $A^{ab}{}_\mu(x)$ is
1510.06699_F00195	9	1.000	`B_{\mu}(x)`	$B_{\mu}(x)$	tions and $B_\mu(x)$ is the gauge-field corresponding to local
1510.06699_F00196	9	1.000	`\Sigma_{a b}=-\Sigma_{a b}`	$\Sigma_{a b}=-\Sigma_{a b}$	dilations. The matrices $\Sigma_{ab} = -\Sigma_{ab}$ are the generators of
1510.06699_F00197	9	0.483	`(2, C)`	$(2, C)$	the $\mathrm{SL}(2, C)$ representation $\mathsf{S}(\Lambda)$ to which $\varphi(x)$ belongs,
1510.06699_F00198	9	0.483	`(\Lambda)`	(Λ)	the $\mathrm{SL}(2, C)$ representation $\mathsf{S}(\Lambda)$ to which $\varphi(x)$ belongs,
1510.06699_F00199	9	1.000	`\Sigma_{a b}`	$\Sigma_{a b}$	Given the antisymmetry of the generator matrices Σ_{ab},
1510.06699_F00200	9	1.000	`A^{a b}{ }_{\mu}=`	$A^{a b}{ }_{\mu}=$	to share this antisymmetry property, such that $A^{ab}{}_\mu =$
1510.06699_F00201	9	0.890	`-A^{b a}{ }_{\mu}`	$-A^{b a}{ }_{\mu}$	$-A^{ba}{}_\mu$. The aim of introducing the A and B gauge fields
1510.06699_F00202	9	0.890	`B`	B	$-A^{ba}{}_\mu$. The aim of introducing the A and B gauge fields
1510.06699_F00203	9	1.000	`w=0`	$w=0$	local dilations (i.e. has Weyl weight $w = 0$).
1510.06699_F00204	10	1.000	`\partial_{\mu}^{*} \varphi(x)`	$\partial_{\mu}^{*} \varphi(x)$	It is straightforward to show that $\partial^*_\mu\varphi(x)$ itself also trans-
1510.06699_F00205	10	0.999	`D_{\mu}^{*} \varphi`	$D_{\mu}^{*} \varphi$	derivative, linearly related to $D^*_\mu\varphi$, by introducing a new,
1510.06699_F00206	10	1.000	`h_{a}{ }^{\mu}(x)`	$h_{a}{ }^{\mu}(x)$	'translational' gauge field $h_a{}^\mu(x)$ that transforms via
1510.06699_F00207	10	1.000	`w=-1`	$w=-1$	Lorentz four-vector, and has Weyl weight $w = -1$ under
1510.06699_F00208	10	1.000	`b^{a}{ }_{\mu}(x)`	$b^{a}{ }_{\mu}(x)$	$b^a{}_\mu(x)$ (which has Weyl weight $w = 1$) and satisfies

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1510.06699_F00209	10	1.000	`w=1`	$w=1$	$b^{a}{}_{\mu}(x)$ (which has Weyl weight $w=1$) and satisfies
1510.06699_F00210	10	1.000	`\partial \varphi`	$\partial \varphi$	mations in the same way as $\partial\varphi$ does under global Weyl
1510.06699_F00211	10	1.000	`x \rightarrow x^{\prime}`	$x \rightarrow x^{\prime}$	$w=-4$. However, since $x \rightarrow x'$ is a GCT, we need a
1510.06699_F00212	10	0.997	`L_{\mathrm{M}}\left(\varphi ; \mathcal{D}^{*} \varphi \right)`	$L_{\mathrm{M}}\left(\varphi ; \mathcal{D}^{*} \varphi \right)$	$L_{\mathrm{M}}(\varphi;\mathcal{D}^{*}\varphi)$ by a factor which transforms by the appro-
1510.06699_F00213	10	1.000	`w=4`	$w=4$	weight $w=4$. From the transformation properties of the
1510.06699_F00214	10	1.000	`\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}} \left(\varphi, \mathcal{D}_{a}^{*} \varphi\right)`	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}} \left(\varphi, \mathcal{D}_{a}^{*} \varphi\right)$	by the Lagrangian density $\mathcal{L}_{\mathrm{M}} \equiv h^{-1}L_{\mathrm{M}}(\varphi,\mathcal{D}_{a}^{*}\varphi)$. We
1510.06699_F00215	10	0.999	`h, A`	h, A	note that the gravitational gauge fields h, A and B con-
1510.06699_F00216	10	0.613	`L_{\mathcal{T} * 2}`	$L_{\mathcal{T} * 2}$	weight $w=-4$ and so may be added to L_{M} (where $L_{\mathcal{T}^{*2}}$
1510.06699_F00217	10	0.999	`D_{\mu}^{*}`	D_{μ}^{*}	choice, one may extend the (Λ,ρ)-covariant derivative D_{μ}^{*}
1510.06699_F00218	10	1.000	`\left(\Sigma_{c d}^{1}\right)^{a}{ }_{b}=\left( \delta_{c}^{a} \eta_{d b}-\delta_{d}^{a} \eta_{c b} \right)`	$\left(\Sigma_{c d}^{1}\right)^{a}{ }_{b}=\left(\delta_{c}^{a} \eta_{d b}-\delta_{d}^{a} \eta_{c b} \right)$	where $(\Sigma_{cd}^{1})^{a}{}_{b}=(\delta_{c}^{a}\eta_{db}-\delta_{d}^{a}\eta_{cb})$ are the generators of the
1510.06699_F00219	10	0.509	`C`	C	der GCT, in addition to $\mathrm{SL}(2,C)$ behaviour. In this case,
1510.06699_F00220	11	1.000	`{ }^{0} \nabla_{\mu}^{*} \equiv{ }^{0} \nabla_{\mu}+w B_{\mu},{ }^{0} \Gamma^{\lambda}{ }_{\mu \nu} \equiv \frac{1}{2} \gamma^{\lambda \rho}\left(\partial_{\mu} \gamma_{\nu \rho}+\partial_{\nu} \gamma_{\mu \rho}- \right.`	${ }^{0} \nabla_{\mu}^{*} \equiv{ }^{0} \nabla_{\mu}+w B_{\mu},{ }^{0} \Gamma^{\lambda}{ }_{\mu \nu} \equiv \frac{1}{2} \gamma^{\lambda \rho}\left(\partial_{\mu} \gamma_{\nu \rho}+\partial_{\nu} \gamma_{\mu \rho}- \right.$	where ${}^{0}\nabla_{\mu}^{*} \equiv {}^{0}\nabla_{\mu}+wB_{\mu}$, ${}^{0}\Gamma^{\lambda}{}_{\mu\nu} \equiv \frac{1}{2}\gamma^{\lambda\rho}(\partial_{\mu}\gamma_{\nu\rho}+\partial_{\nu}\gamma_{\mu\rho}-$
1510.06699_F00221	11	0.709	`\partial_{\rho} \gamma_{\mu \nu}`	$\partial_{\rho} \gamma_{\mu \nu}$	$\partial_{\rho}\gamma_{\mu\nu})$ is the metric connection corresponding to the
1510.06699_F00222	11	0.991	`\mathrm{X}^{\rho}{ }_{\sigma}`	$\mathrm{X}^{\rho}{ }_{\sigma}$	and $\mathrm{X}^{\rho}{}_{\sigma}$ are the $\mathrm{GL}(4,R)$ generator matrices appropriate
1510.06699_F00223	11	0.991	`\operatorname{GL}(4, R)`	$\operatorname{GL}(4, R)$	and $\mathrm{X}^{\rho}{}_{\sigma}$ are the $\mathrm{GL}(4,R)$ generator matrices appropriate
1510.06699_F00224	11	1.000	`{ }^{0} \Gamma^{\sigma}{ }_{\rho \mu}`	${ }^{0} \Gamma^{\sigma}{ }_{\rho \mu}$	spacetime, the connection coefficients ${}^{0}\Gamma^{\sigma}{}_{\rho\mu}$ are not dy-
1510.06699_F00225	11	1.000	`{ }^{0} \nabla_{\mu}`	${ }^{0} \nabla_{\mu}$	spacetime, meaning that the derivatives ${}^{0}\nabla_{\mu}$ and ${}^{0}\nabla_{\nu}$
1510.06699_F00226	11	1.000	`{ }^{0} \nabla_{\nu}`	${ }^{0} \nabla_{\nu}$	spacetime, meaning that the derivatives ${}^{0}\nabla_{\mu}$ and ${}^{0}\nabla_{\nu}$
1510.06699_F00227	11	1.000	`\boldsymbol{h}`	$\boldsymbol{h}$	rank-2 tensor field, which we will denote by $\boldsymbol{h}$, defined on
1510.06699_F00228	11	1.000	`h_{a}{ }^{\mu} \equiv \boldsymbol{h}\left(\hat{ \boldsymbol{e}}_{a}, \boldsymbol{e}^{\mu}\right)`	$h_{a}{ }^{\mu} \equiv \boldsymbol{h}\left(\hat{ \boldsymbol{e}}_{a}, \boldsymbol{e}^{\mu}\right)$	by $h_{a}{}^{\mu} \equiv \boldsymbol{h}(\hat{\boldsymbol{e}}_{a},\boldsymbol{e}^{\mu})$. Similarly, the inverse of $\boldsymbol{h}$, which
1510.06699_F00229	11	1.000	`\boldsymbol{b}`	$\boldsymbol{b}$	we denote by $\boldsymbol{b}$, is also a rank-2 tensor field defined on
1510.06699_F00230	11	1.000	`b^{a}{ }_{\mu} \equiv \boldsymbol{b}\left(\hat{ \boldsymbol{e}}^{a}, \boldsymbol{e}_{\mu}\right)`	$b^{a}{ }_{\mu} \equiv \boldsymbol{b}\left(\hat{ \boldsymbol{e}}^{a}, \boldsymbol{e}_{\mu}\right)$	considered so far are given by $b^{a}{}_{\mu} \equiv \boldsymbol{b}(\hat{\boldsymbol{e}}^{a},\boldsymbol{e}_{\mu})$. Other
1510.06699_F00231	11	1.000	`\boldsymbol{J}`	$\boldsymbol{J}$	field $\boldsymbol{J}$ defined on the Minkowski spacetime, which may
1510.06699_F00232	11	1.000	`\boldsymbol{J}=J^{\mu} \boldsymbol{e}_{\mu}=J_{\mu} \boldsymbol{e}^{\mu}`	$\boldsymbol{J}=J^{\mu} \boldsymbol{e}_{\mu}=J_{\mu} \boldsymbol{e}^{\mu}$	basis or its dual as $\boldsymbol{J}=J^{\mu}\boldsymbol{e}_{\mu}=J_{\mu}\boldsymbol{e}^{\mu}$. Replacing the
1510.06699_F00233	11	1.000	`\mathcal{J}_{a} \equiv h_{a}{ }^{\mu} J_{\mu}=h_{a \mu} J^{\mu}`	$\mathcal{J}_{a} \equiv h_{a}{ }^{\mu} J_{\mu}=h_{a \mu} J^{\mu}$	$\boldsymbol{h}$, we may obtain the quantities $\mathcal{J}_{a} \equiv h_{a}{}^{\mu}J_{\mu}=h_{a\mu}J^{\mu}$ or
1510.06699_F00234	11	1.000	`\mathcal{J}^{a} \equiv h^{a}{ }_{\mu} J^{\mu}=h^{a \mu} J_{\mu}`	$\mathcal{J}^{a} \equiv h^{a}{ }_{\mu} J^{\mu}=h^{a \mu} J_{\mu}$	$\mathcal{J}^{a} \equiv h^{a}{}_{\mu}J^{\mu}=h^{a\mu}J_{\mu}$. Adopting the viewpoint that $\boldsymbol{h}$ is
1510.06699_F00235	11	0.967	`\mathcal{J}=\boldsymbol{h}(\cdot, \boldsymbol{J})^{67}`	$\mathcal{J}=\boldsymbol{h}(\cdot, \boldsymbol{J})^{67}$	basis of a *new* vector field $\boldsymbol{\mathcal{J}}=\boldsymbol{h}(\cdot,\boldsymbol{J})^{67}$. It is also worth
1510.06699_F00236	11	0.820	`\boldsymbol{h}(\mathcal{J}, \cdot)`	$\boldsymbol{h}(\mathcal{J}, \cdot)$	the vector $\boldsymbol{h}(\boldsymbol{\mathcal{J}},\cdot)$, which has contravariant components
1510.06699_F00237	11	1.000	`h_{a}{ }^{\mu} \mathcal{J}^{a}`	$h_{a}{ }^{\mu} \mathcal{J}^{a}$	in the coordinate basis given by $h_{a}{}^{\mu}\mathcal{J}^{a}$; these are the
1510.06699_F00238	11	1.000	`\mathcal{J}`	$\mathcal{J}$	example, if we place $\boldsymbol{\mathcal{J}}$ into the first 'slot' of $\boldsymbol{b}$, we re-
1510.06699_F00239	11	0.690	`\boldsymbol{J}=\boldsymbol{b}(\boldsymbol{\mathcal { J }}, \cdot)`	$\boldsymbol{J}=\boldsymbol{b}(\boldsymbol{\mathcal { J }}, \cdot)$	cover the *original* vector $\boldsymbol{J}=\boldsymbol{b}(\boldsymbol{\mathcal{J}},\cdot)$, since its covari-
1510.06699_F00240	11	1.000	`b^{a}{ }_{\mu} \mathcal{J}_{a}=J_{\mu}`	$b^{a}{ }_{\mu} \mathcal{J}_{a}=J_{\mu}$	ant components in the coordinate basis are $b^{a}{}_{\mu}\mathcal{J}_{a}=J_{\mu}$,
1510.06699_F00241	11	1.000	`J_{\mu}=b_{a \mu} \mathcal{J}^{a}`	$J_{\mu}=b_{a \mu} \mathcal{J}^{a}$	$J_{\mu}=b_{a\mu}\mathcal{J}^{a}$, and their contravariant counterparts as
1510.06699_F00242	11	1.000	`J^{\mu}=b_{a}{ }^{\mu} \mathcal{J}^{a}=b^{a \mu} \mathcal{J}_{a}`	$J^{\mu}=b_{a}{ }^{\mu} \mathcal{J}^{a}=b^{a \mu} \mathcal{J}_{a}$	$J^{\mu}=b_{a}{}^{\mu}\mathcal{J}^{a}=b^{a\mu}\mathcal{J}_{a}$. In all these cases, the compo-
1510.06699_F00243	11	1.000	`\boldsymbol{b}(\cdot, \boldsymbol{J})`	$\boldsymbol{b}(\cdot, \boldsymbol{J})$	tion: consider, for example, the vector $\boldsymbol{b}(\cdot,\boldsymbol{J})$; this has
1510.06699_F00244	11	1.000	`b^{a}{ }_{\mu} J^{\mu}`	$b^{a}{ }_{\mu} J^{\mu}$	$b^{a}{}_{\mu}J^{\mu}$, but these are *not*, in general, equal to $\mathcal{J}^{a}$.

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1510.06699_F00245	11	1.000	`\mathcal{J}^{a}`	$\mathcal{J}^{a}$	$b^a{}_\mu J^\mu$, but these are *not*, in general, equal to $\mathcal{J}^a$.
1510.06699_F00246	11	0.999	`\boldsymbol{T}`	$\boldsymbol{T}$	indices: for example, if $\boldsymbol{T}$ is a rank-2 tensor field, then
1510.06699_F00247	11	0.968	`\mathcal{T}`	$\mathcal{T}$	one obtains the *new* rank-2 tensor field $\mathcal{T}$ with covari-
1510.06699_F00248	11	1.000	`\mathcal{T}_{a b}=`	$\mathcal{T}_{ab} =$	ant components (say) in the tetrad basis given by $\mathcal{T}_{ab} =$
1510.06699_F00249	11	1.000	`h_{a}{ }^{\mu} h_{b}{ }^{\nu} T_{\mu \nu}`	$h_a{}^\mu h_b{}^\nu T_{\mu\nu}$	$h_a{}^\mu h_b{}^\nu T_{\mu\nu}$. Using the reciprocity relations (28), one may
1510.06699_F00250	11	1.000	`T_{\mu \nu}=b^{a}{ }_{\mu} b^{b}{ }_{\nu} \mathcal{T}_{a b}`	$T_{\mu\nu} = b^a{}_\mu b^b{}_\nu \mathcal{T}_{ab}$	therefore also write, for example, $T_{\mu\nu} = b^a{}_\mu b^b{}_\nu \mathcal{T}_{ab}$. One
1510.06699_F00251	11	0.999	`\hat{\boldsymbol{e}}_{a}`	$\hat{\boldsymbol{e}}_a$	frame vectors $\hat{\boldsymbol{e}}_a$; in this case, the components $h_a{}^\mu J_\mu$,
1510.06699_F00252	11	0.999	`h_{a}{ }^{\mu} J_{\mu}`	$h_a{}^\mu J_\mu$	frame vectors $\hat{\boldsymbol{e}}_a$; in this case, the components $h_a{}^\mu J_\mu$,
1510.06699_F00253	11	1.000	`J_{a}`	J_a	for example, are interpreted as the components J_a of the
1510.06699_F00254	12	1.000	`R^{a b}{ }_{\mu \nu}`	$R^{ab}{}_{\mu\nu}$	Both $R^{ab}{}_{\mu\nu}$ and $H_{\mu\nu}$ transform covariantly under GCT
1510.06699_F00255	12	1.000	`H_{\mu \nu}`	$H_{\mu\nu}$	Both $R^{ab}{}_{\mu\nu}$ and $H_{\mu\nu}$ transform covariantly under GCT
1510.06699_F00256	12	1.000	`\mathcal{D}_{a}^{*} \varphi=`	$\mathcal{D}_a^* \varphi =$	two generalised covariant derivatives. Since $\mathcal{D}_a^*\varphi =$
1510.06699_F00257	12	1.000	`h_{a}{ }^{\mu} D_{\mu}^{*} \varphi`	$h_a{}^\mu D_\mu^* \varphi$	$h_a{}^\mu D_\mu^*\varphi$, this commutator differs from (38) by an addi-
1510.06699_F00258	12	1.000	`\mathcal{R}^{a b}{ }_{c d} \equiv h_{a}{ }^{\mu} h_{b}{ }^{\nu} R^{a b}{ }_{\mu \nu}`	$\mathcal{R}^{ab}{}_{cd} \equiv h_a{}^\mu h_b{}^\nu R^{ab}{}_{\mu\nu}$	where $\mathcal{R}^{ab}{}_{cd} \equiv h_a{}^\mu h_b{}^\nu R^{ab}{}_{\mu\nu}$ and $\mathcal{H}_{cd} = h_c{}^\mu h_d{}^\nu H_{\mu\nu}$, and
1510.06699_F00259	12	1.000	`\mathcal{H}_{c d}=h_{c}{ }^{\mu} h_{d}{ }^{\nu} H_{\mu \nu}`	$\mathcal{H}_{cd} = h_c{}^\mu h_d{}^\nu H_{\mu\nu}$	where $\mathcal{R}^{ab}{}_{cd} \equiv h_a{}^\mu h_b{}^\nu R^{ab}{}_{\mu\nu}$ and $\mathcal{H}_{cd} = h_c{}^\mu h_d{}^\nu H_{\mu\nu}$, and
1510.06699_F00260	12	0.669	`\mathcal{B}_{a}=h_{a}{ }^{\mu} B_{\mu}`	$\mathcal{B}_a = h_a{}^\mu B_\mu$	where $\mathcal{B}_a = h_a{}^\mu B_\mu$. It is also convenient to define $\mathcal{T}_b^* \equiv$
1510.06699_F00261	12	0.669	`\mathcal{T}_{b}^{*} \equiv`	$\mathcal{T}_b^* \equiv$	where $\mathcal{B}_a = h_a{}^\mu B_\mu$. It is also convenient to define $\mathcal{T}_b^* \equiv$
1510.06699_F00262	12	1.000	`\mathcal{T}^{* a}{ }_{b a}=\mathcal{T}_{b}+3 \mathcal{B}_{b}`	$\mathcal{T}^{*a}{}_{ba} = \mathcal{T}_b + 3\mathcal{B}_b$	$\mathcal{T}^{*a}{}_{ba} = \mathcal{T}_b + 3\mathcal{B}_b$, where $\mathcal{T}_b$ is the corresponding PGT
1510.06699_F00263	12	1.000	`\mathcal{T}_{b}`	$\mathcal{T}_b$	$\mathcal{T}^{*a}{}_{ba} = \mathcal{T}_b + 3\mathcal{B}_b$, where $\mathcal{T}_b$ is the corresponding PGT
1510.06699_F00264	12	1.000	`\mathcal{R}^{a b}{ }_{c d}, \mathcal{H}_{c d}`	$\mathcal{R}^{ab}{}_{cd}, \mathcal{H}_{cd}$	It is easy to show that $\mathcal{R}^{ab}{}_{cd}$, $\mathcal{H}_{cd}$ and $\mathcal{T}^{*a}{}_{cd}$ trans-
1510.06699_F00265	12	1.000	`\mathcal{T}^{* a}{ }_{c d}`	$\mathcal{T}^{*a}{}_{cd}$	It is easy to show that $\mathcal{R}^{ab}{}_{cd}$, $\mathcal{H}_{cd}$ and $\mathcal{T}^{*a}{}_{cd}$ trans-
1510.06699_F00266	12	1.000	`w\left(\mathcal{R}^{a b}{ }_{c d}\right)=w\left(\mathcal{H}_{c d}\right)=-2`	$w\left(\mathcal{R}^{ab}{}_{cd}\right) = w\left(\mathcal{H}_{cd}\right) = -2$	$w(\mathcal{R}^{ab}{}_{cd}) = w(\mathcal{H}_{cd}) = -2$ and $w(\mathcal{T}^{*a}{}_{cd}) = -1$ respec-
1510.06699_F00267	12	1.000	`w\left(\mathcal{T}^{* a}{ }_{c d}\right)=-1`	$w\left(\mathcal{T}^{*a}{}_{cd}\right) = -1$	$w(\mathcal{R}^{ab}{}_{cd}) = w(\mathcal{H}_{cd}) = -2$ and $w(\mathcal{T}^{*a}{}_{cd}) = -1$ respec-
1510.06699_F00268	12	1.000	`\mathcal{R}^{a b}{ }_{c d}, \mathcal{H}_{a b}`	$\mathcal{R}^{ab}{}_{cd}, \mathcal{H}_{ab}$	strengths $\mathcal{R}^{ab}{}_{cd}$, $\mathcal{H}_{ab}$ and $\mathcal{T}^{*a}{}_{bc}$ are related to the *curva-*
1510.06699_F00269	12	1.000	`A_{a b \mu}=`	$A_{ab\mu} =$	the relevant relationship is easily found to be $A_{ab\mu} =$
1510.06699_F00270	12	1.000	`b^{c}{ }_{\mu} \mathcal{A}_{a b c}`	$b^c{}_\mu \mathcal{A}_{abc}$	$b^c{}_\mu \mathcal{A}_{abc}$, where
1510.06699_F00271	12	1.000	`\mathcal{T}_{a b c}^{*}`	$\mathcal{T}_{abc}^*$	equations of motion result in $\mathcal{T}_{abc}^*$ being independent of
1510.06699_F00272	12	1.000	`\mathcal{A}_{a b c}`	$\mathcal{A}_{abc}$	$\mathcal{A}_{abc}$, (46) gives an explicit expression for the A-field in
1510.06699_F00273	12	1.000	`{ }^{0} \mathcal{A}_{a b c}^{*} \equiv \frac{1}{2} \left(c_{a b c}^{*}+\right.`	${}^0\mathcal{A}_{abc}^* \equiv \frac{1}{2}\left(c_{abc}^* +\right.$	in which we have defined the quantities ${}^0\mathcal{A}_{abc}^* \equiv \frac{1}{2}(c_{abc}^* +$
1510.06699_F00274	12	0.506	`\left.c_{b c a}^{*}-c_{c a b}^{*}\right)`	$\left.c_{bca}^* - c_{cab}^*\right)$	$c_{bca}^* - c_{cab}^*)$ that, as indicated, are fully determined by the
1510.06699_F00275	12	1.000	`\mathcal{K}_{a b c}^{*} \equiv-\frac{1}{2}\left(\mathcal{T}_{a b c}^{*}+\mathcal{T}_{b c a}^{*}-\mathcal{T}_{c a b}^{*}\right)`	$\mathcal{K}_{abc}^* \equiv -\frac{1}{2}\left(\mathcal{T}_{abc}^* + \mathcal{T}_{bca}^* - \mathcal{T}_{cab}^*\right)$	defined $\mathcal{K}_{abc}^* \equiv -\frac{1}{2}(\mathcal{T}_{abc}^* + \mathcal{T}_{bca}^* - \mathcal{T}_{cab}^*)$, which when re-
1510.06699_F00276	12	0.982	`{ }^{0} \mathcal{A}_{a b c}^{*}`	${}^0\mathcal{A}_{abc}^*$	that both ${}^0\mathcal{A}_{abc}^*$ and $\mathcal{K}_{abc}^*$ are antisymmetric in their first
1510.06699_F00277	12	0.982	`\mathcal{K}_{a b c}^{*}`	$\mathcal{K}_{abc}^*$	that both ${}^0\mathcal{A}_{abc}^*$ and $\mathcal{K}_{abc}^*$ are antisymmetric in their first
1510.06699_F00278	12	0.991	`\mathcal{A}_{a b c}^{*}`	$\mathcal{A}_{abc}^*$	${}^0\mathcal{A}_{abc}^*$ transform in the same way as $\mathcal{A}_{abc}^*$, whereas $\mathcal{K}_{abc}^*$
1510.06699_F00279	12	0.961	`{ }^{0} \mathcal{D}_{a}^{*} \equiv h_{a}{ }^{\mu}{ }^{0} D_{\mu}^{*}`	${}^0\mathcal{D}_a^* \equiv h_a{}^\mu{}^0 D_\mu^*$	covariant derivative ${}^0\mathcal{D}_a^* \equiv h_a{}^\mu\,{}^0D_\mu^*$, which consequently
1510.06699_F00280	12	1.000	`\mathcal{D}_{a}^{*}`	$\mathcal{D}_a^*$	torsion (and hence contortion) to zero, then $\mathcal{D}_a^*$ reduces
1510.06699_F00281	12	1.000	`{ }^{0} \mathcal{D}_{a}^{*}`	${}^0\mathcal{D}_a^*$	to ${}^0\mathcal{D}_a^*$.
1510.06699_F00282	12	1.000	`\mathcal{R}^{a b}{ }_{c d}, \mathcal{T}^{* a}{ }_{b c}`	$\mathcal{R}^{ab}{}_{cd}, \mathcal{T}^{*a}{}_{bc}$	gauge field strengths $\mathcal{R}^{ab}{}_{cd}$, $\mathcal{T}^{*a}{}_{bc}$ and $\mathcal{H}_{ab}$ in WGT.
1510.06699_F00283	13	0.947	`\mathcal{T}_{a b c}^{*}=`	$\mathcal{T}_{abc}^* =$	the 'torsion' is totally antisymmetric, such that $\mathcal{T}_{abc}^* =$
1510.06699_F00284	13	0.990	`\epsilon_{a b c d} t^{d}`	$\epsilon_{abcd} t^d$	$\epsilon_{abcd} t^d$ for some vector t^d, then the second term on the

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1510.06699_F00285	13	0.990	`t^{d}`	t^{d}	$\epsilon_{abcd}t^d$ for some vector t^d, then the second term on the
1510.06699_F00286	13	0.999	`d`	d	By contracting over the indices a and d in the '$\mathcal{R}$-
1510.06699_F00287	13	1.000	`b`	b	Contracting again over b and e, one then finds the twice-
1510.06699_F00288	13	1.000	`e`	e	Contracting again over b and e, one then finds the twice-
1510.06699_F00289	13	0.770	`\mathcal{T}_{a}^{*}`	$\mathcal{T}_{a}^{*}$	torsion trace $\mathcal{T}_a^*$ vanishes, which clearly includes the spe-
1510.06699_F00290	13	1.000	`\mathcal{H}`	$\mathcal{H}$	It is clear that the '$\mathcal{H}$-identity' (53) has no non-trivial
1510.06699_F00291	13	1.000	`\mathcal{R}_{a b c d}, \mathcal{T}_{a b c}^{*}`	$\mathcal{R}_{a b c d}, \mathcal{T}_{a b c}^{*}$	itational gauge fields. From $\mathcal{R}_{abcd}$, $\mathcal{T}^*_{abc}$ and $\mathcal{H}_{ab}$, one can
1510.06699_F00292	13	1.000	`\mathcal{L}_{\mathrm{G}} \equiv h^{-1} L_{\mathrm{G}}`	$\mathcal{L}_{\mathrm{G}} \equiv h^{-1} L_{\mathrm{G}}$	sity $\mathcal{L}_{\mathrm{G}} \equiv h^{-1}L_{\mathrm{G}}$ has weight zero (i.e scale invari-
1510.06699_F00293	13	1.000	`\mathcal{R}_{a b c d}, \mathcal{H}_{a b}`	$\mathcal{R}_{a b c d}, \mathcal{H}_{a b}$	ant). It is trivial to show that $\mathcal{R}_{abcd}$, $\mathcal{H}_{ab}$ and $\mathcal{T}^*_{abc}$
1510.06699_F00294	13	1.000	`w\left(\mathcal{R}_{a b c d}\right)=w\left(\mathcal{H}_{a b}\right)=-2`	$w\left(\mathcal{R}_{a b c d}\right)=w\left(\mathcal{H}_{a b}\right)=-2$	weights $w(\mathcal{R}_{abcd}) = w(\mathcal{H}_{ab}) = -2$ and $w(\mathcal{T}^*_{abc}) = -1$
1510.06699_F00295	13	1.000	`w\left(\mathcal{T}_{a b c}^{*}\right)=-1`	$w\left(\mathcal{T}_{a b c}^{*}\right)=-1$	weights $w(\mathcal{R}_{abcd}) = w(\mathcal{H}_{ab}) = -2$ and $w(\mathcal{T}^*_{abc}) = -1$
1510.06699_F00296	13	0.988	`{ }^{\underline{70}}`	${ }^{\underline{70}}$	respectively[70]. Hence L_{G} can be quadratic in $\mathcal{R}_{abcd}$ and
1510.06699_F00297	13	0.994	`L_{\mathcal{H}^{2}}`	$L_{\mathcal{H}^{2}}$	where the expressions for $L_{\mathcal{R}^2}$ and $L_{\mathcal{H}^2}$ are given in (4).
1510.06699_F00298	13	1.000	`D \leq 4`	$D \leq 4$	contributes a total derivative to the action (in $D \leq 4$
1510.06699_F00299	13	0.994	`\mathcal{T}_{a b c} \rightarrow \mathcal{T}_{a b c}^{*}`	$\mathcal{T}_{a b c} \rightarrow \mathcal{T}_{a b c}^{*}$	$\mathcal{T}_{abc} \rightarrow \mathcal{T}^*_{abc}$. However, since these terms depend upon the
1510.06699_F00300	13	1.000	`\mathcal{L}_{\mathrm{G}}`	$\mathcal{L}_{\mathrm{G}}$	sity $\mathcal{L}_{\mathrm{G}}$. As an illustration, in Appendix D we give a
1510.06699_F00301	13	1.000	`L_{\mathrm{G}}=L_{\mathcal{H}^{2}}`	$L_{\mathrm{G}}=L_{\mathcal{H}^{2}}$	$L_{\mathrm{G}} = L_{\mathcal{H}^2}$ and L_{M} contains a term proportional to $\phi^2\mathcal{R}$.
1510.06699_F00302	13	1.000	`S_{\mathrm{T}}`	S_{T}	The total action S_{T} is simply the sum of the matter and
1510.06699_F00303	13	1.000	`h, A, \partial A`	$h, A, \partial A$	h, A, ∂A and ∂B (suppressing indices for brevity). Note,
1510.06699_F00304	13	1.000	`\partial B`	∂B	h, A, ∂A and ∂B (suppressing indices for brevity). Note,
1510.06699_F00305	13	1.000	`\varphi, \partial \varphi, h, A`	$\varphi, \partial \varphi, h, A$	φ induce a dependence on φ, $\partial\varphi$, h, A and B. We will
1510.06699_F00306	13	1.000	`\partial A`	∂A	$\phi^2\mathcal{R}$, then this brings an additional dependence on ∂A.
1510.06699_F00307	13	0.901	`\phi^{2} L_{\mathcal{\mathcal { T } _ { * 2 }}}`	$\phi^{2} L_{\mathcal{T}_{*\in}}$	Finally, if one includes terms of the generic form $\phi^2 L_{\mathcal{T}^{*2}}$,
1510.06699_F00308	13	1.000	`\partial h`	∂h	then it produces an additional dependence on ∂h. Con-
1510.06699_F00309	14	1.000	`\mathcal{L}_{\mathrm{T}}`	$\mathcal{L}_{\mathrm{T}}$	Since $\mathcal{L}_{\mathrm{T}}$ is at most quadratic in the field strength ten-
1510.06699_F00310	14	1.000	`\mathcal{R}_{a b c d}(h, A, \partial A), \mathcal{T}_{a b c}^{*}(h, \partial h, A, B)`	$\mathcal{R}_{a b c d}(h, A, \partial A), \mathcal{T}_{a b c}^{*}(h, \partial h, A, B)$	sors $\mathcal{R}_{abcd}(h,A,\partial A)$, $\mathcal{T}^*_{abc}(h,\partial h,A,B)$ and $\mathcal{H}_{ab}(h,\partial B)$,
1510.06699_F00311	14	1.000	`t^{a}{ }_{\mu} \equiv`	$t^{a}{ }_{\mu} \equiv$	In the free gravitational sector, we have defined $t^a{}_\mu \equiv$
1510.06699_F00312	14	0.999	`\delta \mathcal{L}_{\mathrm{G}} / \delta h_{a}{ }^{\mu}=\partial \mathcal{L}_{\mathrm{G}} / \partial h_{a}{ }^{\mu}, s_{a b}{ }^{\mu} \equiv \delta \mathcal{L}_{\mathrm{G}} / \delta A^{a b}{ }_{\mu}`	$\delta \mathcal{L}_{\mathrm{G}} / \delta h_{a}{ }^{\mu}=\partial \mathcal{L}_{\mathrm{G}} / \partial h_{a}{ }^{\mu}, s_{a b}{ }^{\mu} \equiv \delta \mathcal{L}_{\mathrm{G}} / \delta A^{a b}{ }_{\mu}$	$\delta\mathcal{L}_{\mathrm{G}}/\delta h_a{}^\mu = \partial\mathcal{L}_{\mathrm{G}}/\partial h_a{}^\mu$, $s_{ab}{}^\mu \equiv \delta\mathcal{L}_{\mathrm{G}}/\delta A^{ab}{}_\mu$ and
1510.06699_F00313	14	1.000	`j^{\mu} \equiv \delta \mathcal{L}_{\mathrm{G}} / \delta B_{\mu}`	$j^{\mu} \equiv \delta \mathcal{L}_{\mathrm{G}} / \delta B_{\mu}$	$j^\mu \equiv \delta\mathcal{L}_{\mathrm{G}}/\delta B_\mu$, respectively. In the matter sector,
1510.06699_F00314	14	1.000	`\tau^{a}{ }_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_{a}{ }^{\mu}`	$\tau^{a}{ }_{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta h_{a}{ }^{\mu}$	the energy-momentum $\tau^a{}_\mu \equiv \delta\mathcal{L}_{\mathrm{M}}/\delta h_a{}^\mu$, spin-angular-
1510.06699_F00315	14	0.692	`\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu}=\partial \mathcal{L}_{\mathrm{M}} / \partial B_{\mu}`	$\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu}=\partial \mathcal{L}_{\mathrm{M}} / \partial B_{\mu}$	$\zeta^\mu \equiv \delta\mathcal{L}_{\mathrm{M}}/\delta B_\mu = \partial\mathcal{L}_{\mathrm{M}}/\partial B_\mu$ act as sources[72]. It is easy
1510.06699_F00316	14	0.692	`{ }^{\underline{72}}`	${ }^{\underline{72}}$	$\zeta^\mu \equiv \delta\mathcal{L}_{\mathrm{M}}/\delta B_\mu = \partial\mathcal{L}_{\mathrm{M}}/\partial B_\mu$ act as sources[72]. It is easy
1510.06699_F00317	14	1.000	`w=1, w=0`	$w=1, w=0$	$w=1$, $w=0$ and $w=0$, respectively. It is, in fact, more
1510.06699_F00318	14	1.000	`t^{a}{ }_{b} \equiv t^{a}{ }_{\mu} h_{b}{ }^{\mu}, s_{a b}{ }^{c} \equiv s_{a b}{ }^{\mu} b^{c}{ }_{\mu}`	$t^{a}{ }_{b} \equiv t^{a}{ }_{\mu} h_{b}{ }^{\mu}, s_{a b}{ }^{c} \equiv s_{a b}{ }^{\mu} b^{c}{ }_{\mu}$	where $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu$, $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu$ and $j^a \equiv j^\mu b^a{}_\mu$, and
1510.06699_F00319	14	1.000	`j^{a} \equiv j^{\mu} b^{a}{ }_{\mu}`	$j^{a} \equiv j^{\mu} b^{a}{ }_{\mu}$	where $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu$, $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu$ and $j^a \equiv j^\mu b^a{}_\mu$, and
1510.06699_F00320	14	0.992	`\bar{\partial} \mathcal{L}_{\mathrm{M}} / \partial \varphi \equiv\left[\partial \mathcal{L}_{\mathrm{M}} \left(\varphi, \mathcal{D}_{\mu}^{*} u, \ldots\right) / \partial \varphi\right]_{u=\varphi}`	$\bar{\partial} \mathcal{L}_{\mathrm{M}} / \partial \varphi \equiv\left[\partial \mathcal{L}_{\mathrm{M}} \left(\varphi, \mathcal{D}_{\mu}^{*} u, \ldots\right) / \partial \varphi\right]_{u=\varphi}$	where $\bar\partial\mathcal{L}_{\mathrm{M}}/\partial\varphi \equiv [\partial\mathcal{L}_{\mathrm{M}}(\varphi,\mathcal{D}^*_\mu u,\ldots)/\partial\varphi]_{u=\varphi}$, so that φ
1510.06699_F00321	14	1.000	`\mathcal{D}_{\mu}^{*} \varphi`	$\mathcal{D}_{\mu}^{*} \varphi$	and $\mathcal{D}^*_\mu\varphi$ are treated as independent variables, rather
1510.06699_F00322	14	1.000	`\mathcal{L}_{\mathrm{M}}=h^{-1} L_{\mathrm{M}}`	$\mathcal{L}_{\mathrm{M}}=h^{-1} L_{\mathrm{M}}$	where we have defined $\mathcal{L}_{\mathrm{M}} = h^{-1}L_{\mathrm{M}}$ and $\bar\partial\mathcal{L}_{\mathrm{M}}/\partial\varphi \equiv$

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1510.06699_F00323	14	1.000	`\bar{\partial} L_{\mathrm{M}} / \partial \varphi \equiv`	$\bar{\partial} L_{\mathrm{M}} / \partial \varphi \equiv$	where we have defined $\mathcal{L}_{\mathrm{M}} = h^{-1}L_{\mathrm{M}}$ and $\bar{\partial}L_{\mathrm{M}}/\partial\varphi \equiv$
1510.06699_F00324	14	0.974	`\left[\partial L_{\mathrm{M}}\left(\varphi, \mathcal{D}_{a}^{*} u, \ldots\right) / \partial \varphi \right]_{u=\varphi}`	$\left[\partial L_{\mathrm{M}}\left(\varphi, \mathcal{D}_{a}^{*} u, \ldots\right) / \partial \varphi \right]_{u=\varphi}$	$[\partial L_{\mathrm{M}}(\varphi, \mathcal{D}_a^* u, \ldots)/\partial\varphi]_{u=\varphi}$, so that φ and $\mathcal{D}_a^*\varphi$ are treated
1510.06699_F00325	14	1.000	`S_{\mathrm{M}}`	S_{M}	the minimal-coupling procedure applied to the action S_{M},
1510.06699_F00326	14	0.999	`{ }^{9,74}`	${ }^{9,74}$	equations directly[9,74], which would be given by (67a) and
1510.06699_F00327	14	0.963	`2, C`	$2, C$	S_{G} under (infinitesimal) local Lorentz (or SL$(2,C)$) rota-
1510.06699_F00328	15	0.718	`(69 \mathrm{c})`	$(69 \mathrm{c})$	(69c)
1510.06699_F00329	15	1.000	`w=-\frac{3}{2}`	$w=-\frac{3}{2}$	As we show in Section II L, however, this interpretation
1510.06699_F00330	15	1.000	`\psi \rightarrow \psi e^{i \alpha}`	$\psi \rightarrow \psi e^{i \alpha}$	that the equation:
1510.06699_F00331	15	1.000	`\gamma`	γ	where, on each γ-matrix, one has simply replaced the
1510.06699_F00332	15	1.000	`\bar{\psi}`	$\bar{\psi}$	that, since both ψ and $\bar{\psi}$ have Weyl weight $w = -\frac{3}{2}$, the
1510.06699_F00333	15	1.000	`\mathcal{D}_{a}`	$\mathcal{D}_{a}$	term in (71) can be replaced by its PGT counterpart $\mathcal{D}_a$,
1510.06699_F00334	15	1.000	`\mathcal{D}_{a}^{*} \psi`	$\mathcal{D}_{a}^{*} \psi$	varying the action with respect to ψ. Since $\mathcal{D}_a^*\psi$ and $\mathcal{T}_a^*$
1510.06699_F00335	15	1.000	`w=-\frac{5}{2}`	$w=-\frac{5}{2}$	$w = -\frac{5}{2}$ and $w = -1$ respectively, we see immediately
1510.06699_F00336	15	1.000	`\mathcal{D}_{a} \psi=\mathcal{D}_{a}^{*} \psi+\frac{3}{2} \mathcal{B}_{a} \psi`	$\mathcal{D}_{a} \psi=\mathcal{D}_{a}^{*} \psi+\frac{3}{2} \mathcal{B}_{a} \psi$	directly by recalling that $\mathcal{D}_a\psi = \mathcal{D}_a^*\psi + \frac{3}{2}\mathcal{B}_a\psi$ and $\mathcal{T}_a =$
1510.06699_F00337	15	1.000	`\mathcal{T}_{a}=`	$\mathcal{T}_{a}=$	directly by recalling that $\mathcal{D}_a\psi = \mathcal{D}_a^*\psi + \frac{3}{2}\mathcal{B}_a\psi$ and $\mathcal{T}_a =$
1510.06699_F00338	15	1.000	`\mathcal{T}_{a}^{*}-3 \mathcal{B}_{a}`	$\mathcal{T}_{a}^{*}-3 \mathcal{B}_{a}$	$\mathcal{T}_a^* - 3\mathcal{B}_a$ to recover (73). We note that (73) and (74) are
1510.06699_F00339	16	1.000	`\mathcal{T}_{a}`	$\mathcal{T}_{a}$	density $\mathcal{L}_{\mathrm{G}}$, but the expression for $\mathcal{T}_a$, or $\mathcal{T}_a^*$, will depend
1510.06699_F00340	16	1.000	`\Sigma_{a b}=\frac{1}{4}\left[\gamma_{a}, \gamma_{b} \right]`	$\Sigma_{a b}=\frac{1}{4}\left[\gamma_{a}, \gamma_{b} \right]$	given by $\Sigma_{ab} = \frac{1}{4}[\gamma_a, \gamma_b]$, after a short calculation one
1510.06699_F00341	16	0.945	`\mathcal{T}_{[a b c]}^{*}=0`	$\mathcal{T}_{[a b c]}^{*}=0$	of the Dirac field ψ. In other words, if $\mathcal{T}^*_{[abc]} = 0$ then
1510.06699_F00342	16	1.000	`\tau=h^{-1} \mu \phi \bar{\psi} \psi`	$\tau=h^{-1} \mu \phi \bar{\psi} \psi$	ric and has the trace $\tau = h^{-1}\mu\phi\bar{\psi}\psi$, whereas the spin-
1510.06699_F00343	16	1.000	`\sigma_{a b c}=\sigma_{[a b c]}`	$\sigma_{a b c}=\sigma_{[a b c]}$	that $\sigma_{abc} = \sigma_{[abc]}$. Consequently, all contractions of the
1510.06699_F00344	16	1.000	`\mathcal{D}_{a} \rightarrow \mathcal{D}_{a}^{*}`	$\mathcal{D}_{a} \rightarrow \mathcal{D}_{a}^{*}$	ply making the replacement $\mathcal{D}_a \rightarrow \mathcal{D}_a^*$ throughout; the
1510.06699_F00345	16	1.000	`\mu, \nu, \lambda`	μ, ν, λ	where μ, ν, λ and a are dimensionless constants (usually
1510.06699_F00346	16	0.669	`\beta_{1}, \beta_{2}`	β_{1}, β_{2}	stants β_1, β_2 and β_3 in $L_{\mathcal{T}^{*2}}$. Since none of the additional
1510.06699_F00347	16	0.669	`\beta_{3}`	β_{3}	stants β_1, β_2 and β_3 in $L_{\mathcal{T}^{*2}}$. Since none of the additional
1510.06699_F00348	16	1.000	`F_{\mu \nu} \equiv`	$F_{\mu \nu} \equiv$	where the electromagnetic field strength tensor $F_{\mu\nu} \equiv$
1510.06699_F00349	16	1.000	`\partial_{\mu} A_{\nu}-\partial_{\nu} A_{\mu}`	$\partial_{\mu} A_{\nu}-\partial_{\nu} A_{\mu}$	$\partial_\mu A_\nu - \partial_\nu A_\mu$ and J^μ is some source current density. We
1510.06699_F00350	16	1.000	`A_{\mu} \rightarrow A_{\mu}+\partial_{\mu} \chi`	$A_{\mu} \rightarrow A_{\mu}+\partial_{\mu} \chi$	formation $A_\mu \rightarrow A_\mu + \partial_\mu\chi$, where χ is any scalar field,
1510.06699_F00351	16	1.000	`\chi`	χ	formation $A_\mu \rightarrow A_\mu + \partial_\mu\chi$, where χ is any scalar field,
1510.06699_F00352	16	1.000	`\partial_{\mu} J^{\mu}=0`	$\partial_{\mu} J^{\mu}=0$	ity equation $\partial_\mu J^\mu = 0$, which embodies the conservation
1510.06699_F00353	16	1.000	`\mathcal{A}_{a}=h_{a}{ }^{\mu} A_{\mu}`	$\mathcal{A}_{a}=h_{a}{ }^{\mu} A_{\mu}$	field $\mathcal{A}_a = h_a{}^\mu A_\mu$. Similarly, we introduce the covari-
1510.06699_F00354	16	1.000	`\mathcal{J}_{a}=h_{a}{ }^{\mu} J_{\mu}`	$\mathcal{J}_{a}=h_{a}{ }^{\mu} J_{\mu}$	ant current density $\mathcal{J}_a = h_a{}^\mu J_\mu$. This leads us, however,
1510.06699_F00355	16	1.000	`w\left(\mathcal{A}_{a}\right)=-1`	$w\left(\mathcal{A}_{a}\right)=-1$	weights $w(\mathcal{A}_a) = -1$ and $w(\mathcal{J}_a) = -3$; consequently, the
1510.06699_F00356	16	1.000	`w\left(\mathcal{J}_{a}\right)=-3`	$w\left(\mathcal{J}_{a}\right)=-3$	weights $w(\mathcal{A}_a) = -1$ and $w(\mathcal{J}_a) = -3$; consequently, the
1510.06699_F00357	16	1.000	`w\left(A_{\mu}\right)=0`	$w\left(A_{\mu}\right)=0$	fields carrying a Greek index have weights $w(A_\mu) = 0$
1510.06699_F00358	16	1.000	`w\left(J_{\mu}\right)=-2`	$w\left(J_{\mu}\right)=-2$	and $w(J_\mu) = -2$.
1510.06699_F00359	16	1.000	`\widehat{\mathcal{F}}_{a b}^{*} \equiv \mathcal{D}_{a}^{*} \mathcal{A}_{b}-\mathcal{D}_{b}^{*} \mathcal{A}_{a}`	$\widehat{\mathcal{F}}_{a b}^{*} \equiv \mathcal{D}_{a}^{*} \mathcal{A}_{b}-\mathcal{D}_{b}^{*} \mathcal{A}_{a}$	where we have defined $\widehat{\mathcal{F}}^*_{ab} \equiv \mathcal{D}_a^*\mathcal{A}_b - \mathcal{D}_b^*\mathcal{A}_a$. Since
1510.06699_F00360	16	1.000	`w\left(\widehat{\mathcal{F}}_{a b}^{*}\right)=-2, \mathcal{L}_{\mathrm{M}}`	$w\left(\widehat{\mathcal{F}}_{a b}^{*}\right)=-2, \mathcal{L}_{\mathrm{M}}$	$w(\widehat{\mathcal{F}}^*_{ab}) = -2$, $\mathcal{L}_{\mathrm{M}}$ has an overall Weyl weight of zero,
1510.06699_F00361	17	1.000	`\mathcal{F}_{a b} \equiv h_{a}{ }^{\mu} h_{b}{ }^{\nu} F_{\mu \nu}`	$\mathcal{F}_{a b} \equiv h_{a}{ }^{\mu} h_{b}{ }^{\nu} F_{\mu \nu}$	where we have defined $\mathcal{F}_{ab} \equiv h_a{}^\mu h_b{}^\nu F_{\mu\nu}$, in which
1510.06699_F00362	17	1.000	`F_{\mu \nu} \equiv D_{\mu}^{*} A_{\nu}-D_{\nu}^{*} A_{\mu}=\partial_{\mu} A_{\nu}-\partial_{\nu} A_{\mu}`	$F_{\mu \nu} \equiv D_{\mu}^{*} A_{\nu}-D_{\nu}^{*} A_{\mu}=\partial_{\mu} A_{\nu}-\partial_{\nu} A_{\mu}$	$F_{\mu\nu} \equiv D_\mu^* A_\nu - D_\nu^* A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$, and $\mathcal{T}^{*c}{}_{ab}$ is

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1510.06699_F00363	17	1.000	`\mathcal{T}^{* c}{ }_{a b}`	$\mathcal{T}^{* c}{ }_{a b}$	$F_{\mu\nu} \equiv D^*_\mu A_\nu - D^*_\nu A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$, and $\mathcal{T}^{*c}{}_{ab}$ is
1510.06699_F00364	17	1.000	`\mathcal{F}_{a b}`	$\mathcal{F}_{a b}$	gauge field, as defined in (42). It is apparent that $\mathcal{F}_{ab}$
1510.06699_F00365	17	1.000	`F_{\mu \nu}`	$F_{\mu \nu}$	ance of $F_{\mu\nu}$, whereas the second term on the RHS of (82)
1510.06699_F00366	17	1.000	`\widehat{\mathcal{F}}_{a b}^{*}`	$\widehat{\mathcal{F}}_{a b}^{*}$	transformation, and so neither is $\widehat{\mathcal{F}}^*_{ab}$. Moreover, $\mathcal{F}_{ab}$ has
1510.06699_F00367	17	1.000	`w\left(\mathcal{F}_{a b}\right)=-2`	$w\left(\mathcal{F}_{a b}\right)=-2$	a Weyl weight $w(\mathcal{F}_{ab}) = -2$. Thus, substituting (82)
1510.06699_F00368	17	1.000	`\mathcal{L}_{\mathrm{M}}`	$\mathcal{L}_{\mathrm{M}}$	for $\mathcal{L}_{\mathrm{M}}$ is identical to that obtained in PGT, as should be
1510.06699_F00369	17	0.999	`\delta \mathcal{L}_{\mathrm{EM}} / \delta A_{\mu}=0`	$\delta \mathcal{L}_{\mathrm{EM}} / \delta A_{\mu}=0$	is given by $\delta\mathcal{L}_{\mathrm{EM}}/\delta A_\mu = 0$, which is easily shown to be
1510.06699_F00370	17	0.997	`\delta \mathcal{L}_{\mathrm{EM}} / \delta \mathcal{A}_{a}=0`	$\delta \mathcal{L}_{\mathrm{EM}} / \delta \mathcal{A}_{a}=0$	equivalent to $\delta\mathcal{L}_{\mathrm{EM}}/\delta\mathcal{A}_a = 0$, and in manifestly WGT-
1510.06699_F00371	17	1.000	`\mathcal{D}_{a} \mathcal{F}^{a c}=\left( \mathcal{D}_{a}^{*}+\right.`	$\mathcal{D}_{a} \mathcal{F}^{a c}=\left(\mathcal{D}_{a}^{*}+\right.$	is easily checked directly by noting that $\mathcal{D}_a\mathcal{F}^{ac} = (\mathcal{D}^*_a +$
1510.06699_F00372	17	1.000	`\left.2 \mathcal{B}_{a}\right) \mathcal{F}^{a c}`	$\left.2 \mathcal{B}_{a}\right) \mathcal{F}^{a c}$	$2\mathcal{B}_a)\mathcal{F}^{ac}$ and $\mathcal{T}_a = \mathcal{T}^*{}_a - 3\mathcal{B}_a$, which on substitution
1510.06699_F00373	17	1.000	`\mathcal{T}_{a}=\mathcal{T}^{*}{ }_{a}-3 \mathcal{B}_{a}`	$\mathcal{T}_{a}=\mathcal{T}^{*}{ }_{a}-3 \mathcal{B}_{a}$	$2\mathcal{B}_a)\mathcal{F}^{ac}$ and $\mathcal{T}_a = \mathcal{T}^*{}_a - 3\mathcal{B}_a$, which on substitution
1510.06699_F00374	17	1.000	`\mathcal{J}^{a}=0`	$\mathcal{J}^{a}=0$	angular-momentum tensors. Setting $\mathcal{J}^a = 0$ in (83), one
1510.06699_F00375	17	1.000	`\sigma_{a b}{ }^{c}=0`	$\sigma_{a b}{ }^{c}=0$	whereas $\sigma_{ab}{}^c = 0$ immediately. We see that $\tau^a{}_b$ is the
1510.06699_F00376	17	1.000	`\tau^{a}{ }_{b}`	$\tau^{a}{ }_{b}$	whereas $\sigma_{ab}{}^c = 0$ immediately. We see that $\tau^a{}_b$ is the
1510.06699_F00377	17	1.000	`\tau_{a b}`	$\tau_{a b}$	gravity. Moreover, τ_{ab} is symmetric and traceless, as
1510.06699_F00378	17	1.000	`\mathcal{J}^{a}=q \bar{\psi} \gamma^{a} \psi`	$\mathcal{J}^{a}=q \bar{\psi} \gamma^{a} \psi$	current density of the Dirac field is simply $\mathcal{J}^a = q\bar{\psi}\gamma^a\psi$,
1510.06699_F00379	17	1.000	`w\left(\mathcal{J}^{a}\right)=-3`	$w\left(\mathcal{J}^{a}\right)=-3$	which has the required Weyl weight $w(\mathcal{J}^a) = -3$. Thus,
1510.06699_F00380	17	1.000	`\phi \neq 0`	$\phi \neq 0$	(provided $\phi \neq 0$), which is known as the Einstein gauge.
1510.06699_F00381	17	1.000	`\phi(x)=1`	$\phi(x)=1$	Although it is most common simply to set $\phi(x) = 1$, this
1510.06699_F00382	17	1.000	`\phi(x)=\phi_{0}`	$\phi(x)=\phi_{0}$	can be misleading and so we instead set $\phi(x) = \phi_0$, where
1510.06699_F00383	17	1.000	`\phi_{0}`	ϕ_{0}	ϕ_0 is a constant with the same units as ϕ. Adopting the
1510.06699_F00384	18	0.638	`(91 c)`	$(91 c)$	(91c)
1510.06699_F00385	18	1.000	`\widehat{\mathcal{A}}^{a b}{ }_{c} \equiv \widehat{h}_{a}{ }^{\mu} \widehat{A}^{a b}{ }_{\mu}`	$\widehat{\mathcal{A}}^{a b}{ }_{c} \equiv \widehat{h}_{a}{ }^{\mu} \widehat{A}^{a b}{ }_{\mu}$	ables $\widehat{\mathcal{A}}^{ab}{}_c \equiv \widehat{h}_a{}^\mu \widehat{A}^{ab}{}_\mu$ and $\widehat{\mathcal{B}}_a \equiv \widehat{h}_a{}^\mu \widehat{B}_\mu$.
1510.06699_F00386	18	1.000	`\widehat{\mathcal{B}}_{a} \equiv \widehat{h}_{a}{ }^{\mu} \widehat{B}_{\mu}`	$\widehat{\mathcal{B}}_{a} \equiv \widehat{h}_{a}{ }^{\mu} \widehat{B}_{\mu}$	ables $\widehat{\mathcal{A}}^{ab}{}_c \equiv \widehat{h}_a{}^\mu \widehat{A}^{ab}{}_\mu$ and $\widehat{\mathcal{B}}_a \equiv \widehat{h}_a{}^\mu \widehat{B}_\mu$.
1510.06699_F00387	18	1.000	`\mathcal{D}_{a}^{*} \phi=-\left(\phi^{2} / \phi_{0} \right) \widehat{\mathcal{B}}_{a}`	$\mathcal{D}_{a}^{*} \phi=-\left(\phi^{2} / \phi_{0} \right) \widehat{\mathcal{B}}_{a}$	the scalar field ϕ may be written as $\mathcal{D}^*_a\phi = -(\phi^2/\phi_0)\widehat{\mathcal{B}}_a$,
1510.06699_F00388	18	1.000	`\widehat{\chi}=\left(\phi / \phi_{0}\right)^{-w} \chi`	$\widehat{\chi}=\left(\phi / \phi_{0}\right)^{-w} \chi$	where $\widehat{\chi} = (\phi/\phi_0)^{-w}\chi$ and we have defined the derivative
1510.06699_F00389	18	0.996	`\widehat{\mathcal{D}}_{a} \equiv \widehat{h}_{a}{ }^{\mu} D_{\mu}`	$\widehat{\mathcal{D}}_{a} \equiv \widehat{h}_{a}{ }^{\mu} D_{\mu}$	operators $\widehat{\mathcal{D}}_a \equiv \widehat{h}_a{}^\mu D_\mu$ and $\widehat{\mathcal{D}}^*_a = \widehat{\mathcal{D}}_a + w\widehat{\mathcal{B}}_a$, the latter of
1510.06699_F00390	18	0.996	`\widehat{\mathcal{D}}_{a}^{*}=\widehat{\mathcal{D}}_{a}+w \widehat{\mathcal{B}}_{a}`	$\widehat{\mathcal{D}}_{a}^{*}=\widehat{\mathcal{D}}_{a}+w \widehat{\mathcal{B}}_{a}$	operators $\widehat{\mathcal{D}}_a \equiv \widehat{h}_a{}^\mu D_\mu$ and $\widehat{\mathcal{D}}^*_a = \widehat{\mathcal{D}}_a + w\widehat{\mathcal{B}}_a$, the latter of
1510.06699_F00391	18	1.000	`\left(\phi / \phi_{0}\right)^{1-w}`	$\left(\phi / \phi_{0}\right)^{1-w}$	all multiplicative factor $(\phi/\phi_0)^{1-w}$, the scale-invariant
1510.06699_F00392	18	1.000	`\widehat{\mathcal{D}}_{a}^{*} \widehat{\chi}`	$\widehat{\mathcal{D}}_{a}^{*} \widehat{\chi}$	quantity $\widehat{\mathcal{D}}^*_a\widehat{\chi}$ has the *same* functional dependency on $\widehat{\chi}$,
1510.06699_F00393	18	1.000	`\widehat{\chi}`	$\widehat{\chi}$	quantity $\widehat{\mathcal{D}}^*_a\widehat{\chi}$ has the *same* functional dependency on $\widehat{\chi}$,
1510.06699_F00394	18	0.999	`\widehat{h}_{a}{ }^{\mu}, \widehat{A}^{a b}{ }_{\mu}`	$\widehat{h}_{a}{ }^{\mu}, \widehat{A}^{a b}{ }_{\mu}$	$\widehat{h}_a{}^\mu$, $\widehat{A}^{ab}{}_\mu$ and $\widehat{B}_\mu$, respectively, as $\mathcal{D}^*_a\chi$ does on χ, $h_a{}^\mu$,
1510.06699_F00395	18	0.999	`\widehat{B}_{\mu}`	$\widehat{B}_{\mu}$	$\widehat{h}_a{}^\mu$, $\widehat{A}^{ab}{}_\mu$ and $\widehat{B}_\mu$, respectively, as $\mathcal{D}^*_a\chi$ does on χ, $h_a{}^\mu$,
1510.06699_F00396	18	0.999	`\mathcal{D}_{a}^{*} \chi`	$\mathcal{D}_{a}^{*} \chi$	$\widehat{h}_a{}^\mu$, $\widehat{A}^{ab}{}_\mu$ and $\widehat{B}_\mu$, respectively, as $\mathcal{D}^*_a\chi$ does on χ, $h_a{}^\mu$,
1510.06699_F00397	18	0.999	`\chi, h_{a}{ }^{\mu}`	$\chi, h_{a}{ }^{\mu}$	$\widehat{h}_a{}^\mu$, $\widehat{A}^{ab}{}_\mu$ and $\widehat{B}_\mu$, respectively, as $\mathcal{D}^*_a\chi$ does on χ, $h_a{}^\mu$,
1510.06699_F00398	18	1.000	`\mathcal{R}^{a b}{ }_{c d}=\left(\phi / \phi_{0} \right)^{2} \widehat{\mathcal{R}}^{a b}{ }_{c d}, \mathcal{T}^{* a}{ }_{b c}=\left(\phi / \phi_{0}\right) \widehat{\mathcal{T}}^{* a}{ }_{b c}`	$\mathcal{R}^{a b}{ }_{c d}=\left(\phi / \phi_{0} \right)^{2} \widehat{\mathcal{R}}^{a b}{ }_{c d}, \mathcal{T}^{* a}{ }_{b c}=\left(\phi / \phi_{0}\right) \widehat{\mathcal{T}}^{* a}{ }_{b c}$	ten as $\mathcal{R}^{ab}{}_{cd} = (\phi/\phi_0)^2\widehat{\mathcal{R}}^{ab}{}_{cd}$, $\mathcal{T}^{*a}{}_{bc} = (\phi/\phi_0)\widehat{\mathcal{T}}^{*a}{}_{bc}$ and
1510.06699_F00399	18	1.000	`\mathcal{H}_{a b}=\left(\phi / \phi_{0}\right)^{2} \widehat{\mathcal{H}}_{a b}`	$\mathcal{H}_{a b}=\left(\phi / \phi_{0}\right)^{2} \widehat{\mathcal{H}}_{a b}$	$\mathcal{H}_{ab} = (\phi/\phi_0)^2\widehat{\mathcal{H}}_{ab}$, where each quantity with a caret is

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1510.06699_F00400	18	1.000	`h^{-1}=\left(\phi / \phi_{0}\right)^{-4} \widehat{h}^{-1}`	$h^{-1}=\left(\phi / \phi_{0}\right)^{-4} \widehat{h}^{-1}$	Noting that $h^{-1} = (\phi/\phi_0)^{-4}\widehat{h}^{-1}$, one is thus led to the
1510.06699_F00401	18	1.000	`\varphi, h_{a}{ }^{\mu}, A^{a b}{ }_{\mu}`	$\varphi, h_{a}{ }^{\mu}, A^{a b}{ }_{\mu}$	Thus, if χ represents φ, $h_a{}^\mu$, $A^{ab}{}_\mu$ or B_μ, one may im-
1510.06699_F00402	18	1.000	`\delta \mathcal{L}_{\mathrm{T}} / \delta \widehat{\chi}=0`	$\delta \mathcal{L}_{\mathrm{T}} / \delta \widehat{\chi}=0$	motion $\delta\mathcal{L}_{\mathrm{T}}/\delta\widehat{\chi} = 0$ has the *same* functional form as
1510.06699_F00403	18	1.000	`\delta \mathcal{L}_{\mathrm{T}} /\left.\delta \chi \right|_{\phi=\phi_{0}}=0`	$\delta \mathcal{L}_{\mathrm{T}} /\left.\delta \chi \right\|_{\phi=\phi_{0}}=0$	(the corresponding term in) $\delta\mathcal{L}_{\mathrm{T}}/\delta\chi\|_{\phi=\phi_0} = 0$, but with
1510.06699_F00404	18	1.000	`\delta \mathcal{L}_{\mathrm{T}} / \delta \phi=\delta \mathcal{L}_{\mathrm{M}} / \delta \phi`	$\delta \mathcal{L}_{\mathrm{T}} / \delta \phi=\delta \mathcal{L}_{\mathrm{M}} / \delta \phi$	variables, one begins by noting that $\delta\mathcal{L}_{\mathrm{T}}/\delta\phi = \delta\mathcal{L}_{\mathrm{M}}/\delta\phi$,
1510.06699_F00405	18	1.000	`\delta \mathcal{L}_{\mathrm{T}} / \delta \varphi=\delta \mathcal{L}_{\mathrm{M}} / \delta \varphi=0`	$\delta \mathcal{L}_{\mathrm{T}} / \delta \varphi=\delta \mathcal{L}_{\mathrm{M}} / \delta \varphi=0$	$\delta\mathcal{L}_{\mathrm{T}}/\delta\varphi = \delta\mathcal{L}_{\mathrm{M}}/\delta\varphi = 0$ and set $\phi = \phi_0$ in the result-
1510.06699_F00406	18	1.000	`\partial \mathcal{L}_{\mathrm{M}} / \partial \widehat{B}_{\mu}`	$\partial \mathcal{L}_{\mathrm{M}} / \partial \widehat{B}_{\mu}$	may replace $\partial\mathcal{L}_{\mathrm{M}}/\partial\widehat{B}_\mu$ by $\delta\mathcal{L}_{\mathrm{M}}/\delta\widehat{B}_\mu$ in the above equa-
1510.06699_F00407	18	1.000	`\delta \mathcal{L}_{\mathrm{M}} / \delta \widehat{B}_{\mu}`	$\delta \mathcal{L}_{\mathrm{M}} / \delta \widehat{B}_{\mu}$	may replace $\partial\mathcal{L}_{\mathrm{M}}/\partial\widehat{B}_\mu$ by $\delta\mathcal{L}_{\mathrm{M}}/\delta\widehat{B}_\mu$ in the above equa-
1510.06699_F00408	19	1.000	`m \rightarrow \mu \phi`	$m \rightarrow \mu \phi$	making the replacement $m \to \mu\phi$, as in (70). One begins
1510.06699_F00409	19	1.000	`\frac{1}{2}`	$\frac{1}{2}$	by constructing the appropriate action for a spin-$\frac{1}{2}$ point
1510.06699_F00410	19	1.000	`p_{\mu} \rightarrow p_{a} \equiv h_{a}{ }^{\mu} p_{\mu}`	$p_{\mu} \rightarrow p_{a} \equiv h_{a}{ }^{\mu} p_{\mu}$	needs only to make the replacements $p_\mu \to p_a \equiv h_a{}^\mu p_\mu$,
1510.06699_F00411	19	1.000	`\dot{x}^{\mu} \rightarrow v^{a} \equiv b^{a}{ }_{\mu} \dot{x}^{\mu}`	$\dot{x}^{\mu} \rightarrow v^{a} \equiv b^{a}{ }_{\mu} \dot{x}^{\mu}$	$\dot{x}^\mu \to v^a \equiv b^a{}_\mu \dot{x}^\mu$ and $m \to \mu\phi$, which gives[75]
1510.06699_F00412	19	1.000	`p^{a}(\lambda)`	$p^{a}(\lambda)$	momentum $p^a(\lambda)$, 4-velocity $v^a(\lambda)$ and the einbein $e(\lambda)$
1510.06699_F00413	19	1.000	`v^{a}(\lambda)`	$v^{a}(\lambda)$	momentum $p^a(\lambda)$, 4-velocity $v^a(\lambda)$ and the einbein $e(\lambda)$
1510.06699_F00414	19	1.000	`e(\lambda)`	$e(\lambda)$	momentum $p^a(\lambda)$, 4-velocity $v^a(\lambda)$ and the einbein $e(\lambda)$
1510.06699_F00415	19	1.000	`\lambda`	λ	along the worldline, which is parameterised by λ. We
1510.06699_F00416	19	1.000	`w\left(p^{a}\right)=-1, w\left(v^{a}\right)=0, w(e)=1, w(\lambda)=1`	$w\left(p^{a}\right)=-1, w\left(v^{a}\right)=0, w(e)=1, w(\lambda)=1$	(96) are $w(p^a) = -1$, $w(v^a) = 0$, $w(e) = 1$, $w(\lambda) = 1$ and
1510.06699_F00417	19	1.000	`p^{2} \equiv p_{a} p^{a}`	$p^{2} \equiv p_{a} p^{a}$	where $p^2 \equiv p_a p^a$ and the quantities $c^c{}_{ab}$ are the PGT
1510.06699_F00418	19	1.000	`c^{c}{ }_{a b}`	$c^{c}{ }_{a b}$	where $p^2 \equiv p_a p^a$ and the quantities $c^c{}_{ab}$ are the PGT
1510.06699_F00419	19	1.000	`{ }^{0} \mathcal{D}_{c}^{*}`	${ }^{0} \mathcal{D}_{c}^{*}$	where ${}^0\mathcal{D}^*_c$ is the reduced WGT covariant derivative, de-
1510.06699_F00420	19	1.000	`e=1 /(\mu \phi)`	$e=1 /(\mu \phi)$	one chooses $e = 1/(\mu\phi)$, in which case $v^2 = 1$ and λ cor-
1510.06699_F00421	19	1.000	`v^{2}=1`	$v^{2}=1$	one chooses $e = 1/(\mu\phi)$, in which case $v^2 = 1$ and λ cor-
1510.06699_F00422	19	1.000	`\tau`	τ	responds to the proper time τ along the worldline; in this
1510.06699_F00423	19	0.348	`{ }^{\mathbf{7 6 - 7 8}}`	${ }^{\mathbf{76-78}}$	terms of the reduced PGT-covariant derivative as[76–78]
1510.06699_F00424	19	0.996	`{ }^{0} A^{a b}{ }_{\mu}`	${ }^{0} A^{a b}{ }_{\mu}$	Appendix B), the quantities ${}^0A^{ab}{}_\mu$ appearing in ${}^0\mathcal{D}_b$ con-
1510.06699_F00425	19	0.996	`{ }^{0} \mathcal{D}_{b}`	${ }^{0} \mathcal{D}_{b}$	Appendix B), the quantities ${}^0A^{ab}{}_\mu$ appearing in ${}^0\mathcal{D}_b$ con-
1510.06699_F00426	19	0.906	`v^{b} \mathcal{D}_{b} v_{a}=0`	$v^{b} \mathcal{D}_{b} v_{a}=0$	the equation of motion $v^b\mathcal{D}_b v_a = 0$). We note that
1510.06699_F00427	19	1.000	`{ }^{79,80}`	${ }^{79,80}$	this conclusion differs from some previous studies[79,80],
1510.06699_F00428	19	1.000	`\left(q_{\mathrm{W}} / m\right) \mathcal{H}_{a b} v^{b}`	$\left(q_{\mathrm{W}} / m\right) \mathcal{H}_{a b} v^{b}$	$(q_{\mathrm{W}}/m)\mathcal{H}_{ab}v^b$, where q_{W} is the particle's Weyl charge
1510.06699_F00429	19	1.000	`q_{\mathrm{W}}`	q_{W}	$(q_{\mathrm{W}}/m)\mathcal{H}_{ab}v^b$, where q_{W} is the particle's Weyl charge
1510.06699_F00430	19	1.000	`\mu=0`	$\mu=0$	By setting $\mu = 0$ in the action (96) and choosing the
1510.06699_F00431	19	1.000	`e=1`	$e=1$	einbein $e = 1$, such that $v^2 = 0$, one finds that the
1510.06699_F00432	19	1.000	`v^{2}=0`	$v^{2}=0$	einbein $e = 1$, such that $v^2 = 0$, one finds that the
1510.06699_F00433	20	1.000	`\widehat{p}_{a}, \widehat{v}^{a}`	$\widehat{p}_{a}, \widehat{v}^{a}$	where $\widehat{p}_a$, $\widehat{v}^a$ and $\widehat{e}$ are to be considered as functions of
1510.06699_F00434	20	1.000	`\widehat{e}`	$\widehat{e}$	where $\widehat{p}_a$, $\widehat{v}^a$ and $\widehat{e}$ are to be considered as functions of
1510.06699_F00435	20	0.990	`\widehat{\lambda}`	$\widehat{\lambda}$	$\widehat{\lambda}$, such that for example $\widehat{p}_a(\widehat{\lambda}) \equiv \widehat{h}_a{}^\mu p_\mu(\lambda(\widehat{\lambda}))$. It is then
1510.06699_F00436	20	0.990	`\widehat{p}_{a}(\widehat{\lambda}) \equiv \widehat{h}_{a}{ }^{\mu} p_{\mu}(\lambda(\widehat{\lambda}))`	$\widehat{p}_{a}(\widehat{\lambda}) \equiv \widehat{h}_{a}{ }^{\mu} p_{\mu}(\lambda(\widehat{\lambda}))$	$\widehat{\lambda}$, such that for example $\widehat{p}_a(\widehat{\lambda}) \equiv \widehat{h}_a{}^\mu p_\mu(\lambda(\widehat{\lambda}))$. It is then
1510.06699_F00437	20	1.000	`\widehat{\epsilon}=\left[1-\widehat{\lambda}\left(d x^{\mu} / d \widehat{\lambda}\right) \partial_{\mu} \ln \left(\phi / \phi_{0}\right)\right] \widehat{e}`	$\widehat{\epsilon}=\left[1-\widehat{\lambda}\left(d x^{\mu} / d \widehat{\lambda}\right) \partial_{\mu} \ln \left(\phi / \phi_{0}\right)\right] \widehat{e}$	bein $\widehat{\epsilon} = [1 - \widehat{\lambda}(dx^\mu/d\widehat{\lambda})\partial_\mu \ln(\phi/\phi_0)]\widehat{e}$. Thus, in terms

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1510.06699_F00438	20	1.000	`\widehat{\lambda}, \widehat{p}_{a}, \widehat{v}^{a}`	$\widehat{\lambda}, \widehat{p}_{a}, \widehat{v}^{a}$	of the scale-invariant variables $\widehat{\lambda}$, $\widehat{p}_a$, $\widehat{v}^a$ and $\widehat{\epsilon}$, respec-
1510.06699_F00439	20	1.000	`\widehat{\epsilon}`	$\widehat{\epsilon}$	of the scale-invariant variables $\widehat{\lambda}$, $\widehat{p}_a$, $\widehat{v}^a$ and $\widehat{\epsilon}$, respec-
1510.06699_F00440	20	1.000	`\lambda, p_{a}, v^{a}`	λ, p_{a}, v^{a}	it does in terms of the original variables λ, p_a, v^a and ϵ
1510.06699_F00441	20	0.999	`\widehat{h}_{a}{ }^{\mu}`	$\widehat{h}_{a}{ }^{\mu}$	counterpart $\widehat{h}_a{}^\mu$ or $\widehat{b}^a{}_\mu$. Thus, the calculation presented
1510.06699_F00442	20	0.999	`\widehat{b}^{a}{ }_{\mu}`	$\widehat{b}^{a}{ }_{\mu}$	counterpart $\widehat{h}_a{}^\mu$ or $\widehat{b}^a{}_\mu$. Thus, the calculation presented
1510.06699_F00443	20	1.000	`{ }^{0} \widehat{\mathcal{D}}_{a} \equiv`	${ }^{0} \widehat{\mathcal{D}}_{a} \equiv$	where we have defined the derivative operator ${}^0\widehat{\mathcal{D}}_a \equiv$
1510.06699_F00444	20	0.977	`\widehat{h}_{a}{ }^{\mu 0} \mathcal{D}_{\mu}`	$\widehat{h}_{a}{ }^{\mu 0} \mathcal{D}_{\mu}$	$\widehat{h}_a{}^\mu\,{}^0\mathcal{D}_\mu$.
1510.06699_F00445	20	1.000	`{ }^{0} D_{\mu}^{*}`	${ }^{0} D_{\mu}^{*}$	variant derivative operator ${}^0D^*_\mu$ in (48), to which the
1510.06699_F00446	20	0.837	`\mathcal{T}_{a b c}^{*} \equiv 0`	$\mathcal{T}_{a b c}^{*} \equiv 0$	$\mathcal{T}^*_{abc} \equiv 0$. Hence such quantities may be written entirely
1510.06699_F00447	20	1.000	`{ }^{0} R^{* a b}{ }_{\mu \nu}\left(h, \partial h, \partial^{2} h, B, \partial B\right)`	${ }^{0} R^{* a b}{ }_{\mu \nu}\left(h, \partial h, \partial^{2} h, B, \partial B\right)$	sor ${}^0R^{*ab}{}_{\mu\nu}(h,\partial h,\partial^2 h,B,\partial B)$, which is again given by
1510.06699_F00448	20	1.000	`{ }^{0} A^{* a b}{ }_{\mu}`	${ }^{0} A^{* a b}{ }_{\mu}$	the formula (39), but with $A^{ab}{}_\mu$ replaced by ${}^0A^{*ab}{}_\mu$, and
1510.06699_F00449	20	0.998	`{ }^{0} \mathcal{R}^{* a b}{ }_{c d}=h_{c}{ }^{\mu} h_{d}{ }^{\mu}{ }^{0} R^{* a b}{ }_{\mu \nu}`	${ }^{0} \mathcal{R}^{* a b}{ }_{c d}=h_{c}{ }^{\mu} h_{d}{ }^{\mu}{ }^{0} R^{* a b}{ }_{\mu \nu}$	where ${}^0\mathcal{R}^{*ab}{}_{cd} = h_c{}^\mu h_d{}^\mu\,{}^0R^{*ab}{}_{\mu\nu}$. Unlike (41), the com-
1510.06699_F00450	20	1.000	`{ }^{0} \mathcal{T}^{* a}{ }_{b c} \equiv`	${ }^{0} \mathcal{T}^{* a}{ }_{b c} \equiv$	lational field strength of the h gauge field, since ${}^0\mathcal{T}^{*a}{}_{bc} \equiv$
1510.06699_F00451	20	1.000	`h_{b}{ }^{\mu} h_{c}{ }^{\nu}\left({ }^{0} D_{\mu}^{*} b^{a}{ }_{\nu}-{ }^{0} D_{\nu}^{*} b^{a}{ }_{\mu}\right)=0`	$h_{b}{ }^{\mu} h_{c}{ }^{\nu}\left({ }^{0} D_{\mu}^{*} b^{a}{ }_{\nu}-{ }^{0} D_{\nu}^{*} b^{a}{ }_{\mu}\right)=0$	$h_b{}^\mu h_c{}^\nu({}^0D^*_\mu b^a{}_\nu - {}^0D^*_\nu b^a{}_\mu) = 0$. As one might expect, the
1510.06699_F00452	20	0.913	`\mathcal{D}_{a}^{*} \rightarrow^{0} \mathcal{D}_{a}^{*}, \mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{*}`	$\mathcal{D}_{a}^{*} \rightarrow^{0} \mathcal{D}_{a}^{*}, \mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{*}$	with the replacements $\mathcal{D}^*_a \to {}^0\mathcal{D}^*_a$, $\mathcal{R}_{abcd} \to {}^0\mathcal{R}^*_{abcd}$ and
1510.06699_F00453	20	0.982	`\mathcal{T}_{a b c}^{*} \rightarrow{ }^{0} \mathcal{T}_{a b c}^{*} \equiv 0`	$\mathcal{T}_{a b c}^{*} \rightarrow{ }^{0} \mathcal{T}_{a b c}^{*} \equiv 0$	$\mathcal{T}^*_{abc} \to {}^0\mathcal{T}^*_{abc} \equiv 0$.
1510.06699_F00454	20	1.000	`{ }^{0} \mathcal{R}^{* a b}{ }_{c d}`	${ }^{0} \mathcal{R}^{* a b}{ }_{c d}$	${}^0\mathcal{R}^{*ab}{}_{cd}$.
1510.06699_F00455	20	1.000	`\mathcal{L}_{\mathrm{G}}=h^{-1} L_{\mathrm{G}}`	$\mathcal{L}_{\mathrm{G}}=h^{-1} L_{\mathrm{G}}$	density $\mathcal{L}_{\mathrm{G}} = h^{-1}L_{\mathrm{G}}$, where by analogy with (58),
1510.06699_F00456	21	0.516	`{ }^{0} \mathcal{R}_{\text {abcd }}^{*}`	${ }^{0} \mathcal{R}_{\text {abcd }}^{*}$	dimensions), since ${}^0\mathcal{R}^*_{abcd}$ and its contractions also sat-
1510.06699_F00457	21	0.999	`\mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{*}`	$\mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{*}$	$\mathcal{R}_{abcd} \to {}^0\mathcal{R}^*_{abcd}$. Thus, one can set to zero any one of the
1510.06699_F00458	21	0.996	`\mathcal{D}_{a}^{*} \rightarrow{ }^{0} \mathcal{D}_{a}^{*}`	$\mathcal{D}_{a}^{*} \rightarrow{ }^{0} \mathcal{D}_{a}^{*}$	$\mathcal{D}^*_a \to {}^0\mathcal{D}^*_a$ and $\mathcal{T}^*_a \to {}^0\mathcal{T}^*_a \equiv 0$. One should also note,
1510.06699_F00459	21	0.996	`\mathcal{T}_{a}^{*} \rightarrow{ }^{0} \mathcal{T}_{a}^{*} \equiv 0`	$\mathcal{T}_{a}^{*} \rightarrow{ }^{0} \mathcal{T}_{a}^{*} \equiv 0$	$\mathcal{D}^*_a \to {}^0\mathcal{D}^*_a$ and $\mathcal{T}^*_a \to {}^0\mathcal{T}^*_a \equiv 0$. One should also note,
1510.06699_F00460	21	0.756	`\mathcal{D}_{a}^{*} \rightarrow^{0} \mathcal{D}_{a}^{*}, \mathcal{T}_{a b c}^{*} \rightarrow{ }^{0} \mathcal{T}_{a b c}^{*} \equiv 0`	$\mathcal{D}_{a}^{*} \rightarrow^{0} \mathcal{D}_{a}^{*}, \mathcal{T}_{a b c}^{*} \rightarrow{ }^{0} \mathcal{T}_{a b c}^{*} \equiv 0$	with the replacements $\mathcal{D}^*_a \to {}^0\mathcal{D}^*_a$, $\mathcal{T}^*_{abc} \to {}^0\mathcal{T}^*_{abc} \equiv 0$ and
1510.06699_F00461	21	1.000	`s_{a b c} \equiv 0 \equiv \sigma_{a b c}`	$s_{a b c} \equiv 0 \equiv \sigma_{a b c}$	$s_{abc} \equiv 0 \equiv \sigma_{abc}$. Furthermore, by performing analogous
1510.06699_F00462	21	0.975	`\mathcal{T}_{a b c}^{*}=0`	$\mathcal{T}_{a b c}^{*}=0$	condition $\mathcal{T}^*_{abc} = 0$ in the field equations of WGT. In the
1510.06699_F00463	21	1.000	`{ }^{82-84}`	${ }^{82-84}$	only for the Lovelock action[82–84], which coincides with
1510.06699_F00464	21	1.000	`D=4`	$D=4$	term) in $D = 4$ dimensions.
1510.06699_F00465	21	0.741	`{ }^{\underline{85}}`	${ }^{\underline{85}}$	frame vectors $\boldsymbol{e}_\mu(x)$ are related by[85]
1510.06699_F00466	21	0.968	`\hat{\boldsymbol{e}}^{a}(x)`	$\hat{\boldsymbol{e}}^{a}(x)$	sis vectors $\hat{\boldsymbol{e}}^a(x)$ and $\boldsymbol{e}^\mu(x)$ in each set. For any other
1510.06699_F00467	21	0.968	`\boldsymbol{e}^{\mu}(x)`	$\boldsymbol{e}^{\mu}(x)$	sis vectors $\hat{\boldsymbol{e}}^a(x)$ and $\boldsymbol{e}^\mu(x)$ in each set. For any other
1510.06699_F00468	21	0.769	`J_{\mu} \boldsymbol{e}^{\mu}`	$J_{\mu} \boldsymbol{e}^{\mu}$	vector $\boldsymbol{J}$, written in the coordinate basis as (say) $J_\mu\boldsymbol{e}^\mu$,
1510.06699_F00469	21	1.000	`J_{a}=h_{a}{ }^{\mu} J_{\mu}`	$J_{a}=h_{a}{ }^{\mu} J_{\mu}$	one then identifies the quantities $J_a = h_a{}^\mu J_\mu$, for ex-
1510.06699_F00470	21	1.000	`h_{a}{ }^{\mu} J^{a}=J^{\mu}`	$h_{a}{ }^{\mu} J^{a}=J^{\mu}$	that $h_a{}^\mu J^a = J^\mu$ and $b^a{}_\mu J^\mu = J^a$. Moreover, any con-
1510.06699_F00471	21	1.000	`b^{a}{ }_{\mu} J^{\mu}=J^{a}`	$b^{a}{ }_{\mu} J^{\mu}=J^{a}$	that $h_a{}^\mu J^a = J^\mu$ and $b^a{}_\mu J^\mu = J^a$. Moreover, any con-
1510.06699_F00472	21	1.000	`g_{\mu \nu}`	$g_{\mu \nu}$	a more general spacetime with metric $g_{\mu\nu}$. Conversely,
1510.06699_F00473	21	1.000	`h^{-1}=\sqrt{-g}`	$h^{-1}=\sqrt{-g}$	From (115), one also finds that $h^{-1} = \sqrt{-g}$ (where we
1510.06699_F00474	22	1.000	`w\left(g_{\mu \nu}\right)=2`	$w\left(g_{\mu \nu}\right)=2$	have Weyl weights $w(g_{\mu\nu}) = 2$ and $w(h_a{}^\mu) = -1$ respec-

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1510.06699_F00475	22	1.000	`w\left(h_{a}{ }^{\mu}\right)=-1`	$w\left(h_{a}{ }^{\mu}\right)=-1$	have Weyl weights $w(g_{\mu\nu})=2$ and $w(h_a{}^\mu)=-1$ respec-
1510.06699_F00476	22	1.000	`w\left(\eta_{a b}\right)=0`	$w\left(\eta_{a b}\right)=0$	tively, and so (115) and (116) imply that $w(\eta_{ab})=0$, as
1510.06699_F00477	22	1.000	`e_{a}{ }^{\mu}`	$e_{a}{ }^{\mu}$	$h_a{}^\mu$ and $b^a{}_\mu$ as $e_a{}^\mu$ and $e^a{}_\mu$, respectively.
1510.06699_F00478	22	1.000	`e^{a}{ }_{\mu}`	$e^{a}{ }_{\mu}$	$h_a{}^\mu$ and $b^a{}_\mu$ as $e_a{}^\mu$ and $e^a{}_\mu$, respectively.
1510.06699_F00479	22	1.000	`x+\delta x`	$x+\delta x$	and $x+\delta x$, which is accompanied by a local change in
1510.06699_F00480	22	1.000	`J^{a}`	J^{a}	for some vector J^a of weight w is defined as
1510.06699_F00481	22	1.000	`J^{a}(x)`	$J^{a}(x)$	which is required to compare vectors $J^a(x)$ and $J^a(x+\delta x)$
1510.06699_F00482	22	1.000	`J^{a}(x+\delta x)`	$J^{a}(x+\delta x)$	which is required to compare vectors $J^a(x)$ and $J^a(x+\delta x)$
1510.06699_F00483	22	1.000	`\hat{\boldsymbol{e}}_{a}(x+\delta x)`	$\hat{\boldsymbol{e}}_{a}(x+\delta x)$	tetrad frames $\hat{\boldsymbol{e}}_a(x)$ and $\hat{\boldsymbol{e}}_a(x+\delta x)$ respectively. Hence,
1510.06699_F00484	22	1.000	`\partial_{\mu}^{*}`	∂_{μ}^{*}	where we have used the partial derivative operator ∂^*_μ
1510.06699_F00485	22	1.000	`\eta_{a b}`	$\eta_{a b}$	the Lorentz metric η_{ab} at each point. Then demanding
1510.06699_F00486	22	1.000	`A^{a b}{ }_{\mu}=-A^{b a}{ }_{\mu}`	$A^{a b}{ }_{\mu}=-A^{b a}{ }_{\mu}$	antisymmetric, i.e. $A^{ab}{}_\mu=-A^{ba}{}_\mu$, as previously.
1510.06699_F00487	22	0.528	`\nabla_{\mu}^{*}=\partial_{\mu}^{*}+\Gamma^{\sigma}{ }_{\rho \mu} \mathrm{X}^{\rho}{ }_{\sigma}`	$\nabla_{\mu}^{*}=\partial_{\mu}^{*}+\Gamma^{\sigma}{ }_{\rho \mu} \mathrm{X}^{\rho}{ }_{\sigma}$	where $\nabla^*_\mu=\partial^*_\mu+\Gamma^\sigma{}_{\rho\mu}\mathrm{X}^\rho{}_\sigma$ and $\mathrm{X}^\rho{}_\sigma$ are the GL(4, R) gen-
1510.06699_F00488	22	1.000	`\Delta_{\mu}^{*}`	Δ_{μ}^{*}	of the field to which Δ^*_μ is applied. If a field ψ carries
1510.06699_F00489	22	1.000	`\nabla_{\mu}^{*} \psi=\partial_{\mu}^{*} \psi`	$\nabla_{\mu}^{*} \psi=\partial_{\mu}^{*} \psi$	only Latin indices, then $\nabla^*_\mu\psi=\partial^*_\mu\psi$ and so $\Delta^*_\mu\psi=D^*_\mu\psi$;
1510.06699_F00490	22	1.000	`\Delta_{\mu}^{*} \psi=D_{\mu}^{*} \psi`	$\Delta_{\mu}^{*} \psi=D_{\mu}^{*} \psi$	only Latin indices, then $\nabla^*_\mu\psi=\partial^*_\mu\psi$ and so $\Delta^*_\mu\psi=D^*_\mu\psi$;
1510.06699_F00491	22	1.000	`D_{\mu}^{*} \psi=\partial_{\mu}^{*} \psi`	$D_{\mu}^{*} \psi=\partial_{\mu}^{*} \psi$	$D^*_\mu\psi=\partial^*_\mu\psi$ and so $\Delta^*_\mu\psi=\nabla^*_\mu\psi$. When acting on an ob-
1510.06699_F00492	22	1.000	`\Delta_{\mu}^{*} \psi=\nabla_{\mu}^{*} \psi`	$\Delta_{\mu}^{*} \psi=\nabla_{\mu}^{*} \psi$	$D^*_\mu\psi=\partial^*_\mu\psi$ and so $\Delta^*_\mu\psi=\nabla^*_\mu\psi$. When acting on an ob-
1510.06699_F00493	22	1.000	`\Gamma^{\sigma}{ }_{\rho \mu}`	$\Gamma^{\sigma}{ }_{\rho \mu}$	connection coefficients $\Gamma^\sigma{}_{\rho\mu}$ are themselves dynamical
1510.06699_F00494	22	1.000	`J^{a}+\delta J^{a}`	$J^{a}+\delta J^{a}$	ported from x to $x+\delta x$, be equal to $J^a+\delta J^a$, i.e.
1510.06699_F00495	22	1.000	`\Gamma`	Γ	which relates A and Γ (and B); in particular, we note that
1510.06699_F00496	22	1.000	`w\left(\Gamma^{\sigma}{ }_{\nu \mu}\right)=0`	$w\left(\Gamma^{\sigma}{ }_{\nu \mu}\right)=0$	$w(\Gamma^\sigma{}_{\nu\mu})=0$. The relation (121) is sometimes known as
1510.06699_F00497	22	1.000	`\nabla_{\sigma}^{*} g_{\mu \nu}=0`	$\nabla_{\sigma}^{*} g_{\mu \nu}=0$	Using (115) and (121), one finds that $\nabla^*_\sigma g_{\mu\nu}=0$, and
1510.06699_F00498	22	1.000	`T^{* a}{ }_{\mu \nu}`	$T^{* a}{ }_{\mu \nu}$	the gauge field strengths $R^{ab}{}_{\mu\nu}$ and $T^{*a}{}_{\mu\nu}$, one finds that
1510.06699_F00499	22	1.000	`R^{\rho}{ }_{\sigma \mu \nu}=e_{a}{ }^{\rho} e^{b}{ }_{\sigma} R^{a}{ }_{b \mu \nu}`	$R^{\rho}{ }_{\sigma \mu \nu}=e_{a}{ }^{\rho} e^{b}{ }_{\sigma} R^{a}{ }_{b \mu \nu}$	where $R^\rho{}_{\sigma\mu\nu}=e_a{}^\rho e^b{}_\sigma R^a{}_{b\mu\nu}$ and $T^{*\lambda}{}_{\mu\nu}=e_a{}^\lambda T^{*a}{}_{\mu\nu}$.
1510.06699_F00500	22	1.000	`T^{* \lambda}{ }_{\mu \nu}=e_{a}{ }^{\lambda} T^{* a}{ }_{\mu \nu}`	$T^{* \lambda}{ }_{\mu \nu}=e_{a}{ }^{\lambda} T^{* a}{ }_{\mu \nu}$	where $R^\rho{}_{\sigma\mu\nu}=e_a{}^\rho e^b{}_\sigma R^a{}_{b\mu\nu}$ and $T^{*\lambda}{}_{\mu\nu}=e_a{}^\lambda T^{*a}{}_{\mu\nu}$.
1510.06699_F00501	22	1.000	`T^{* \lambda}{ }_{\mu \nu}`	$T^{* \lambda}{ }_{\mu \nu}$	Thus, although we recognise $T^{*\lambda}{}_{\mu\nu}$ as (minus) the tor-
1510.06699_F00502	22	1.000	`R^{\rho}{ }_{\sigma \mu \nu}`	$R^{\rho}{ }_{\sigma \mu \nu}$	sion tensor of the Y_4 spacetime, we see that $R^\rho{}_{\sigma\mu\nu}$ is not
1510.06699_F00503	22	1.000	`\widetilde{R}_{\rho \sigma \mu \nu}`	$\widetilde{R}_{\rho \sigma \mu \nu}$	One should note that, although $\widetilde{R}_{\rho\sigma\mu\nu}$ is antisymmet-
1510.06699_F00504	22	0.774	`\mu, \nu`	μ, ν	ric in (μ,ν), it is no longer antisymmetric in (ρ,σ) (in-
1510.06699_F00505	22	0.774	`\rho, \sigma`	ρ, σ	ric in (μ,ν), it is no longer antisymmetric in (ρ,σ) (in-
1510.06699_F00506	22	0.764	`\widetilde{R}_{(\rho \sigma) \mu \nu}=g_{\rho \sigma} H_{\mu \nu}`	$\widetilde{R}_{(\rho \sigma) \mu \nu}=g_{\rho \sigma} H_{\mu \nu}$	deed $\widetilde{R}_{(\rho\sigma)\mu\nu}=g_{\rho\sigma}H_{\mu\nu}$) and does not satisfy the fa-
1510.06699_F00507	22	1.000	`V_{4}`	V_{4}	sor in a Riemannian V_4 spacetime. One may also show
1510.06699_F00508	22	1.000	`\widetilde{R}^{\rho}{ }_{\sigma \mu \nu}`	$\widetilde{R}^{\rho}{ }_{\sigma \mu \nu}$	that, with the given arrangements of indices, both $\widetilde{R}^\rho{}_{\sigma\mu\nu}$
1510.06699_F00509	22	1.000	`\widetilde{R}_{\mu \nu} \equiv \widetilde{R}_{\mu \lambda \nu}{ }^{\lambda}=R_{\mu \nu}-H_{\mu \nu}`	$\widetilde{R}_{\mu \nu} \equiv \widetilde{R}_{\mu \lambda \nu}{ }^{\lambda}=R_{\mu \nu}-H_{\mu \nu}$	$\widetilde{R}_{\mu\nu}\equiv\widetilde{R}_{\mu\lambda\nu}{}^\lambda=R_{\mu\nu}-H_{\mu\nu}$ and $\widetilde{R}\equiv\widetilde{R}^\mu{}_\mu=R$. As one
1510.06699_F00510	22	1.000	`\widetilde{R} \equiv \widetilde{R}^{\mu}{ }_{\mu}=R`	$\widetilde{R} \equiv \widetilde{R}^{\mu}{ }_{\mu}=R$	$\widetilde{R}_{\mu\nu}\equiv\widetilde{R}_{\mu\lambda\nu}{}^\lambda=R_{\mu\nu}-H_{\mu\nu}$ and $\widetilde{R}\equiv\widetilde{R}^\mu{}_\mu=R$. As one
1510.06699_F00511	23	1.000	`J^{\rho}`	J^{ρ}	operators acting on a vector J^ρ (say) of Weyl weight w,
1510.06699_F00512	23	1.000	`{ }^{0} \Gamma^{\lambda}{ }_{\mu \nu} \equiv \frac{1}{2} g^{\lambda \rho}\left(\partial_{\mu} g_{\nu \rho}+\partial_{\nu} g_{\mu \rho}-\partial_{\rho} g_{\mu \nu}\right)`	${ }^{0} \Gamma^{\lambda}{ }_{\mu \nu} \equiv \frac{1}{2} g^{\lambda \rho}\left(\partial_{\mu} g_{\nu \rho}+\partial_{\nu} g_{\mu \rho}-\partial_{\rho} g_{\mu \nu}\right)$	in which ${}^0\Gamma^\lambda{}_{\mu\nu}\equiv\frac{1}{2}g^{\lambda\rho}(\partial_\mu g_{\nu\rho}+\partial_\nu g_{\mu\rho}-\partial_\rho g_{\mu\nu})$ is the
1510.06699_F00513	23	1.000	`K^{* \lambda}{ }_{\mu \nu}`	$K^{* \lambda}{ }_{\mu \nu}$	standard metric (Christoffel) connection and $K^{*\lambda}{}_{\mu\nu}$ is
1510.06699_F00514	23	1.000	`K_{\lambda \mu \nu}^{*}=-K_{\mu \lambda \nu}^{*}`	$K_{\lambda \mu \nu}^{*}=-K_{\mu \lambda \nu}^{*}$	for which it is worth noting that $K^*_{\lambda\mu\nu}=-K^*_{\mu\lambda\nu}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_F00515	23	1.000	`{ }^{0} \Gamma^{* \sigma}{ }_{\rho \mu}`	${ }^{0} \Gamma^{* \sigma}{ }_{\rho \mu}$	connection ${}^{0}\Gamma^{*\sigma}{}_{\rho\mu}$ represent the same geometrical object
1510.06699_F00516	23	0.661	`\Gamma^{\sigma}{ }_{\nu \mu} \rightarrow^{0} \Gamma^{* \sigma}{ }_{\nu \mu}`	$\Gamma^{\sigma}{ }_{\nu \mu} \rightarrow^{0} \Gamma^{* \sigma}{ }_{\nu \mu}$	again hold with the replacements $\Gamma^{\sigma}{}_{\nu\mu} \rightarrow {}^{0}\Gamma^{*\sigma}{}_{\nu\mu}$ and
1510.06699_F00517	23	1.000	`A^{a}{ }_{b \mu} \rightarrow{ }^{0} A^{* a}{ }_{b \mu}`	$A^{a}{ }_{b \mu} \rightarrow{ }^{0} A^{* a}{ }_{b \mu}$	$A^{a}{}_{b\mu} \rightarrow {}^{0}A^{*a}{}_{b\mu}$, from which one can directly derive (131).
1510.06699_F00518	23	1.000	`{ }^{0} \nabla_{\sigma}^{*} g_{\mu \nu}=0`	${ }^{0} \nabla_{\sigma}^{*} g_{\mu \nu}=0$	One also obtains the metricity condition ${}^{0}\nabla^{*}_{\sigma} g_{\mu\nu} = 0$. Fi-
1510.06699_F00519	23	0.983	`R^{\rho}{ }_{\sigma \mu \nu} \rightarrow{ }^{0} R^{* \rho}{ }_{\sigma \mu \nu}`	$R^{\rho}{ }_{\sigma \mu \nu} \rightarrow{ }^{0} R^{* \rho}{ }_{\sigma \mu \nu}$	but with $R^{\rho}{}_{\sigma\mu\nu} \rightarrow {}^{0}R^{*\rho}{}_{\sigma\mu\nu}$ and $\Gamma^{\sigma}{}_{\nu\mu} \rightarrow {}^{0}\Gamma^{*\sigma}{}_{\nu\mu}$, whereas
1510.06699_F00520	23	0.983	`\Gamma^{\sigma}{ }_{\nu \mu} \rightarrow{ }^{0} \Gamma^{* \sigma}{ }_{\nu \mu}`	$\Gamma^{\sigma}{ }_{\nu \mu} \rightarrow{ }^{0} \Gamma^{* \sigma}{ }_{\nu \mu}$	but with $R^{\rho}{}_{\sigma\mu\nu} \rightarrow {}^{0}R^{*\rho}{}_{\sigma\mu\nu}$ and $\Gamma^{\sigma}{}_{\nu\mu} \rightarrow {}^{0}\Gamma^{*\sigma}{}_{\nu\mu}$, whereas
1510.06699_F00521	23	1.000	`{ }^{0} T^{* \lambda}{ }_{\mu \nu}=0`	${ }^{0} T^{* \lambda}{ }_{\mu \nu}=0$	(126) becomes simply ${}^{0}T^{*\lambda}{}_{\mu\nu} = 0$, indicating the ab-
1510.06699_F00522	23	0.941	`\nabla_{\mu}^{*} \rightarrow{ }^{0} \nabla_{\mu}^{*}, R^{\rho}{ }_{\sigma \mu \nu} \rightarrow{ }^{0} R^{* \rho}{ }_{\sigma \mu \nu}`	$\nabla_{\mu}^{*} \rightarrow{ }^{0} \nabla_{\mu}^{*}, R^{\rho}{ }_{\sigma \mu \nu} \rightarrow{ }^{0} R^{* \rho}{ }_{\sigma \mu \nu}$	also valid, but with $\nabla^{*}_{\mu} \rightarrow {}^{0}\nabla^{*}_{\mu}$, $R^{\rho}{}_{\sigma\mu\nu} \rightarrow {}^{0}R^{*\rho}{}_{\sigma\mu\nu}$ and
1510.06699_F00523	23	0.976	`T^{* \sigma}{ }_{\mu \nu} \rightarrow{ }^{0} T^{* \sigma}{ }_{\mu \nu}=0`	$T^{* \sigma}{ }_{\mu \nu} \rightarrow{ }^{0} T^{* \sigma}{ }_{\mu \nu}=0$	$T^{*\sigma}{}_{\mu\nu} \rightarrow {}^{0}T^{*\sigma}{}_{\mu\nu} = 0$.
1510.06699_F00524	23	1.000	`P_{\nu} \equiv \partial_{\nu} \rho, \mathcal{P}_{a} \equiv h_{a}{ }^{\nu} P_{\nu}`	$P_{\nu} \equiv \partial_{\nu} \rho, \mathcal{P}_{a} \equiv h_{a}{ }^{\nu} P_{\nu}$	where $P_{\nu} \equiv \partial_{\nu}\rho$, $\mathcal{P}_{a} \equiv h_{a}{}^{\nu}P_{\nu}$ and θ is an arbitrary pa-
1510.06699_F00525	23	1.000	`\theta`	θ	where $P_{\nu} \equiv \partial_{\nu}\rho$, $\mathcal{P}_{a} \equiv h_{a}{}^{\nu}P_{\nu}$ and θ is an arbitrary pa-
1510.06699_F00526	23	1.000	`w=-2`	$w=-2$	forms covariantly with weight $w = -2$ and also acts as
1510.06699_F00527	24	1.000	`\mathcal{T}^{a}{ }_{b c}=0`	$\mathcal{T}^{a}{ }_{b c}=0$	field originally has vanishing PGT torsion $\mathcal{T}^{a}{}_{bc} = 0$, the
1510.06699_F00528	24	1.000	`\mathcal{T}^{\prime a}{ }_{b c} \neq 0`	$\mathcal{T}^{\prime a}{ }_{b c} \neq 0$	means that the local dilation leads to $\mathcal{T}'^{a}{}_{bc} \neq 0$.
1510.06699_F00529	24	1.000	`\mathcal{T}^{\dagger a}{ }_{b c}`	$\mathcal{T}^{\dagger a}{ }_{b c}$	Thus, for general values of θ, neither $\mathcal{R}^{ab}{}_{cd}$ nor $\mathcal{T}^{\dagger a}{}_{bc}$
1510.06699_F00530	24	1.000	`\theta=0`	$\theta=0$	transforms covariantly, but for $\theta = 0$ one recovers the
1510.06699_F00531	24	1.000	`\theta=1`	$\theta=1$	curvature under local dilations, whereas for $\theta = 1$ one
1510.06699_F00532	24	1.000	`{ }^{0} \mathcal{R}^{a b}{ }_{c d}`	${ }^{0} \mathcal{R}^{a b}{ }_{c d}$	${}^{0}\mathcal{R}^{ab}{}_{cd}$ and its contractions (see Appendix B) are given
1510.06699_F00533	24	0.971	`{ }^{0} \mathcal{T}^{a}{ }_{b c}`	${ }^{0} \mathcal{T}^{a}{ }_{b c}$	in Section V B; the 'reduced' PGT torsion ${}^{0}\mathcal{T}^{a}{}_{bc}$ vanishes
1510.06699_F00534	24	1.000	`D_{\mu}^{\dagger} \varphi`	$D_{\mu}^{\dagger} \varphi$	tive $D^{\dagger}_{\mu}\varphi$ by introducing a dilation vector gauge field
1510.06699_F00535	24	1.000	`V_{\mu}`	V_{μ}	(called V_{μ} to distinguish it from the dilation gauge field
1510.06699_F00536	24	0.999	`\mathcal{V}_{a}=h_{a}{ }^{\mu} V_{\mu}`	$\mathcal{V}_{a}=h_{a}{ }^{\mu} V_{\mu}$	where $\mathcal{V}_{a} = h_{a}{}^{\mu}V_{\mu}$. Clearly, by construction, $A^{\dagger ab}{}_{\mu}$ is
1510.06699_F00537	24	0.999	`A^{\dagger a b}{ }_{\mu}`	$A^{\dagger a b}{ }_{\mu}$	where $\mathcal{V}_{a} = h_{a}{}^{\mu}V_{\mu}$. Clearly, by construction, $A^{\dagger ab}{}_{\mu}$ is
1510.06699_F00538	24	1.000	`\partial_{\mu} \varphi^{\prime}`	$\partial_{\mu} \varphi^{\prime}$	covariant derivative to cancel terms arising from $\partial_{\mu}\varphi'$
1510.06699_F00539	24	1.000	`T_{\mu}=`	$T_{\mu}=$	in which w is the Weyl weight of the field φ and $T_{\mu} =$
1510.06699_F00540	24	0.999	`b^{a}{ }_{\mu} \mathcal{T}_{a}`	$b^{a}{ }_{\mu} \mathcal{T}_{a}$	$b^{a}{}_{\mu}\mathcal{T}_{a}$, where $\mathcal{T}_{a}$ is the trace of the PGT torsion (see
1510.06699_F00541	24	1.000	`T_{\mu}`	T_{μ}	equation (66). Here we consider T_{μ} merely as a short-
1510.06699_F00542	24	1.000	`T_{\mu}^{\prime}=T_{\mu}+3(1-\theta) P_{\mu}`	$T_{\mu}^{\prime}=T_{\mu}+3(1-\theta) P_{\mu}$	$T'_{\mu} = T_{\mu} + 3(1-\theta)P_{\mu}$ and so (143) does indeed trans-
1510.06699_F00543	24	0.898	`A^{\dagger a b}{ }_{\mu}=A^{\dagger a b}{ }_{\mu}`	$A^{\dagger a b}{ }_{\mu}=A^{\dagger a b}{ }_{\mu}$	tions (136)–(137), i.e. $A^{\dagger\prime ab}{}_{\mu} = A^{\dagger ab}{}_{\mu}$ and so it has a
1510.06699_F00544	24	1.000	`V`	V	covariant derivative (143) and the V-field transformation
1510.06699_F00545	25	1.000	`V_{\mu}=0`	$V_{\mu}=0$	that one can covariantly set $V_{\mu} = 0$, so that $A^{\dagger ab}{}_{\mu} = A^{ab}{}_{\mu}$
1510.06699_F00546	25	1.000	`A^{\dagger a b}{ }_{\mu}=A^{a b}{ }_{\mu}`	$A^{\dagger a b}{ }_{\mu}=A^{a b}{ }_{\mu}$	that one can covariantly set $V_{\mu} = 0$, so that $A^{\dagger ab}{}_{\mu} = A^{ab}{}_{\mu}$
1510.06699_F00547	25	1.000	`\left.D_{\mu}^{\dagger} \varphi\right|_{\theta=0}=\left(D_{\mu}-\frac{1}{3} w T_{\mu}\right) \varphi`	$\left.D_{\mu}^{\dagger} \varphi\right\|_{\theta=0}=\left(D_{\mu}-\frac{1}{3} w T_{\mu}\right) \varphi$	as $D^{\dagger}_{\mu}\varphi\|_{\theta=0} = (D_{\mu} - \frac{1}{3}wT_{\mu})\varphi$, in which the role played
1510.06699_F00548	25	1.000	`T_{\mu}=0`	$T_{\mu}=0$	variantly set the PGT torsion to zero, and hence $T_{\mu} = 0$,
1510.06699_F00549	25	1.000	`\left.D_{\mu}^{\dagger} \varphi\right|_{\theta=1} \equiv\left(\partial_{\mu}+\frac{1}{2} A^{\dagger a b}{ }_{\mu} \Sigma_{a b}-w V_{\mu}\right) \varphi`	$\left.D_{\mu}^{\dagger} \varphi\right\|_{\theta=1} \equiv\left(\partial_{\mu}+\frac{1}{2} A^{\dagger a b}{ }_{\mu} \Sigma_{a b}-w V_{\mu}\right) \varphi$	$D^{\dagger}_{\mu}\varphi\|_{\theta=1} \equiv (\partial_{\mu} + \frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab} - wV_{\mu})\varphi$.
1510.06699_F00550	25	1.000	`\partial_{\mu}^{\dagger} \varphi(x)`	$\partial_{\mu}^{\dagger} \varphi(x)$	(151) Thus the new minimally-coupled matter Lagrangian is a
1510.06699_F00551	25	1.000	`w-1`	$w-1$	which transforms covariantly with weight $w-1$, as de-
1510.06699_F00552	25	0.999	`\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}} \left(\varphi, \mathcal{D}^{\dagger} \varphi\right)`	$\mathcal{L}_{\mathrm{M}} \equiv h^{-1} L_{\mathrm{M}} \left(\varphi, \mathcal{D}^{\dagger} \varphi\right)$	action by the Lagrangian density $\mathcal{L}_{\mathrm{M}} \equiv h^{-1}L_{\mathrm{M}}(\varphi, \mathcal{D}^{\dagger}\varphi)$.

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1510.06699_F00553	25	1.000	`\phi^{2} \mathcal{R}^{\dagger}`	$\phi^{2} \mathcal{R}^{\dagger}$	tional to $\phi^2\mathcal{R}^\dagger$ or $\phi^2 L_{\mathcal{T}^{\dagger 2}}$ are extended Weyl-covariant
1510.06699_F00554	25	1.000	`\phi^{2} L_{\mathcal{T}^{\dagger 2}}`	$\phi^{2} L_{\mathcal{T}^{\dagger 2}}$	tional to $\phi^2\mathcal{R}^\dagger$ or $\phi^2 L_{\mathcal{T}^{\dagger 2}}$ are extended Weyl-covariant
1510.06699_F00555	25	0.968	`\operatorname{SU}(2,2)`	$\operatorname{SU}(2,2)$	of $\mathrm{SU}(2,2)$, to which $\mathrm{O}(4,2)$ is locally isomorphic. In
1510.06699_F00556	25	0.968	`\mathrm{O}(4,2)`	$\mathrm{O}(4,2)$	of $\mathrm{SU}(2,2)$, to which $\mathrm{O}(4,2)$ is locally isomorphic. In
1510.06699_F00557	26	1.000	`e_{b \mu} b_{a}-e_{a \mu} b_{b}`	$e_{b \mu} b_{a}-e_{a \mu} b_{b}$	in which a piece $e_{b\mu}b_a - e_{a\mu}b_b$ is added to the spin connec-
1510.06699_F00558	26	1.000	`\omega_{\mu a b}`	$\omega_{\mu a b}$	tion $\omega_{\mu ab}$ (their b-field is the 'dilatation field', which in
1510.06699_F00559	26	0.999	`\mathcal{V}`	$\mathcal{V}$	our case corresponds to $\mathcal{V}$, while their $e_{a\mu}$ is a version of
1510.06699_F00560	26	0.999	`e_{a \mu}`	$e_{a \mu}$	our case corresponds to $\mathcal{V}$, while their $e_{a\mu}$ is a version of
1510.06699_F00561	26	1.000	`c_{b \mu} x_{a}-c_{a \mu} x_{b}`	$c_{b \mu} x_{a}-c_{a \mu} x_{b}$	$c_{b\mu}x_a - c_{a\mu}x_b$, where here $c_{a\mu}$ is the extra 'vierbein-type'
1510.06699_F00562	26	1.000	`c_{a \mu}`	$c_{a \mu}$	$c_{b\mu}x_a - c_{a\mu}x_b$, where here $c_{a\mu}$ is the extra 'vierbein-type'
1510.06699_F00563	26	1.000	`x_{a}`	x_{a}	tions. Here x_a is a global surrogate for $\mathcal{V}_a$ (noting the fact
1510.06699_F00564	26	1.000	`\mathcal{V}_{a}`	$\mathcal{V}_{a}$	tions. Here x_a is a global surrogate for $\mathcal{V}_a$ (noting the fact
1510.06699_F00565	26	1.000	`x \cdot \nabla`	$x \cdot \nabla$	that the generator of global dilations is $x\cdot\nabla$), and so this
1510.06699_F00566	26	1.000	`{ }^{0} \nabla_{\mu}^{\dagger} \equiv{ }^{0} \nabla_{\mu}-w\left(V_{\mu}+\frac{1}{3} T_{\mu}\right),{ }^{0} \Gamma^{\lambda}{ }_{\mu \nu} \equiv \frac{1}{2} \gamma^{\lambda \rho}\left(\partial_{\mu} \gamma_{\nu \rho}+\right.`	${ }^{0} \nabla_{\mu}^{\dagger} \equiv{ }^{0} \nabla_{\mu}-w\left(V_{\mu}+\frac{1}{3} T_{\mu}\right),{ }^{0} \Gamma^{\lambda}{ }_{\mu \nu} \equiv \frac{1}{2} \gamma^{\lambda \rho}\left(\partial_{\mu} \gamma_{\nu \rho}+\right.$	where ${}^0\nabla^\dagger_\mu \equiv {}^0\nabla_\mu - w(V_\mu+\frac{1}{3}T_\mu)$, ${}^0\Gamma^\lambda{}_{\mu\nu} \equiv \frac{1}{2}\gamma^{\lambda\rho}(\partial_\mu\gamma_{\nu\rho}+$
1510.06699_F00567	26	0.817	`\partial_{\nu} \gamma_{\mu \rho}-\partial_{\rho} \gamma_{\mu \nu}`	$\partial_{\nu} \gamma_{\mu \rho}-\partial_{\rho} \gamma_{\mu \nu}$	$\partial_\nu\gamma_{\mu\rho} - \partial_\rho\gamma_{\mu\nu})$ is the metric connection corresponding to
1510.06699_F00568	26	0.801	`D \mu`	$D \mu$	$D^\dagger_\mu$ is applied, and w is its Weyl weight.
1510.06699_F00569	26	1.000	`T^{a}{ }_{\mu \nu}`	$T^{a}{ }_{\mu \nu}$	vature) in PGT (and WGT), $T^a{}_{\mu\nu}$ is the PGT transla-
1510.06699_F00570	26	1.000	`D_{\mu}`	D_{μ}	tional gauge field (torsion), and D_μ is the PGT covariant
1510.06699_F00571	26	1.000	`R^{\dagger a b}{ }_{\mu \nu}`	$R^{\dagger a b}{ }_{\mu \nu}$	As in WGT, both $R^{\dagger ab}{}_{\mu\nu}$ and $H^\dagger_{\mu\nu}$ transform covariantly
1510.06699_F00572	26	1.000	`H_{\mu \nu}^{\dagger}`	$H_{\mu \nu}^{\dagger}$	As in WGT, both $R^{\dagger ab}{}_{\mu\nu}$ and $H^\dagger_{\mu\nu}$ transform covariantly
1510.06699_F00573	26	1.000	`\mathcal{R}^{\dagger a b}{ }_{c d}=h_{c}{ }^{\mu} h_{d}{ }^{\nu} R^{\dagger a b}{ }_{\mu \nu}`	$\mathcal{R}^{\dagger a b}{ }_{c d}=h_{c}{ }^{\mu} h_{d}{ }^{\nu} R^{\dagger a b}{ }_{\mu \nu}$	where $\mathcal{R}^{\dagger ab}{}_{cd} = h_c{}^\mu h_d{}^\nu R^{\dagger ab}{}_{\mu\nu}$ and $\mathcal{H}^\dagger_{cd} = h_c{}^\mu h_d{}^\nu H^\dagger_{\mu\nu}$,
1510.06699_F00574	26	1.000	`\mathcal{H}_{c d}^{\dagger}=h_{c}{ }^{\mu} h_{d}{ }^{\nu} H_{\mu \nu}^{\dagger}`	$\mathcal{H}_{c d}^{\dagger}=h_{c}{ }^{\mu} h_{d}{ }^{\nu} H_{\mu \nu}^{\dagger}$	where $\mathcal{R}^{\dagger ab}{}_{cd} = h_c{}^\mu h_d{}^\nu R^{\dagger ab}{}_{\mu\nu}$ and $\mathcal{H}^\dagger_{cd} = h_c{}^\mu h_d{}^\nu H^\dagger_{\mu\nu}$,
1510.06699_F00575	27	1.000	`\mathcal{R}^{\dagger a b}{ }_{c d}, \mathcal{H}_{c d}^{\dagger}`	$\mathcal{R}^{\dagger a b}{ }_{c d}, \mathcal{H}_{c d}^{\dagger}$	It is easy to show that, similarly to WGT, $\mathcal{R}^{\dagger ab}{}_{cd}$, $\mathcal{H}^\dagger_{cd}$
1510.06699_F00576	27	1.000	`\mathcal{T}^{\dagger a}{ }_{c d}`	$\mathcal{T}^{\dagger a}{ }_{c d}$	and $\mathcal{T}^{\dagger a}{}_{cd}$ transform covariantly under extended local
1510.06699_F00577	27	1.000	`w\left(\mathcal{R}^{\dagger a b}{ }_{c d}\right)=w\left(\mathcal{H}_{c d}^{\dagger}\right)=-2`	$w\left(\mathcal{R}^{\dagger a b}{ }_{c d}\right)=w\left(\mathcal{H}_{c d}^{\dagger}\right)=-2$	dilations with weights $w(\mathcal{R}^{\dagger ab}{}_{cd}) = w(\mathcal{H}^\dagger_{cd}) = -2$ and
1510.06699_F00578	27	0.999	`w\left(\mathcal{T}^{\dagger a}{ }_{c d}\right)=-1`	$w\left(\mathcal{T}^{\dagger a}{ }_{c d}\right)=-1$	$w(\mathcal{T}^{\dagger a}{}_{cd}) = -1$ respectively.
1510.06699_F00579	27	1.000	`\mathcal{R}^{\dagger a b}{ }_{c d}`	$\mathcal{R}^{\dagger a b}{ }_{c d}$	plicit forms of $\mathcal{R}^{\dagger ab}{}_{cd}$ and its contractions are given in
1510.06699_F00580	27	1.000	`\mathcal{H}_{a b}^{\dagger}`	$\mathcal{H}_{a b}^{\dagger}$	strength $\mathcal{H}^\dagger_{ab}$ is antisymmetric in a and b and hence also
1510.06699_F00581	27	0.725	`\mathcal{R}^{\dagger a b}{ }_{c d}(h, A, \partial A, V, \partial V), \mathcal{T}^{\dagger a}{ }_{b c}(h, \partial h, A)`	$\mathcal{R}^{\dagger a b}{ }_{c d}(h, A, \partial A, V, \partial V), \mathcal{T}^{\dagger a}{ }_{b c}(h, \partial h, A)$	tives are $\mathcal{R}^{\dagger ab}{}_{cd}(h,A,\partial A,V,\partial V)$, $\mathcal{T}^{\dagger a}{}_{bc}(h,\partial h,A)$ and
1510.06699_F00582	27	1.000	`\mathcal{H}_{a b}^{\dagger}\left(h, \partial h, \partial^{2} h, A, \partial A, \partial V\right)`	$\mathcal{H}_{a b}^{\dagger}\left(h, \partial h, \partial^{2} h, A, \partial A, \partial V\right)$	$\mathcal{H}^\dagger_{ab}(h,\partial h,\partial^2 h,A,\partial A,\partial V)$. It is important to note that
1510.06699_F00583	27	1.000	`\partial h, \partial^{2} h, A`	$\partial h, \partial^{2} h, A$	strength $\mathcal{H}^\dagger_{ab}$, which depends on ∂h, $\partial^2 h$, A and ∂A, in
1510.06699_F00584	27	1.000	`\partial^{2} h`	$\partial^{2} h$	tive $\partial^2 h$ is a particularly unusual feature as compared
1510.06699_F00585	27	1.000	`{ }^{0} \mathcal{A}_{a b c}^{\dagger} \equiv \frac{1}{2}\left(c_{a b c}^{\dagger}+\right.`	${ }^{0} \mathcal{A}_{a b c}^{\dagger} \equiv \frac{1}{2}\left(c_{a b c}^{\dagger}+\right.$	that both ${}^0\mathcal{A}^*_{abc}$ and $\mathcal{K}^*_{abc}$ are antisymmetric in their first
1510.06699_F00586	27	0.973	`\left.c_{b c a}^{\dagger}-c_{c a b}^{\dagger}\right)`	$\left.c_{b c a}^{\dagger}-c_{c a b}^{\dagger}\right)$	two indices[69].

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1510.06699_F00587	27	0.973	`c^{\dagger c}{ }_{a b} \equiv h_{a}{ }^{\mu} h_{b}{ }^{\nu}\left(\partial_{\mu}^{\dagger} b^{c}{ }_{\nu}-\partial_{\nu}^{\dagger} b^{c}{ }_{\mu}\right)`	$c^{\dagger c}{}_{ab} \equiv h_a{}^\mu h_b{}^\nu\left(\partial^\dagger_\mu b^c{}_\nu - \partial^\dagger_\nu b^c{}_\mu\right)$	two indices[69].
1510.06699_F00588	27	0.981	`\mathcal{K}_{a b c}^{\dagger} \equiv-\frac{1}{2}\left(\mathcal{T}_{a b c}^{\dagger}+\mathcal{T}_{b c a}^{\dagger}-\mathcal{T}_{c a b}^{\dagger}\right)`	$\mathcal{K}^\dagger_{abc} \equiv -\frac{1}{2}\left(\mathcal{T}^\dagger_{abc}+\mathcal{T}^\dagger_{bca}-\mathcal{T}^\dagger_{cab}\right)$	Under a local Weyl transformation, the quantities $^0\mathcal{A}^*_{\cdot}$ transform in the same way as $\mathcal{A}^*_{\cdot}$, whereas $\mathcal{K}^*_{\cdot}$
1510.06699_F00589	27	0.981	`{ }^{0} \mathcal{A}_{a b c}^{\dagger}`	$^0\mathcal{A}^\dagger_{abc}$	Under a local Weyl transformation, the quantities $^0\mathcal{A}^*_{\cdot}$ transform in the same way as $\mathcal{A}^*_{\cdot}$, whereas $\mathcal{K}^*_{\cdot}$
1510.06699_F00590	27	0.840	`\mathcal{K}_{a b c}^{\dagger}`	$\mathcal{K}^\dagger_{abc}$	transform as the components of a local tensor with weight
1510.06699_F00591	27	0.568	`{ }^{0} \mathcal{A}_{\text {abc }}^{\dagger}`	$^0\mathcal{A}^\dagger_{\text{abc}}$	well as $\mathcal{A}^*_{abc}$ in the construction of a covariant derivative.
1510.06699_F00592	27	0.551	`\mathcal{K}_{\text {abc }}^{\dagger}`	$\mathcal{K}^\dagger_{\text{abc}}$	We may therefore construct the 'reduced' (Λ, ρ)-covariant
1510.06699_F00593	27	0.992	`\mathcal{A}_{a b c}^{\dagger}`	$\mathcal{A}^\dagger_{abc}$	
1510.06699_F00594	27	1.000	`{ }^{0} D_{\mu}^{\dagger} \varphi`	$^0D^\dagger_\mu\varphi$	in eWGT, however, is that $^0D^\dagger_\mu\varphi$ still depends on the
1510.06699_F00595	27	0.523	`{ }^{0} \mathcal{D}_{a}^{\dagger} \equiv h_{a}{ }^{\mu} 0^{0} D_{\mu}^{\dagger}`	$^0\mathcal{D}^\dagger_a \equiv h_a{}^\mu 0^0 D^\dagger_\mu$	$^0\mathcal{D}^\dagger_a \equiv h_a{}^\mu\,{}^0D^\dagger_\mu$, which consequently transforms in the
1510.06699_F00596	27	1.000	`\mathcal{D}_{a}^{\dagger}`	$\mathcal{D}^\dagger_a$	same way as $\mathcal{D}^\dagger_a$, as required. The 'full' generalised co-
1510.06699_F00597	27	1.000	`{ }^{0} \mathcal{D}_{a}^{\dagger}`	$^0\mathcal{D}^\dagger_a$	to $^0\mathcal{D}^\dagger_a$.
1510.06699_F00598	27	1.000	`\mathcal{R}^{\dagger a b}{ }_{c d}, \mathcal{T}^{\dagger a}{ }_{b c}`	$\mathcal{R}^{\dagger ab}{}_{cd}, \mathcal{T}^{\dagger a}{}_{bc}$	the gravitational gauge field strengths $\mathcal{R}^{\dagger ab}{}_{cd}$, $\mathcal{T}^{\dagger a}{}_{bc}$ and
1510.06699_F00599	28	0.999	`\mathcal{T}_{a b c}^{\dagger}=\epsilon_{a b c d} t^{d}`	$\mathcal{T}^\dagger_{abc} = \epsilon_{abcd}t^d$	sion' is totally antisymmetric, such that $\mathcal{T}^\dagger_{abc} = \epsilon_{abcd}t^d$
1510.06699_F00600	28	0.800	`\mathcal{R}_{a b c d}^{\dagger}`	$\mathcal{R}^\dagger_{abcd}$	and discussed in Section II G. In principle, from $\mathcal{R}^\dagger_{abcd}$,
1510.06699_F00601	28	0.899	`\mathcal{T}_{a b c}^{\dagger}`	$\mathcal{T}^\dagger_{abc}$	$\mathcal{T}^\dagger_{abc}$ and $\mathcal{H}^\dagger_{ab}$, one can construct a free gravitational ac-
1510.06699_F00602	28	1.000	`\mathcal{R}^{\dagger} \equiv`	$\mathcal{R}^\dagger \equiv$	are not allowed. Similarly, higher-order terms in $\mathcal{R}_{abcd}$
1510.06699_F00603	28	1.000	`\mathcal{R}^{\dagger a b}{ }_{a b}`	$\mathcal{R}^{\dagger ab}{}_{ab}$	and $\mathcal{H}_{ab}$ are forbidden; in principle one could include
1510.06699_F00604	28	1.000	`L_{\mathcal{R}^{\dagger 2}}`	$L_{\mathcal{R}^{\dagger 2}}$	where the expressions for $L_{\mathcal{R}^{\dagger 2}}$ and $L_{\mathcal{H}^{\dagger 2}}$ are given in
1510.06699_F00605	28	1.000	`L_{\mathcal{H}^{\dagger 2}}`	$L_{\mathcal{H}^{\dagger 2}}$	where the expressions for $L_{\mathcal{R}^{\dagger 2}}$ and $L_{\mathcal{H}^{\dagger 2}}$ are given in
1510.06699_F00606	28	1.000	`\mathcal{R}_{a b c d} \rightarrow \mathcal{R}_{a b c d}^{\dagger}`	$\mathcal{R}_{abcd} \rightarrow \mathcal{R}^\dagger_{abcd}$	i.e. making the replacements $\mathcal{R}_{abcd} \to \mathcal{R}^\dagger_{abcd}$ and $\mathcal{H}_{ab} \to$
1510.06699_F00607	28	1.000	`\mathcal{H}_{a b} \rightarrow`	$\mathcal{H}_{ab} \rightarrow$	i.e. making the replacements $\mathcal{R}_{abcd} \to \mathcal{R}^\dagger_{abcd}$ and $\mathcal{H}_{ab} \to$
1510.06699_F00608	28	0.829	`{ }^{\underline{71}}`	$^{\underline{71}}$	$\mathcal{H}^\dagger_{ab}$. Performing a similar derivation to the WGT case[71],
1510.06699_F00609	28	0.556	`\phi^{2} L_{\mathcal{T} \dagger^{2}}`	$\phi^2 L_{\mathcal{T}\dagger^2}$	form $\phi^2\mathcal{R}^\dagger$ and $\phi^2 L_{\mathcal{T}^{\dagger 2}}$, where $L_{\mathcal{T}^{\dagger 2}}$ is given in (3) with
1510.06699_F00610	28	0.556	`L_{\mathcal{T} \dagger^{2}}`	$L_{\mathcal{T}\dagger^2}$	form $\phi^2\mathcal{R}^\dagger$ and $\phi^2 L_{\mathcal{T}^{\dagger 2}}$, where $L_{\mathcal{T}^{\dagger 2}}$ is given in (3) with
1510.06699_F00611	28	0.999	`\mathcal{T}_{a b c} \rightarrow \mathcal{T}_{a b c}^{\dagger}`	$\mathcal{T}_{abc} \rightarrow \mathcal{T}^\dagger_{abc}$	$\mathcal{T}_{abc} \to \mathcal{T}^\dagger_{abc}$ (but with no term proportional to β_3,
1510.06699_F00612	28	1.000	`\mathcal{H}_{a b}^{\dagger} \mathcal{H}^{\dagger a b}`	$\mathcal{H}^\dagger_{ab}\mathcal{H}^{\dagger ab}$	tional to $\mathcal{H}^\dagger_{ab}\mathcal{H}^{\dagger ab}$ in (166). In the matter sector, covari-
1510.06699_F00613	28	1.000	`\varphi, \partial \varphi, h, \partial h, A`	$\varphi, \partial\varphi, h, \partial h, A$	on φ, $\partial\varphi$, h, ∂h, A and V; the dependence on ∂h is new
1510.06699_F00614	28	0.998	`\mathcal{D}_{a}^{\dagger} \varphi`	$\mathcal{D}^\dagger_a\varphi$	$\mathcal{T}_a$ in $\mathcal{D}^\dagger_a\varphi$. We will also consider here the inclusion of
1510.06699_F00615	28	1.000	`\partial V`	∂V	tional dependence on ∂A and ∂V; the latter is new com-
1510.06699_F00616	29	0.626	`\mathcal{R}_{a b c d}^{\dagger}(h, A, \partial A, V, \partial V), \mathcal{T}_{a b c}^{\dagger}(h, \partial h, A)`	$\mathcal{R}^\dagger_{abcd}(h, A, \partial A, V, \partial V), \mathcal{T}^\dagger_{abc}(h, \partial h, A)$	ever, that if one substitutes for $\mathcal{A}_{abc}$ using (47), one ob-
1510.06699_F00617	29	1.000	`\mathcal{H}^{\dagger a b} \mathcal{H}_{a b}^{\dagger}`	$\mathcal{H}^{\dagger ab}\mathcal{H}^\dagger_{ab}$	Variation of $S_{\rm T}$ with respect to the gauge fields $h_a{}^\mu$,
1510.06699_F00618	29	1.000	`t^{a}{ }_{\mu} \equiv \delta \mathcal{L}_{\mathrm{G}} / \delta h_{a}{ }^{\mu}, s_{a b}{ }^{\mu} \equiv \delta \mathcal{L}_{\mathrm{G}} / \delta A^{a b}{ }_{\mu}`	$t^a{}_\mu \equiv \delta\mathcal{L}_{\mathrm{G}}/\delta h_a{}^\mu, s_{ab}{}^\mu \equiv \delta\mathcal{L}_{\mathrm{G}}/\delta A^{ab}{}_\mu$	accordance with their respective index structures, and
1510.06699_F00619	29	1.000	`j^{\mu} \equiv`	$j^\mu \equiv$	accordance with their respective index structures, and
1510.06699_F00620	29	1.000	`\delta \mathcal{L}_{\mathrm{G}} / \delta V_{\mu}`	$\delta\mathcal{L}_{\mathrm{G}}/\delta V_\mu$	are also covariant under local dilations with Weyl weights

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1510.06699_F00621	29	1.000	`\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta V_{\mu}`	$\zeta^{\mu} \equiv \delta \mathcal{L}_{\mathrm{M}} / \delta V_{\mu}$	convenient to work with quantities carrying only Latin
1510.06699_F00622	29	0.998	`t^{a}{ }_{b}`	$t^{a}{ }_{b}$	(62a)–(62c) are clearly invariant under GCT transform
1510.06699_F00623	29	1.000	`t^{a}{ }_{a}`	$t^{a}{ }_{a}$	although it is worth noting that the traces $t^a{}_a$ and $\tau^a{}_a$
1510.06699_F00624	29	1.000	`\tau^{a}{ }_{a}`	$\tau^{a}{ }_{a}$	although it is worth noting that the traces $t^a{}_a$ and $\tau^a{}_a$
1510.06699_F00625	29	1.000	`t_{a b}`	$t_{a b}$	momentum tensor. Nonetheless, although neither t_{ab} nor
1510.06699_F00626	29	1.000	`\tau^{\dagger a}{ }_{a}=\tau^{a}{ }_{a}`	$\tau^{\dagger a}{ }_{a}=\tau^{a}{ }_{a}$	is invariant). It is also worth noting that $\tau^{\dagger a}{}_a = \tau^a{}_a$.
1510.06699_F00627	29	0.999	`t_{a b}^{\dagger}`	$t_{a b}^{\dagger}$	One may construct a covariant $t^\dagger_{ab}$ in a similar way. In-
1510.06699_F00628	29	1.000	`\mathcal{T}_{a}^{\dagger} \equiv 0`	$\mathcal{T}_{a}^{\dagger} \equiv 0$	procedure can be applied to the ϕ field, and so the mat-
1510.06699_F00629	29	1.000	`\bar{\partial} L_{\mathrm{M}} / \partial \varphi \equiv\left[\partial L_{\mathrm{M}}\left(\varphi, \mathcal{D}_{a}^{\dagger} u, \phi, \mathcal{D}_{a}^{\dagger} \phi, \ldots\right) / \partial \varphi \right]_{u=\varphi}`	$\bar{\partial} L_{\mathrm{M}} / \partial \varphi \equiv\left[\partial L_{\mathrm{M}}\left(\varphi, \mathcal{D}_{a}^{\dagger} u, \phi, \mathcal{D}_{a}^{\dagger} \phi, \ldots\right) / \partial \varphi \right]_{u=\varphi}$	where $\bar{\partial} L_{\mathrm{M}}/\partial\varphi \equiv [\partial L_{\mathrm{M}}(\varphi, \mathcal{D}^\dagger_a u, \phi, \mathcal{D}^\dagger_a\phi,\ldots)/\partial\varphi]_{u=\varphi}$, so
1510.06699_F00630	30	1.000	`H_{a b}^{\dagger}`	$H_{a b}^{\dagger}$	strength tensor $H^\dagger_{ab}$ of the vector gauge field (152), but
1510.06699_F00631	30	1.000	`H_{a b}=h_{a}{ }^{\mu} h_{b}{ }^{\nu}\left(\partial_{\mu} V_{\nu}-\partial_{\nu} V_{\mu}\right)`	$H_{a b}=h_{a}{ }^{\mu} h_{b}{ }^{\nu}\left(\partial_{\mu} V_{\nu}-\partial_{\nu} V_{\mu}\right)$	instead the simpler object $H_{ab} = h_a{}^\mu h_b{}^\nu(\partial_\mu V_\nu - \partial_\nu V_\mu)$,
1510.06699_F00632	30	1.000	`j^{\dagger a}`	$j^{\dagger a}$	by $j^{\dagger a}$ rather than simply by j^a. The largest difference
1510.06699_F00633	30	1.000	`j^{a}`	j^{a}	by $j^{\dagger a}$ rather than simply by j^a. The largest difference
1510.06699_F00634	30	1.000	`\varphi, \phi, h_{a}^{\mu}, A^{a b}{ }_{\mu}, V_{\mu}`	$\varphi, \phi, h_{a}^{\mu}, A^{a b}{ }_{\mu}, V_{\mu}$	from the set φ, ϕ, h^μ_a, $A^{ab}{}_\mu$, V_μ and their derivatives,
1510.06699_F00635	30	1.000	`\varphi, \phi`	φ, ϕ	with which we have worked so far, to the new set φ, ϕ,
1510.06699_F00636	30	1.000	`h_{a}^{\mu}, A^{\dagger a b}{ }_{\mu}, V_{\mu}`	$h_{a}^{\mu}, A^{\dagger a b}{ }_{\mu}, V_{\mu}$	h^μ_a, $A^{\dagger ab}{}_\mu$, V_μ and their derivatives (i.e. we replace $A^{ab}{}_\mu$
1510.06699_F00637	30	1.000	`h_{a}^{\mu}, A^{a b}{ }_{\mu}, V_{\mu}`	$h_{a}^{\mu}, A^{a b}{ }_{\mu}, V_{\mu}$	of h^μ_a, $A^{ab}{}_\mu$, V_μ by its defining relationship (142), rather
1510.06699_F00638	30	1.000	`\mathcal{T}_{b}^{\natural} \equiv \mathcal{T}^{\natural a}{ }_{b a}`	$\mathcal{T}_{b}^{\natural} \equiv \mathcal{T}^{\natural a}{ }_{b a}$	and the corresponding trace $\mathcal{T}^\natural_b \equiv \mathcal{T}^{\natural a}{}_{ba}$. It is straight-
1510.06699_F00639	30	1.000	`\mathcal{T}_{b}^{\natural}=\mathcal{T}_{b}+3 \mathcal{V}_{b}`	$\mathcal{T}_{b}^{\natural}=\mathcal{T}_{b}+3 \mathcal{V}_{b}$	forward to show that $\mathcal{T}^\natural_b = \mathcal{T}_b + 3\mathcal{V}_b$, where $\mathcal{T}_b$ is the
1510.06699_F00640	31	0.997	`h, \partial h`	$h, \partial h$	which is expressible wholly in terms of the fields h, ∂h
1510.06699_F00641	31	1.000	`A^{\dagger}`	$A^{\dagger}$	and $A^\dagger$ (suppressing indices). Since (182) does not de-
1510.06699_F00642	31	0.566	`\quad \mathcal{R}_{a b c d}^{\dagger}\left(h, A^{\dagger}, \partial A^{\dagger}\right), \quad \mathcal{T}_{a b c}^{\dagger}\left(h, \partial h, A^{\dagger}\right) \quad`	$\quad \mathcal{R}_{a b c d}^{\dagger}\left(h, A^{\dagger}, \partial A^{\dagger}\right), \quad \mathcal{T}_{a b c}^{\dagger}\left(h, \partial h, A^{\dagger}\right) \quad$	tensors $\mathcal{R}^\dagger_{abcd}(h, A^\dagger, \partial A^\dagger)$, $\mathcal{T}^\dagger_{abc}(h, \partial h, A^\dagger)$ and
1510.06699_F00643	31	1.000	`\mathcal{H}_{a b}^{\dagger}\left(h, \partial h, \partial^{2} h, A^{\dagger}, \partial A^{\dagger}\right)`	$\mathcal{H}_{a b}^{\dagger}\left(h, \partial h, \partial^{2} h, A^{\dagger}, \partial A^{\dagger}\right)$	$\mathcal{H}^\dagger_{ab}(h, \partial h, \partial^2 h, A^\dagger, \partial A^\dagger)$, where once again the last
1510.06699_F00644	31	1.000	`\mathcal{L}`	$\mathcal{L}$	derivative of a Lagrangian $\mathcal{L}$ with respect to any one
1510.06699_F00645	31	1.000	`(\delta \mathcal{L} / \delta \chi)_{\dagger}`	$(\delta \mathcal{L} / \delta \chi)_{\dagger}$	of the fields χ by $(\delta\mathcal{L}/\delta\chi)_\dagger$ to distinguish it from the
1510.06699_F00646	31	1.000	`\delta \mathcal{L} / \delta \chi`	$\delta \mathcal{L} / \delta \chi$	variational derivative $\delta\mathcal{L}/\delta\chi$ obtained previously using
1510.06699_F00647	31	1.000	`\sigma_{a b}{ }^{\mu}`	$\sigma_{a b}{ }^{\mu}$	ther that the spin-angular momentum tensor $\sigma_{ab}{}^\mu$ and
1510.06699_F00648	31	1.000	`\zeta^{\dagger \mu}=0`	$\zeta^{\dagger \mu}=0$	gauge field V_μ or its derivative, then $\zeta^{\dagger\mu} = 0$ and so
1510.06699_F00649	31	1.000	`\phi, h_{a}{ }^{\mu}, A^{a b}{ }_{\mu}, V_{\mu}`	$\phi, h_{a}{ }^{\mu}, A^{a b}{ }_{\mu}, V_{\mu}$	ϕ, $h_a{}^\mu$, $A^{ab}{}_\mu$, V_μ and their derivatives, but we will use the
1510.06699_F00650	32	1.000	`\mathcal{D}_{a}^{*} \rightarrow \mathcal{D}_{a}^{\dagger}`	$\mathcal{D}_{a}^{*} \rightarrow \mathcal{D}_{a}^{\dagger}$	with $\mathcal{D}^*_a \to \mathcal{D}^\dagger_a$. Similar to WGT, since both ψ and $\bar\psi$ have
1510.06699_F00651	32	1.000	`\gamma^{a}\left(\mathcal{D}_{a}+\frac{1}{2} \mathcal{T}_{a}\right) \psi=\gamma^{a} \mathcal{D}_{a}^{\dagger} \psi`	$\gamma^{a}\left(\mathcal{D}_{a}+\frac{1}{2} \mathcal{T}_{a}\right) \psi=\gamma^{a} \mathcal{D}_{a}^{\dagger} \psi$	begin with the special-relativistic Lagrangian (14) for a
1510.06699_F00652	32	1.000	`\gamma^{a} \Sigma_{b a}=\frac{1}{4} \gamma^{a}\left[ \gamma_{b}, \gamma_{a}\right]=-\frac{3}{2} \gamma_{b}`	$\gamma^{a} \Sigma_{b a}=\frac{1}{4} \gamma^{a}\left[\gamma_{b}, \gamma_{a}\right]=-\frac{3}{2} \gamma_{b}$	field ϕ. For the moment, however, we will consider only
1510.06699_F00653	32	0.997	`\mathcal{T}_{[a b c]}=0`	$\mathcal{T}_{[a b c]}=0$	$w = -\frac{3}{2}$ and $w = -1$, respectively; we note that the

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1510.06699_F00654	32	1.000	`{ }^{0} \mathcal{D}_{a} \rightarrow{ }^{0} \mathcal{D}_{a}^{\dagger}`	${ }^{0} \mathcal{D}_{a} \rightarrow{ }^{0} \mathcal{D}_{a}^{\dagger}$	
1510.06699_F00655	32	1.000	`\mathcal{T}_{[a b c]} \rightarrow \mathcal{T}_{[a b c]}^{\dagger}`	$\mathcal{T}_{[a b c]} \rightarrow \mathcal{T}_{[a b c]}^{\dagger}$	$\mathcal{L}_{\mathrm{D}} = h^{-1}L_{\mathrm{D}} = h^{-1}(\frac{1}{2}i\bar{\psi}\gamma^a\overleftrightarrow{\mathcal{D}}^*_a\psi - \mu\phi\bar{\psi}\psi)$, (71)
1510.06699_F00656	32	1.000	`\sigma_{a b}{ }^{c}`	$\sigma_{a b}{ }^{c}$	global phase transformation. The first point to note is
1510.06699_F00657	32	1.000	`\sigma_{a b}{ }^{b}=0`	$\sigma_{a b}{ }^{b}=0$	Recalling that $\sigma_{ab}{}^b = 0$ for Dirac matter, the transfor-
1510.06699_F00658	32	0.444	`\beta_{1}`	β_{1}	unnecessary for the Dirac field, is therefore to convert
1510.06699_F00659	32	0.444	`\beta_{2}`	β_{2}	unnecessary for the Dirac field, is therefore to convert
1510.06699_F00660	32	0.444	`L_{\mathcal{\mathcal { T } ^ { + } 2}}`	$L_{\mathcal{T}^{+\in}}$	unnecessary for the Dirac field, is therefore to convert
1510.06699_F00661	32	0.998	`\widehat{\mathcal{F}}_{a b}^{*} \rightarrow \widehat{\mathcal{F}}_{a b}^{\dagger}, \mathcal{T}^{* c}{ }_{a b} \rightarrow \mathcal{T}^{\dagger c}{ }_{a b}`	$\widehat{\mathcal{F}}_{a b}^{*} \rightarrow \widehat{\mathcal{F}}_{a b}^{\dagger}, \mathcal{T}^{* c}{ }_{a b} \rightarrow \mathcal{T}^{\dagger c}{ }_{a b}$	$\widehat{\mathcal{F}}^*_{ab} \to \widehat{\mathcal{F}}^\dagger_{ab}$, $\mathcal{T}^{*c}{}_{ab} \to \mathcal{T}^{\dagger c}{}_{ab}$ and $\mathcal{D}^*_a \to \mathcal{D}^\dagger_a$, and recalling
1510.06699_F00662	33	1.000	`\tau^{\dagger a}{ }_{b}=\tau^{a}{ }_{b}`	$\tau^{\dagger a}{ }_{b}=\tau^{a}{ }_{b}$	deed, from (170), one sees immediately that $\tau^{\dagger a}{}_b = \tau^a{}_b$,
1510.06699_F00663	33	1.000	`\varphi, \phi, h_{a}^{\mu}`	$\varphi, \phi, h_{a}^{\mu}$	set of field variables φ, ϕ, h^μ_a and $A^{\dagger ab}{}_\mu$ discussed in
1510.06699_F00664	33	0.607	`\widehat{\mathcal{A}}^{\dagger a b}{ }_{c} \equiv \widehat{h}_{a}{ }^{\mu} \widehat{A}^{\dagger a b}{ }_{\mu}`	$\widehat{\mathcal{A}}^{\dagger a b}{ }_{c} \equiv \widehat{h}_{a}{ }^{\mu} \widehat{A}^{\dagger a b}{ }_{\mu}$	(60) may be written as (su
1510.06699_F00665	33	1.000	`\mathcal{D}_{a}^{\dagger} \chi=\left(\mathcal{D}_{a}^{\natural}-\frac{1}{3} w \mathcal{T}_{a}^{\natural}\right) \chi`	$\mathcal{D}_{a}^{\dagger} \chi=\left(\mathcal{D}_{a}^{\natural}-\frac{1}{3} w \mathcal{T}_{a}^{\natural}\right) \chi$	In other words, when written in terms of the scale-
1510.06699_F00666	33	1.000	`\mathcal{D}_{a}^{\natural}`	$\mathcal{D}_{a}^{\natural}$	In other words, when written in terms of the scale-
1510.06699_F00667	33	1.000	`\mathcal{T}_{a}^{\natural}`	$\mathcal{T}_{a}^{\natural}$	invariant field variables, the Lagrangian density $\mathcal{L}_{\mathrm{T}}$ (in-
1510.06699_F00668	33	1.000	`\widehat{\mathcal{D}}_{a}^{\natural}`	$\widehat{\mathcal{D}}_{a}^{\natural}$	deed each term separately) has the *same* functional form
1510.06699_F00669	33	1.000	`\widehat{\mathcal{T}}_{a}^{\natural}`	$\widehat{\mathcal{T}}_{a}^{\natural}$	as it does in terms of the original variables with $\phi = \phi_0$.
1510.06699_F00670	33	1.000	`\widehat{A}^{\dagger a b}{ }_{\mu}`	$\widehat{A}^{\dagger a b}{ }_{\mu}$	mediately conclude that (each term in) the equation of
1510.06699_F00671	33	1.000	`\widehat{\mathcal{D}}_{a}^{\natural}=\widehat{h}_{a}{ }^{\mu} D_{\mu}^{\natural}`	$\widehat{\mathcal{D}}_{a}^{\natural}=\widehat{h}_{a}{ }^{\mu} D_{\mu}^{\natural}$	mediately conclude that (each term in) the equation of
1510.06699_F00672	33	1.000	`\mathcal{D}_{a}^{\dagger} \phi=\frac{1}{3}\left(\phi^{2} / \phi_{0}\right) \widehat{\mathcal{T}}_{a}^{\natural}`	$\mathcal{D}_{a}^{\dagger} \phi=\frac{1}{3}\left(\phi^{2} / \phi_{0}\right) \widehat{\mathcal{T}}_{a}^{\natural}$	For the ϕ-field equation, however, the situation is
1510.06699_F00673	33	0.996	`\widehat{\mathcal{D}}_{a}^{\dagger}=\widehat{\mathcal{D}}_{a}^{\natural}-\frac{1}{3} w \widehat{\mathcal{T}}_{a}^{\natural}`	$\widehat{\mathcal{D}}_{a}^{\dagger}=\widehat{\mathcal{D}}_{a}^{\natural}-\frac{1}{3} w \widehat{\mathcal{T}}_{a}^{\natural}$	law (69c), which may be straightforwardly rewritten as
1510.06699_F00674	33	1.000	`\widehat{\mathcal{D}}_{a}^{\dagger} \widehat{\chi}`	$\widehat{\mathcal{D}}_{a}^{\dagger} \widehat{\chi}$	
1510.06699_F00675	33	1.000	`\widehat{\chi}, \widehat{h}_{a}{ }^{\mu}`	$\widehat{\chi}, \widehat{h}_{a}{ }^{\mu}$	(94) where we have made use of the φ-field equation
1510.06699_F00676	33	1.000	`\mathcal{D}_{a}^{\dagger} \chi`	$\mathcal{D}_{a}^{\dagger} \chi$	$\delta\mathcal{L}_{\mathrm{T}}/\delta\varphi = \delta\mathcal{L}_{\mathrm{M}}/\delta\varphi = 0$ and set $\phi = \phi_0$ in the result-
1510.06699_F00677	33	0.999	`\mathcal{R}^{\dagger a b}{ }_{c d}=\left(\phi / \phi_{0}\right)^{2} \widehat{\mathcal{R}}^{\dagger a b}{ }_{c d}`	$\mathcal{R}^{\dagger a b}{ }_{c d}=\left(\phi / \phi_{0}\right)^{2} \widehat{\mathcal{R}}^{\dagger a b}{ }_{c d}$	$\mathcal{L}_{\mathrm{T}}$ in terms of scale-invariant variables and applying the
1510.06699_F00678	33	1.000	`\mathcal{T}^{\dagger a}{ }_{b c}=\left(\phi / \phi_{0}\right) \widehat{\mathcal{T}}^{\dagger a}{ }_{b c}`	$\mathcal{T}^{\dagger a}{ }_{b c}=\left(\phi / \phi_{0}\right) \widehat{\mathcal{T}}^{\dagger a}{ }_{b c}$	chain rule for partial derivatives, one may show after a
1510.06699_F00679	33	1.000	`\mathcal{H}_{a b}^{\dagger}=\left(\phi / \phi_{0}\right)^{2} \widehat{\mathcal{H}}_{a b}^{\dagger}`	$\mathcal{H}_{a b}^{\dagger}=\left(\phi / \phi_{0}\right)^{2} \widehat{\mathcal{H}}_{a b}^{\dagger}$	chain rule for partial derivatives, one may show after a
1510.06699_F00680	33	1.000	`\varphi, h_{a}{ }^{\mu}`	$\varphi, h_{a}{ }^{\mu}$	if χ represents φ, $h_a{}^\mu$, or $A^{\dagger ab}{}_\mu$, one may immediately
1510.06699_F00681	34	1.000	`\varphi, h`	φ, h	for the φ, h and A (or $A^\dagger$) fields are the only three *in*-
1510.06699_F00682	34	1.000	`\varphi, \phi, h_{a}^{\mu}, A^{a b}{ }_{\mu}`	$\varphi, \phi, h_{a}^{\mu}, A^{a b}{ }_{\mu}$	ables φ, ϕ, h^μ_a, $A^{ab}{}_\mu$ and V_μ, rather than the alternative
1510.06699_F00683	34	1.000	`\widehat{V}_{\mu}=\frac{1}{3} \widehat{T}_{\mu}^{\natural}`	$\widehat{V}_{\mu}=\frac{1}{3} \widehat{T}_{\mu}^{\natural}$	where it is clear from the last definition that $\widehat{V}_\mu = \frac{1}{3}\widehat{T}^\natural_\mu$.

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1510.06699_F00684	34	1.000	`\widehat{\mathcal{V}}_{a} \equiv \widehat{h}_{a}{ }^{\mu} \widehat{V}_{\mu}`	$\widehat{\mathcal{V}}_{a} \equiv \widehat{h}_{a}{ }^{\mu} \widehat{V}_{\mu}$	variables $\widehat{\mathcal{A}}^{\dagger ab}{}_{c} \equiv \widehat{h}_{a}{}^{\mu} \widehat{A}^{\dagger ab}{}_{\mu}$ and $\widehat{\mathcal{V}}_{a} \equiv \widehat{h}_{a}{}^{\mu} \widehat{V}_{\mu}$. The situa-
1510.06699_F00685	34	1.000	`\varphi, h_{a}^{\mu}`	φ, h_{a}^{μ}	Specifically, the equations of motion in terms of φ, h_{a}^{μ},
1510.06699_F00686	34	1.000	`V_{\mu}+\frac{1}{3} T_{\mu}`	$V_{\mu}+\frac{1}{3} T_{\mu}$	tions $A^{\dagger ab}{}_{\mu}$ and $V_{\mu}+\frac{1}{3}T_{\mu}$ (which will always be possible).
1510.06699_F00687	34	1.000	`\widehat{\varphi}, \widehat{h}_{a}{ }^{\mu}, \widehat{A}^{\dagger a b}{ }_{\mu}`	$\widehat{\varphi}, \widehat{h}_{a}{ }^{\mu}, \widehat{A}^{\dagger a b}{ }_{\mu}$	$\widehat{\varphi}$, $\widehat{h}_{a}{}^{\mu}$, $\widehat{A}^{\dagger ab}{}_{\mu}$ and $\widehat{V}_{\mu}$, respectively.
1510.06699_F00688	34	1.000	`\widehat{V}_{\mu}`	$\widehat{V}_{\mu}$	$\widehat{\varphi}$, $\widehat{h}_{a}{}^{\mu}$, $\widehat{A}^{\dagger ab}{}_{\mu}$ and $\widehat{V}_{\mu}$, respectively.
1510.06699_F00689	34	1.000	`v^{c}\left(\mathcal{D}_{c}^{\dagger} p_{a}-\right.`	$v^{c}\left(\mathcal{D}_{c}^{\dagger} p_{a}-\right.$	in a manifestly eWGT-covariant manner as $v^{c}(\mathcal{D}_{c}^{\dagger}p_{a}$ –
1510.06699_F00690	34	0.971	`\left.\mathcal{T}_{\text {cab }}^{\dagger} p^{b}\right)=e \mu^{2} \phi \mathcal{D}_{a}^{\dagger} \phi`	$\left.\mathcal{T}_{\text {cab }}^{\dagger} p^{b}\right)=e \mu^{2} \phi \mathcal{D}_{a}^{\dagger} \phi$	$\mathcal{T}_{cab}^{\dagger}p^{b}) = e\mu^{2}\phi\mathcal{D}_{a}^{\dagger}\phi$. By analogy with our treatment in
1510.06699_F00691	34	0.997	`v^{c}{ }^{0} \mathcal{D}_{c}^{\dagger} p_{a}=e \mu^{2} \phi^{0} \mathcal{D}_{a}^{\dagger} \phi`	$v^{c}{ }^{0} \mathcal{D}_{c}^{\dagger} p_{a}=e \mu^{2} \phi^{0} \mathcal{D}_{a}^{\dagger} \phi$	$v^{c}\,{}^{0}\mathcal{D}_{c}^{\dagger}p_{a} = e\mu^{2}\phi\,{}^{0}\mathcal{D}_{a}^{\dagger}\phi$, which is again manifestly eWGT-
1510.06699_F00692	34	1.000	`{ }^{0} D_{\mu}^{\dagger}`	${ }^{0} D_{\mu}^{\dagger}$	covariant derivative operator ${}^{0}D_{\mu}^{\dagger}$ in (159), to which the
1510.06699_F00693	34	0.718	`\mathcal{T}_{a b c}^{\dagger} \equiv 0`	$\mathcal{T}_{a b c}^{\dagger} \equiv 0$	with $\mathcal{T}_{abc}^{\dagger} \equiv 0$ (and hence $\mathcal{K}_{abc}^{\dagger} \equiv 0$). In contrast to
1510.06699_F00694	34	0.718	`\left.\mathcal{K}_{a b c}^{\dagger} \equiv 0\right)`	$\left.\mathcal{K}_{a b c}^{\dagger} \equiv 0\right)$	with $\mathcal{T}_{abc}^{\dagger} \equiv 0$ (and hence $\mathcal{K}_{abc}^{\dagger} \equiv 0$). In contrast to
1510.06699_F00695	35	1.000	`{ }^{0} R^{\dagger a b}{ }_{\mu \nu}\left(h, \partial h, \partial^{2} h, A, \partial A, V, \partial V\right)`	${ }^{0} R^{\dagger a b}{ }_{\mu \nu}\left(h, \partial h, \partial^{2} h, A, \partial A, V, \partial V\right)$	where we have introduced the new scale-invariant ein-
1510.06699_F00696	35	1.000	`{ }^{0} A^{\dagger a b}{ }_{\mu}`	${ }^{0} A^{\dagger a b}{ }_{\mu}$	bein $\widehat{\epsilon} = [1 - \lambda(dx^{\mu}/d\lambda)\partial_{\mu}\ln(\phi/\phi_{0})]\epsilon$. Thus, in terms
1510.06699_F00697	35	1.000	`{ }^{0} R^{\dagger a b}{ }_{\mu \nu}`	${ }^{0} R^{\dagger a b}{ }_{\mu \nu}$	of the scale-invariant variables $\widehat{\lambda}$, $\widehat{p}_{a}$, $\widehat{v}^{a}$ and $\widehat{\epsilon}$, respec-
1510.06699_F00698	35	1.000	`R^{\dagger a b}{ }_{\mu \nu}(h, A, \partial A, V, \partial V)`	$R^{\dagger a b}{ }_{\mu \nu}(h, A, \partial A, V, \partial V)$	invariant variables (105), each occurrence of the gravita-
1510.06699_F00699	35	1.000	`{ }^{0} \mathcal{R}^{\dagger a b}{ }_{c d}=h_{c}{ }^{\mu} h_{d}{ }^{\mu}{ }^{0} R^{\dagger a b}{ }_{\mu \nu}`	${ }^{0} \mathcal{R}^{\dagger a b}{ }_{c d}=h_{c}{ }^{\mu} h_{d}{ }^{\mu}{ }^{0} R^{\dagger a b}{ }_{\mu \nu}$	where ${}^{0}\mathcal{R}^{\dagger ab}{}_{cd} = h_{c}{}^{\mu}h_{d}{}^{\mu}\,{}^{0}R^{\dagger ab}{}_{\mu\nu}$, and the commuta-
1510.06699_F00700	35	1.000	`{ }^{0} \mathcal{T}^{\dagger a}{ }_{b c} \equiv`	${ }^{0} \mathcal{T}^{\dagger a}{ }_{b c} \equiv$	strength of the h gauge field (or 'torsion'), since ${}^{0}\mathcal{T}^{\dagger a}{}_{bc} \equiv$
1510.06699_F00701	35	1.000	`h_{b}{ }^{\mu} h_{c}{ }^{\nu}\left({ }^{0} D_{\mu}^{\dagger} b^{a}{ }_{\nu}-{ }^{0} D_{\nu}^{\dagger} b^{a}{ }_{\mu}\right)=0`	$h_{b}{ }^{\mu} h_{c}{ }^{\nu}\left({ }^{0} D_{\mu}^{\dagger} b^{a}{ }_{\nu}-{ }^{0} D_{\nu}^{\dagger} b^{a}{ }_{\mu}\right)=0$	$h_{b}{}^{\mu}h_{c}{}^{\nu}({}^{0}D_{\mu}^{\dagger}b^{a}{}_{\nu} - {}^{0}D_{\nu}^{\dagger}b^{a}{}_{\mu}) = 0$. As one might expect, the
1510.06699_F00702	35	0.785	`\mathcal{D}_{a}^{\dagger} \rightarrow^{0} \mathcal{D}_{a}^{\dagger}, \mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{\dagger}`	$\mathcal{D}_{a}^{\dagger} \rightarrow^{0} \mathcal{D}_{a}^{\dagger}, \mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{\dagger}$	with the replacements $\mathcal{D}_{a}^{\dagger} \rightarrow {}^{0}\mathcal{D}_{a}^{\dagger}$, $\mathcal{R}_{abcd} \rightarrow {}^{0}\mathcal{R}_{abcd}^{\dagger}$ and
1510.06699_F00703	35	0.831	`\mathcal{T}_{a b c}^{\dagger} \rightarrow{ }^{0} \mathcal{T}_{a b c}^{\dagger}=0`	$\mathcal{T}_{a b c}^{\dagger} \rightarrow{ }^{0} \mathcal{T}_{a b c}^{\dagger}=0$	$\mathcal{T}_{abc}^{\dagger} \rightarrow {}^{0}\mathcal{T}_{abc}^{\dagger} = 0$.
1510.06699_F00704	35	1.000	`{ }^{0} \mathcal{R}^{a b}{ }_{c d} \rightarrow`	${ }^{0} \mathcal{R}^{a b}{ }_{c d} \rightarrow$	(110c), but with the replacements on the RHS ${}^{0}\mathcal{R}^{ab}{}_{cd} \rightarrow$
1510.06699_F00705	35	1.000	`{ }^{0} \mathcal{R}^{\dagger a b}{ }_{c d},{ }^{0} \mathcal{D}_{a} \rightarrow{ }^{0} \mathcal{D}_{a}^{\dagger}, \mathcal{K}_{a b c} \rightarrow \mathcal{K}_{a b c}^{\dagger}`	${ }^{0} \mathcal{R}^{\dagger a b}{ }_{c d},{ }^{0} \mathcal{D}_{a} \rightarrow{ }^{0} \mathcal{D}_{a}^{\dagger}, \mathcal{K}_{a b c} \rightarrow \mathcal{K}_{a b c}^{\dagger}$	${}^{0}\mathcal{R}^{\dagger ab}{}_{cd}$, ${}^{0}\mathcal{D}_{a} \rightarrow {}^{0}\mathcal{D}_{a}^{\dagger}$, $\mathcal{K}_{abc} \rightarrow \mathcal{K}_{abc}^{\dagger}$ and $\mathcal{T}_{abc} \rightarrow \mathcal{T}_{abc}^{\dagger}$, and
1510.06699_F00706	35	1.000	`\mathcal{R}^{a b}{ }_{c d} \rightarrow{ }^{0} \mathcal{R}^{\dagger a b}{ }_{c d}`	$\mathcal{R}^{a b}{ }_{c d} \rightarrow{ }^{0} \mathcal{R}^{\dagger a b}{ }_{c d}$	gauge fields: although one can substitute for the rota-
1510.06699_F00707	35	1.000	`{ }^{0} \mathcal{R}^{\dagger}{ }_{a b}`	${ }^{0} \mathcal{R}^{\dagger}{ }_{a b}$	tional gauge field to obtain equations that contain higher-
1510.06699_F00708	35	0.999	`{ }^{0} \mathcal{R}_{a b c d}^{\dagger}`	${ }^{0} \mathcal{R}_{a b c d}^{\dagger}$	order derviatives, the theory does not suffer from Os-
1510.06699_F00709	35	1.000	`\mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{\dagger}`	$\mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{\dagger}$	correspond to 'second-order' and 'first-order' variational
1510.06699_F00710	35	0.535	`\mathcal{D}_{a}^{\dagger} \rightarrow^{0} \mathcal{D}_{a}^{\dagger}`	$\mathcal{D}_{a}^{\dagger} \rightarrow^{0} \mathcal{D}_{a}^{\dagger}$	ment $\mathcal{D}_{a}^{\dagger} \rightarrow {}^{0}\mathcal{D}_{a}^{\dagger}$. As in WGT, however, the gravitational
1510.06699_F00711	35	0.720	`\phi^{2}{ }^{0} \mathcal{R}^{\dagger}`	$\phi^{2}{ }^{0} \mathcal{R}^{\dagger}$	$\mathcal{L}_{\rm G}$ and the term proportional to $\phi^{2}\,{}^{0}\mathcal{R}^{\dagger}$ in $\mathcal{L}_{\rm M}$ depends

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1510.06699_F00712	35	0.806	`\mathcal{D}_{a}^{\dagger} \rightarrow{ }^{0} \mathcal{D}_{a}^{\dagger}, \mathcal{R}_{a b c d}^{\dagger} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{\dagger}`	$\mathcal{D}_{a}^{\dagger} \rightarrow{ }^{0} \mathcal{D}_{a}^{\dagger}, \mathcal{R}_{a b c d}^{\dagger} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{\dagger}$	replacements $\mathcal{D}_a^\dagger \rightarrow {}^0\mathcal{D}_a^\dagger$, $\mathcal{R}_{abcd}^\dagger \rightarrow {}^0\mathcal{R}_{abcd}^\dagger$ and $\mathcal{T}_{abc}^\dagger \rightarrow$
1510.06699_F00713	35	0.806	`\mathcal{T}_{a b c}^{\dagger} \rightarrow`	$\mathcal{T}_{a b c}^{\dagger} \rightarrow$	replacements $\mathcal{D}_a^\dagger \rightarrow {}^0\mathcal{D}_a^\dagger$, $\mathcal{R}_{abcd}^\dagger \rightarrow {}^0\mathcal{R}_{abcd}^\dagger$ and $\mathcal{T}_{abc}^\dagger \rightarrow$
1510.06699_F00714	35	0.961	`{ }^{0} \mathcal{T}_{a b c}^{\dagger} \equiv 0`	${ }^{0} \mathcal{T}_{a b c}^{\dagger} \equiv 0$	${}^0\mathcal{T}_{abc}^\dagger \equiv 0$. Furthermore, by performing analogous calcu-
1510.06699_F00715	35	0.891	`\mathcal{T}_{a b c}^{\dagger}=0`	$\mathcal{T}_{a b c}^{\dagger}=0$	erly eWGT-covariant) condition $\mathcal{T}_{abc}^\dagger = 0$ in the field
1510.06699_F00716	36	1.000	`b^{b}{ }_{\nu}`	$b^{b}{ }_{\nu}$	notate $h_a{}^\mu$ and $b^b{}_\nu$ by $e_a{}^\mu$ and $e^b{}_\nu$, respectively.
1510.06699_F00717	36	1.000	`e^{b}{ }_{\nu}`	$e^{b}{ }_{\nu}$	notate $h_a{}^\mu$ and $b^b{}_\nu$ by $e_a{}^\mu$ and $e^b{}_\nu$, respectively.
1510.06699_F00718	36	1.000	`e_{a}{ }^{\mu}, A^{a b}{ }_{\mu}`	$e_{a}{ }^{\mu}, A^{a b}{ }_{\mu}$	(142) of the three gauge fields $e_a{}^\mu$, $A^{ab}{}_\mu$, and V_μ that
1510.06699_F00719	36	1.000	`\partial_{\mu}^{\dagger}`	$\partial_{\mu}^{\dagger}$	ject of weight w, for all these derivative operators the re-
1510.06699_F00720	36	1.000	`A^{\dagger a b}{ }_{\mu}=-A^{\dagger b a}{ }_{\mu}`	$A^{\dagger a b}{ }_{\mu}=-A^{\dagger b a}{ }_{\mu}$	weight w.
1510.06699_F00721	36	0.545	`\nabla_{\mu}^{\dagger}=\partial_{\mu}^{\dagger}+\Gamma^{\sigma}{ }_{\rho \mu} \mathrm{X}^{\rho}{ }_{\sigma}`	$\nabla_{\mu}^{\dagger}=\partial_{\mu}^{\dagger}+\Gamma^{\sigma}{ }_{\rho \mu} \mathrm{X}^{\rho}{ }_{\sigma}$	where $\nabla_\mu^\dagger = \partial_\mu^\dagger + \Gamma^\sigma{}_{\rho\mu}\mathrm{X}^\rho{}_\sigma$ and $\mathrm{X}^\rho{}_\sigma$ are the GL(4, R) gen-
1510.06699_F00722	36	0.545	`4, R`	$4, R$	where $\nabla_\mu^\dagger = \partial_\mu^\dagger + \Gamma^\sigma{}_{\rho\mu}\mathrm{X}^\rho{}_\sigma$ and $\mathrm{X}^\rho{}_\sigma$ are the GL(4, R) gen-
1510.06699_F00723	36	1.000	`\Delta_{\mu}^{\dagger}`	$\Delta_{\mu}^{\dagger}$	of the field to which $\Delta_\mu^\dagger$ is applied. If a field ψ carries
1510.06699_F00724	36	1.000	`\nabla_{\mu}^{\dagger} \psi=\partial_{\mu}^{\dagger} \psi`	$\nabla_{\mu}^{\dagger} \psi=\partial_{\mu}^{\dagger} \psi$	only Latin indices, then $\nabla_\mu^\dagger\psi = \partial_\mu^\dagger\psi$ and so $\Delta_\mu^\dagger\psi = D_\mu^\dagger\psi$;
1510.06699_F00725	36	1.000	`\Delta_{\mu}^{\dagger} \psi=D_{\mu}^{\dagger} \psi`	$\Delta_{\mu}^{\dagger} \psi=D_{\mu}^{\dagger} \psi$	only Latin indices, then $\nabla_\mu^\dagger\psi = \partial_\mu^\dagger\psi$ and so $\Delta_\mu^\dagger\psi = D_\mu^\dagger\psi$;
1510.06699_F00726	36	1.000	`D_{\mu}^{\dagger} \psi=\partial_{\mu}^{\dagger} \psi`	$D_{\mu}^{\dagger} \psi=\partial_{\mu}^{\dagger} \psi$	$D_\mu^\dagger\psi = \partial_\mu^\dagger\psi$ and so $\Delta_\mu^\dagger\psi = \nabla_\mu^\dagger\psi$. When acting on an ob-
1510.06699_F00727	36	1.000	`\Delta_{\mu}^{\dagger} \psi=\nabla_{\mu}^{\dagger} \psi`	$\Delta_{\mu}^{\dagger} \psi=\nabla_{\mu}^{\dagger} \psi$	$D_\mu^\dagger\psi = \partial_\mu^\dagger\psi$ and so $\Delta_\mu^\dagger\psi = \nabla_\mu^\dagger\psi$. When acting on an ob-
1510.06699_F00728	36	1.000	`T`	T	which relates A and Γ (and V and T). Comparing this
1510.06699_F00729	36	1.000	`\partial_{\mu}^{*} \rightarrow \partial_{\mu}^{\dagger}`	$\partial_{\mu}^{*} \rightarrow \partial_{\mu}^{\dagger}$	$\partial_\mu^* \rightarrow \partial_\mu^\dagger$ and $A^{ab}_\mu \rightarrow A^{\dagger ab}{}_\mu$. Consequently, the results
1510.06699_F00730	36	1.000	`A_{\mu}^{a b} \rightarrow A^{\dagger a b}{ }_{\mu}`	$A_{\mu}^{a b} \rightarrow A^{\dagger a b}{ }_{\mu}$	$\partial_\mu^* \rightarrow \partial_\mu^\dagger$ and $A^{ab}_\mu \rightarrow A^{\dagger ab}{}_\mu$. Consequently, the results
1510.06699_F00731	36	1.000	`\nabla_{\sigma}^{\dagger} g_{\mu \nu}=0`	$\nabla_{\sigma}^{\dagger} g_{\mu \nu}=0$	$\nabla_\sigma^\dagger g_{\mu\nu} = 0$, and so this derivative operator commutes
1510.06699_F00732	36	1.000	`R^{\dagger \rho}{ }_{\sigma \mu \nu}=e_{a}{ }^{\rho} e^{b}{ }_{\sigma} R^{\dagger a}{ }_{b \mu \nu}`	$R^{\dagger \rho}{ }_{\sigma \mu \nu}=e_{a}{ }^{\rho} e^{b}{ }_{\sigma} R^{\dagger a}{ }_{b \mu \nu}$	where $R^{\dagger\rho}{}_{\sigma\mu\nu} = e_a{}^\rho e^b{}_\sigma R^{\dagger a}{}_{b\mu\nu}$ and $T^{\dagger\lambda}{}_{\mu\nu} = e_a{}^\lambda T^{\dagger a}{}_{\mu\nu}$.
1510.06699_F00733	36	1.000	`T^{\dagger \lambda}{ }_{\mu \nu}=e_{a}{ }^{\lambda} T^{\dagger a}{ }_{\mu \nu}`	$T^{\dagger \lambda}{ }_{\mu \nu}=e_{a}{ }^{\lambda} T^{\dagger a}{ }_{\mu \nu}$	where $R^{\dagger\rho}{}_{\sigma\mu\nu} = e_a{}^\rho e^b{}_\sigma R^{\dagger a}{}_{b\mu\nu}$ and $T^{\dagger\lambda}{}_{\mu\nu} = e_a{}^\lambda T^{\dagger a}{}_{\mu\nu}$.
1510.06699_F00734	37	1.000	`T^{\dagger \lambda}{ }_{\mu \lambda}=0`	$T^{\dagger \lambda}{ }_{\mu \lambda}=0$	torsion vanishes, $T^{\dagger\lambda}{}_{\mu\lambda} = 0$, so that the affine connection
1510.06699_F00735	37	1.000	`R^{\dagger \rho}{ }_{\sigma \mu \nu}`	$R^{\dagger \rho}{ }_{\sigma \mu \nu}$	From (212), we see that $R^{\dagger\rho}{}_{\sigma\mu\nu}$ is not simply its Riemann
1510.06699_F00736	37	0.999	`{ }^{0} \Gamma^{\dagger \sigma}{ }_{\rho \mu}`	${ }^{0} \Gamma^{\dagger \sigma}{ }_{\rho \mu}$	nection ${}^0\Gamma^{\dagger\sigma}{}_{\rho\mu}$ represent the same geometrical object in
1510.06699_F00737	37	1.000	`\widehat{R}_{\rho \sigma \mu \nu}`	$\widehat{R}_{\rho \sigma \mu \nu}$	One should note that, although $\widehat{R}_{\rho\sigma\mu\nu}$ is antisymmet-
1510.06699_F00738	37	1.000	`\widehat{R}_{(\rho \sigma) \mu \nu}=-g_{\rho \sigma} H_{\mu \nu}^{\dagger}`	$\widehat{R}_{(\rho \sigma) \mu \nu}=-g_{\rho \sigma} H_{\mu \nu}^{\dagger}$	$\widehat{R}_{(\rho\sigma)\mu\nu} = -g_{\rho\sigma}H^\dagger_{\mu\nu}$. Indeed, it does not satisfy the
1510.06699_F00739	37	1.000	`\widehat{R}^{\rho}{ }_{\sigma \mu \nu}`	$\widehat{R}^{\rho}{ }_{\sigma \mu \nu}$	that, with the given arrangements of indices, both $\widehat{R}^\rho{}_{\sigma\mu\nu}$
1510.06699_F00740	37	1.000	`T^{\dagger \lambda}{ }_{\mu \nu}`	$T^{\dagger \lambda}{ }_{\mu \nu}$	(or $R^{\dagger\rho}{}_{\sigma\mu\nu}$) and $T^{\dagger\lambda}{}_{\mu\nu}$ transform covariantly with weight
1510.06699_F00741	37	1.000	`\widehat{R}_{\mu \nu} \equiv \widehat{R}_{\mu \lambda \nu}{ }^{\lambda}=R_{\mu \nu}^{\dagger}+H_{\mu \nu}^{\dagger}`	$\widehat{R}_{\mu \nu} \equiv \widehat{R}_{\mu \lambda \nu}{ }^{\lambda}=R_{\mu \nu}^{\dagger}+H_{\mu \nu}^{\dagger}$	$\widehat{R}_{\mu\nu} \equiv \widehat{R}_{\mu\lambda\nu}{}^\lambda = R^\dagger_{\mu\nu} + H^\dagger_{\mu\nu}$ and $\widehat{R} \equiv \widehat{R}^\mu{}_\mu = R^\dagger$. In a
1510.06699_F00742	37	1.000	`\widehat{R} \equiv \widehat{R}^{\mu}{ }_{\mu}=R^{\dagger}`	$\widehat{R} \equiv \widehat{R}^{\mu}{ }_{\mu}=R^{\dagger}$	$\widehat{R}_{\mu\nu} \equiv \widehat{R}_{\mu\lambda\nu}{}^\lambda = R^\dagger_{\mu\nu} + H^\dagger_{\mu\nu}$ and $\widehat{R} \equiv \widehat{R}^\mu{}_\mu = R^\dagger$. In a
1510.06699_F00743	37	1.000	`{ }^{0} \Delta_{\mu}^{\dagger} e^{a}{ }_{\nu} \equiv \partial_{\mu}^{\dagger} e^{a}{ }_{\nu}-{ }^{0} \Gamma^{\dagger \sigma}{ }_{\nu \mu} e^{a}{ }_{\sigma}+{ }^{0} A^{\dagger a}{ }_{b \mu} e^{b}{ }_{\nu}=0`	${ }^{0} \Delta_{\mu}^{\dagger} e^{a}{ }_{\nu} \equiv \partial_{\mu}^{\dagger} e^{a}{ }_{\nu}-{ }^{0} \Gamma^{\dagger \sigma}{ }_{\nu \mu} e^{a}{ }_{\sigma}+{ }^{0} A^{\dagger a}{ }_{b \mu} e^{b}{ }_{\nu}=0$	${}^0\Delta_\mu^\dagger e^a{}_\nu \equiv \partial_\mu^\dagger e^a{}_\nu - {}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}e^a{}_\sigma + {}^0A^{\dagger a}{}_{b\mu}e^b{}_\nu = 0.$
1510.06699_F00744	37	0.992	`\partial_{\mu}^{*} \rightarrow \partial_{\mu}^{\dagger}, \Gamma^{\sigma}{ }_{\nu \mu} \rightarrow`	$\partial_{\mu}^{*} \rightarrow \partial_{\mu}^{\dagger}, \Gamma^{\sigma}{ }_{\nu \mu} \rightarrow$	again hold with the replacements $\partial_\mu^* \rightarrow \partial_\mu^\dagger$, $\Gamma^\sigma{}_{\nu\mu} \rightarrow$

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1510.06699_F00745	37	1.000	`{ }^{0} \Gamma^{\dagger \sigma}{ }_{\nu \mu}`	${ }^{0} \Gamma^{\dagger \sigma}{ }_{\nu \mu}$	${}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}$ and $A^a{}_{b\mu} \rightarrow {}^0A^{\dagger a}{}_{b\mu}$, from which one can di-
1510.06699_F00746	37	1.000	`A^{a}{ }_{b \mu} \rightarrow{ }^{0} A^{\dagger a}{ }_{b \mu}`	$A^{a}{ }_{b \mu} \rightarrow{ }^{0} A^{\dagger a}{ }_{b \mu}$	${}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}$ and $A^a{}_{b\mu} \rightarrow {}^0A^{\dagger a}{}_{b\mu}$, from which one can di-
1510.06699_F00747	37	1.000	`{ }^{0} \nabla_{\sigma}^{\dagger} g_{\mu \nu}=0`	${ }^{0} \nabla_{\sigma}^{\dagger} g_{\mu \nu}=0$	dition ${}^0\nabla^\dagger_\sigma g_{\mu\nu} = 0$. Finally, the expression (212) for
1510.06699_F00748	37	1.000	`R^{\dagger \rho}{ }_{\sigma \mu \nu} \rightarrow{ }^{0} R^{\dagger \rho}{ }_{\sigma \mu \nu}`	$R^{\dagger \rho}{ }_{\sigma \mu \nu} \rightarrow{ }^{0} R^{\dagger \rho}{ }_{\sigma \mu \nu}$	the curvature also holds, but with $R^{\dagger\rho}{}_{\sigma\mu\nu} \rightarrow {}^0R^{\dagger\rho}{}_{\sigma\mu\nu}$
1510.06699_F00749	37	0.999	`\Gamma^{\sigma}{ }_{\nu \mu} \rightarrow{ }^{0} \Gamma^{\dagger \sigma}{ }_{\nu \mu}`	$\Gamma^{\sigma}{ }_{\nu \mu} \rightarrow{ }^{0} \Gamma^{\dagger \sigma}{ }_{\nu \mu}$	and $\Gamma^\sigma{}_{\nu\mu} \rightarrow {}^0\Gamma^{\dagger\sigma}{}_{\nu\mu}$, whereas (213) becomes simply
1510.06699_F00750	37	1.000	`{ }^{0} T^{\dagger \lambda}{ }_{\mu \nu}=0`	${ }^{0} T^{\dagger \lambda}{ }_{\mu \nu}=0$	${}^0T^{\dagger\lambda}{}_{\mu\nu} = 0$, indicating the absence of torsion, as ex-
1510.06699_F00751	37	0.986	`\nabla_{\mu}^{\dagger} \rightarrow{ }^{0} \nabla_{\mu}^{\dagger}, R^{\rho}{ }_{\sigma \mu \nu} \rightarrow{ }^{0} R^{\dagger \rho}{ }_{\sigma \mu \nu}`	$\nabla_{\mu}^{\dagger} \rightarrow{ }^{0} \nabla_{\mu}^{\dagger}, R^{\rho}{ }_{\sigma \mu \nu} \rightarrow{ }^{0} R^{\dagger \rho}{ }_{\sigma \mu \nu}$	$\nabla^\dagger_\mu \rightarrow {}^0\nabla^\dagger_\mu$, $R^\rho{}_{\sigma\mu\nu} \rightarrow {}^0R^{\dagger\rho}{}_{\sigma\mu\nu}$ and $T^{\dagger\sigma}{}_{\mu\nu} \rightarrow {}^0T^{\dagger\sigma}{}_{\mu\nu} = 0$.
1510.06699_F00752	37	0.986	`T^{\dagger \sigma}{ }_{\mu \nu} \rightarrow{ }^{0} T^{\dagger \sigma}{ }_{\mu \nu}=0`	$T^{\dagger \sigma}{ }_{\mu \nu} \rightarrow{ }^{0} T^{\dagger \sigma}{ }_{\mu \nu}=0$	$\nabla^\dagger_\mu \rightarrow {}^0\nabla^\dagger_\mu$, $R^\rho{}_{\sigma\mu\nu} \rightarrow {}^0R^{\dagger\rho}{}_{\sigma\mu\nu}$ and $T^{\dagger\sigma}{}_{\mu\nu} \rightarrow {}^0T^{\dagger\sigma}{}_{\mu\nu} = 0$.
1510.06699_F00753	37	1.000	`L_{\mathrm{T}}=`	$L_{\mathrm{T}}=$	we now consider in detail has a total Lagrangian $L_{\mathrm{T}} =$
1510.06699_F00754	37	1.000	`L_{\mathrm{G}}+L_{\mathrm{M}}`	$L_{\mathrm{G}}+L_{\mathrm{M}}$	$L_{\mathrm{G}}+L_{\mathrm{M}}$ corresponding to the general eWGT form (167).
1510.06699_F00755	37	1.000	`{ }^{0} \Gamma^{\lambda}{ }_{\mu \nu}`	${ }^{0} \Gamma^{\lambda}{ }_{\mu \nu}$	in which ${}^0\Gamma^\lambda{}_{\mu\nu}$ is the standard metric (Christoffel) con-
1510.06699_F00756	37	1.000	`K^{\dagger \lambda}{ }_{\mu \nu}`	$K^{\dagger \lambda}{ }_{\mu \nu}$	nection and $K^{\dagger\lambda}{}_{\mu\nu}$ is the $\widehat{Y}_4$ contortion tensor
1510.06699_F00757	37	1.000	`L_{G}=L_{\mathcal{R}^{\dagger 2}}+L_{\mathcal{H}^{\dagger 2}}`	$L_{G}=L_{\mathcal{R}^{\dagger 2}}+L_{\mathcal{H}^{\dagger 2}}$	namely $L_G = L_{\mathcal{R}^{\dagger 2}} + L_{\mathcal{H}^{\dagger 2}}$, where
1510.06699_F00758	37	1.000	`L_{\mathrm{M}}=`	$L_{\mathrm{M}}=$	of gravity. In particular, we adopt the form $L_{\mathrm{M}} =$
1510.06699_F00759	38	1.000	`\nu, \lambda, a`	ν, λ, a	in which ν, λ, a and β_i are again dimensionless parame-
1510.06699_F00760	38	1.000	`\nu, \lambda`	ν, λ	ters (with ν, λ and a usually positive) and some factors
1510.06699_F00761	38	1.000	`L_{\varphi}`	L_{φ}	the moment we allow the form of L_φ for the matter field
1510.06699_F00762	38	1.000	`\left(\tau_{\varphi}^{\dagger}\right)^{a}{ }_{b}=h_{b}{ }^{\mu}\left(\delta L_{\varphi} / \delta h_{a}{ }^{\mu} \right)_{\dagger}`	$\left(\tau_{\varphi}^{\dagger}\right)^{a}{ }_{b}=h_{b}{ }^{\mu}\left(\delta L_{\varphi} / \delta h_{a}{ }^{\mu} \right)_{\dagger}$	and in the matter sector, in addition to $(\tau^\dagger_\varphi)^a{}_b = h_b{}^\mu(\delta L_\varphi/\delta h_a{}^\mu)_\dagger$, one has
1510.06699_F00763	38	1.000	`\left(t_{\mathcal{H}^{2}}^{\dagger}\right)^{a}{ }_{b}`	$\left(t_{\mathcal{H}^{2}}^{\dagger}\right)^{a}{ }_{b}$	In particular, we note that $(t^\dagger_{\mathcal{H}^2})^a{}_b$ in (230) is linear
1510.06699_F00764	39	1.000	`\left(\sigma_{\varphi}^{\dagger}\right)_{a b}{ }^{c}=`	$\left(\sigma_{\varphi}^{\dagger}\right)_{a b}{ }^{c}=$	on the scalar field ϕ.
1510.06699_F00765	39	1.000	`b^{c}{ }_{\mu} \delta L_{\varphi} / \delta A^{a b}{ }_{\mu}`	$b^{c}{ }_{\mu} \delta L_{\varphi} / \delta A^{a b}{ }_{\mu}$	
1510.06699_F00766	39	0.860	`\left(s_{\mathcal{H}^{2}}\right)_{a b}{ }^{c}`	$\left(s_{\mathcal{H}^{2}}\right)_{a b}{ }^{c}$	We note that $(s_{\mathcal{H}^2})_{ab}{}^c$ in (236) is linear in third-order
1510.06699_F00767	39	0.554	`\mathcal{T}_{c a b}^{\dagger}+2 \mathcal{T}_{[a b] c}^{\dagger}-3 \mathcal{T}_{[c a b]}^{\dagger}=0`	$\mathcal{T}_{c a b}^{\dagger}+2 \mathcal{T}_{[a b] c}^{\dagger}-3 \mathcal{T}_{[c a b]}^{\dagger}=0$	strength satisfies the identity $\mathcal{T}^\dagger_{cab} + 2\mathcal{T}^\dagger_{[ab]c} - 3\mathcal{T}^\dagger_{[cab]} = 0$.
1510.06699_F00768	39	1.000	`\alpha_{4}`	α_{4}	in the action (namely those in (235) proportional to α_4,
1510.06699_F00769	39	1.000	`\alpha_{5}`	α_{5}	α_5 and α_6, respectively) have a form similar to the LHS
1510.06699_F00770	39	0.606	`\overline{\mathcal{R}}_{a b c d}^{\dagger} \equiv \mathcal{R}_{c d a b}^{\dagger}`	$\overline{\mathcal{R}}_{a b c d}^{\dagger} \equiv \mathcal{R}_{c d a b}^{\dagger}$	$\overline{\mathcal{R}}^\dagger_{abcd} \equiv \mathcal{R}^\dagger_{cdab}$. Since $\overline{\mathcal{R}}^{\dagger a}{}_b = \mathcal{R}^{\dagger a}_b$, one also sees that
1510.06699_F00771	39	0.606	`\overline{\mathcal{R}}^{\dagger a}{ }_{b}=\mathcal{R}_{b}^{\dagger a}`	$\overline{\mathcal{R}}^{\dagger a}{ }_{b}=\mathcal{R}_{b}^{\dagger a}$	$\overline{\mathcal{R}}^\dagger_{abcd} \equiv \mathcal{R}^\dagger_{cdab}$. Since $\overline{\mathcal{R}}^{\dagger a}{}_b = \mathcal{R}^{\dagger a}_b$, one also sees that
1510.06699_F00772	39	1.000	`\alpha_{2}`	α_{2}	terms in (235) proportional to α_2 and α_3, respectively,
1510.06699_F00773	39	1.000	`\alpha_{3}`	α_{3}	terms in (235) proportional to α_2 and α_3, respectively,
1510.06699_F00774	39	1.000	`\alpha_{1}`	α_{1}	proportional to α_1, since $\overline{\mathcal{R}}^\dagger = \mathcal{R}^\dagger$. The term propor-
1510.06699_F00775	39	1.000	`\overline{\mathcal{R}}^{\dagger}=\mathcal{R}^{\dagger}`	$\overline{\mathcal{R}}^{\dagger}=\mathcal{R}^{\dagger}$	proportional to α_1, since $\overline{\mathcal{R}}^\dagger = \mathcal{R}^\dagger$. The term propor-
1510.06699_F00776	39	1.000	`\left(\zeta_{\varphi}\right)_{a}=`	$\left(\zeta_{\varphi}\right)_{a}=$	
1510.06699_F00777	39	1.000	`b_{a \mu} \delta L_{\varphi} / \delta V_{\mu}`	$b_{a \mu} \delta L_{\varphi} / \delta V_{\mu}$	[illegible]
1510.06699_F00778	39	1.000	`c`	c	equation (234) contracted on the indices b and c, in ac-
1510.06699_F00779	39	1.000	`j_{a}^{\dagger}`	$j_{a}^{\dagger}$	so $j^\dagger_a$ and $\zeta^\dagger_a$ vanish identically for each part of the La-

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1510.06699_F00780	39	1.000	`\zeta_{a}^{\dagger}`	$\zeta_{a}^{\dagger}$	so $j_a^\dagger$ and $\zeta_a^\dagger$ vanish identically for each part of the La-
1510.06699_F00781	39	1.000	`\partial_{\phi} \equiv \partial / \partial \phi`	$\partial_{\phi} \equiv \partial / \partial \phi$	where $\partial_\phi \equiv \partial/\partial\phi$ and $\delta_\phi \equiv \delta/\delta\phi$, since only L_ϕ depends
1510.06699_F00782	39	1.000	`\delta_{\phi} \equiv \delta / \delta \phi`	$\delta_{\phi} \equiv \delta / \delta \phi$	where $\partial_\phi \equiv \partial/\partial\phi$ and $\delta_\phi \equiv \delta/\delta\phi$, since only L_ϕ depends
1510.06699_F00783	39	1.000	`L_{\phi}`	L_{ϕ}	where $\partial_\phi \equiv \partial/\partial\phi$ and $\delta_\phi \equiv \delta/\delta\phi$, since only L_ϕ depends
1510.06699_F00784	40	1.000	`-\phi`	$-\phi$	by $-\phi$, in accordance with the general results (177) and
1510.06699_F00785	40	0.999	`\delta_{\varphi} L_{\varphi}=0`	$\delta_{\varphi} L_{\varphi}=0$	(178d), assuming the φ-equation $\delta_\varphi L_\varphi = 0$ is satisfied.
1510.06699_F00786	40	1.000	`\xi=0`	$\xi=0$	tional Lagrangian (by setting $\xi = 0$). Indeed, one can
1510.06699_F00787	40	1.000	`\left.\mathcal{D}_{a}^{\dagger} \phi\right|_{\phi=\phi_{0}}=\phi_{0}\left(\mathcal{V}_{a}+\frac{1}{3} \mathcal{T}_{a}\right)`	$\left.\mathcal{D}_{a}^{\dagger} \phi\right\|_{\phi=\phi_{0}}=\phi_{0}\left(\mathcal{V}_{a}+\frac{1}{3} \mathcal{T}_{a}\right)$	the replacement $\mathcal{D}_a^\dagger\phi\|_{\phi=\phi_0} = \phi_0(\mathcal{V}_a + \frac{1}{3}\mathcal{T}_a)$ throughout,
1510.06699_F00788	40	1.000	`\widehat{\varphi}, \widehat{h}_{a}{ }^{\mu}`	$\widehat{\varphi}, \widehat{h}_{a}{ }^{\mu}$	expressed in terms of the scale-invariant variables $\widehat{\varphi}$, $\widehat{h}_a{}^\mu$,
1510.06699_F00789	41	1.000	`h^{\prime}{ }_{a}{ }^{\mu}=e^{-\rho} h_{a}{ }^{\mu}`	$h^{\prime}{ }_{a}{ }^{\mu}=e^{-\rho} h_{a}{ }^{\mu}$	fields are given by $h'_a{}^\mu = e^{-\rho}h_a{}^\mu$ and $A'^{ab}{}_\mu = A^{ab}{}_\mu +$
1510.06699_F00790	41	1.000	`A^{\prime a b}{ }_{\mu}=A^{a b}{ }_{\mu}+`	$A^{\prime a b}{ }_{\mu}=A^{a b}{ }_{\mu}+$	fields are given by $h'_a{}^\mu = e^{-\rho}h_a{}^\mu$ and $A'^{ab}{}_\mu = A^{ab}{}_\mu +$
1510.06699_F00791	41	1.000	`\theta\left(b^{a}{ }_{\mu} \mathcal{P}^{b}-b^{b}{ }_{\mu} \mathcal{P}^{a}\right)`	$\theta\left(b^{a}{ }_{\mu} \mathcal{P}^{b}-b^{b}{ }_{\mu} \mathcal{P}^{a}\right)$	$\theta(b^a{}_\mu\mathcal{P}^b - b^b{}_\mu\mathcal{P}^a)$, respectively, where $\mathcal{P}_a = h_a{}^\mu\partial_\mu\rho$. Here
1510.06699_F00792	41	1.000	`\mathcal{P}_{a}=h_{a}{ }^{\mu} \partial_{\mu} \rho`	$\mathcal{P}_{a}=h_{a}{ }^{\mu} \partial_{\mu} \rho$	$\theta(b^a{}_\mu\mathcal{P}^b - b^b{}_\mu\mathcal{P}^a)$, respectively, where $\mathcal{P}_a = h_a{}^\mu\partial_\mu\rho$. Here
1510.06699_F00793	41	1.000	`\mathcal{P}^{2} \equiv \mathcal{P}_{a} \mathcal{P}^{a}`	$\mathcal{P}^{2} \equiv \mathcal{P}_{a} \mathcal{P}^{a}$	where $\mathcal{P}^2 \equiv \mathcal{P}_a\mathcal{P}^a$, and the translational gauge field
1510.06699_F00794	41	1.000	`(\theta=0)`	$(\theta=0)$	is covariant with weight $w = -2$ under 'normal' $(\theta = 0)$
1510.06699_F00795	41	1.000	`(\theta=1)`	$(\theta=1)$	$w = -1$ under 'special' transformations $(\theta = 1)$.
1510.06699_F00796	41	1.000	`\theta \neq 0`	$\theta \neq 0$	extended transformation law for the A-field, with $\theta \neq 0$,
1510.06699_F00797	41	1.000	`\alpha_{i}=0`	$\alpha_{i}=0$	forms covariantly (aside from the trivial case $\alpha_i = 0$ for
1510.06699_F00798	41	0.996	`i`	i	all i) and so L_{G} must vanish. Nonetheless, by contrast,
1510.06699_F00799	41	0.998	`L_{\mathcal{T}^{2}}`	$L_{\mathcal{T}^{2}}$	ance under electromagnetic gauge
1510.06699_F00800	42	0.989	`\mathcal{L}_{\mathrm{M}} \equiv \mathcal{L}_{\psi}+\mathcal{L}_{\phi}+\phi^{2} \mathcal{L}_{\mathcal{R}}+\phi^{2} \mathcal{L}_{\mathcal{T}^{2}}`	$\mathcal{L}_{\mathrm{M}} \equiv \mathcal{L}_{\psi}+\mathcal{L}_{\phi}+\phi^{2} \mathcal{L}_{\mathcal{R}}+\phi^{2} \mathcal{L}_{\mathcal{T}^{2}}$	Let us denote (256) by $\mathcal{L}_{\mathrm{M}} \equiv \mathcal{L}_\psi + \mathcal{L}_\phi + \phi^2\mathcal{L}_{\mathcal{R}} + \phi^2\mathcal{L}_{\mathcal{T}^2}$.
1510.06699_F00801	42	0.990	`\mathcal{L}_{\psi}`	$\mathcal{L}_{\psi}$	As discussed in Section III K, $\mathcal{L}_\psi$ is invariant under ex-
1510.06699_F00802	42	1.000	`\nu, a`	ν, a	rameters ν, a and β_i $(i = 1, 2, 3)$ satisfy the conditions
1510.06699_F00803	42	1.000	`\beta_{i}(i=1,2,3)`	$\beta_{i}(i=1,2,3)$	rameters ν, a and β_i $(i = 1, 2, 3)$ satisfy the conditions
1510.06699_F00804	42	1.000	`\nu`	ν	the parameters ν and a must be of opposite sign, which
1510.06699_F00805	42	1.000	`\nu=0=2 \beta_{1}+\beta_{2}+3 \beta_{3}`	$\nu=0=2 \beta_{1}+\beta_{2}+3 \beta_{3}$	$\nu = 0 = 2\beta_1 + \beta_2 + 3\beta_3$, and a 'special' transforma-
1510.06699_F00806	42	0.997	`\nu=0=a`	$\nu=0=a$	tion $(\theta = 1)$, for which $\nu = 0 = a$. Thus, in both special
1510.06699_F00807	42	1.000	`\nu=\beta_{1}=\beta_{2}=\beta_{3}=0`	$\nu=\beta_{1}=\beta_{2}=\beta_{3}=0$	Obukhov[89], who set $\nu = \beta_1 = \beta_2 = \beta_3 = 0$. Adopting
1510.06699_F00808	42	1.000	`2 \beta_{1}+\beta_{2}+3 \beta_{3}=0`	$2 \beta_{1}+\beta_{2}+3 \beta_{3}=0$	with $2\beta_1 + \beta_2 + 3\beta_3 = 0$ in $L_{\mathcal{T}^{*2}}$, which is merely a special
1510.06699_F00809	42	0.706	`\mathcal{D}_{a} \phi \rightarrow\left(\mathcal{D}_{a}+\frac{1}{3} \mathcal{T}_{a}\right) \phi`	$\mathcal{D}_{a} \phi \rightarrow\left(\mathcal{D}_{a}+\frac{1}{3} \mathcal{T}_{a}\right) \phi$	$\mathcal{D}_a\phi \to (\mathcal{D}_a + \frac{1}{3}\mathcal{T}_a)\phi$ (and similarly for $\mathcal{D}^a\phi$) in the kinetic
1510.06699_F00810	42	0.706	`\mathcal{D}^{a} \phi`	$\mathcal{D}^{a} \phi$	$\mathcal{D}_a\phi \to (\mathcal{D}_a + \frac{1}{3}\mathcal{T}_a)\phi$ (and similarly for $\mathcal{D}^a\phi$) in the kinetic
1510.06699_F00811	42	1.000	`\frac{1}{2} \nu+6 a=0`	$\frac{1}{2} \nu+6 a=0$	vided only that $\frac{1}{2}\nu + 6a = 0$ and $2\beta_1 + \beta_2 + 3\beta_3 = 0$
1510.06699_F00812	42	0.994	`2 \beta_{1}+\beta_{2}=0`	$2 \beta_{1}+\beta_{2}=0$	with $2\beta_1 + \beta_2 = 0$ in $L_{\mathcal{T}^{\dagger 2}}$ (in which there is no term
1510.06699_F00813	42	0.994	`L_{\mathcal{T}^{+2}}`	$L_{\mathcal{T}^{+2}}$	with $2\beta_1 + \beta_2 = 0$ in $L_{\mathcal{T}^{\dagger 2}}$ (in which there is no term
1510.06699_F00814	43	1.000	`{ }^{0} \mathcal{D}_{a}`	${ }^{0} \mathcal{D}_{a}$	derivative ${}^0\mathcal{D}_a$, to which the full PGT covariant deriva-
1510.06699_F00815	43	1.000	`{ }^{0} \mathcal{T}^{a}{ }_{b c} \equiv h_{b}{ }^{\mu} h_{c}{ }^{\nu}\left({ }^{0} D_{\mu} b^{a}{ }_{\nu}-{ }^{0} D_{\nu} b^{a}{ }_{\mu}\right)=0`	${ }^{0} \mathcal{T}^{a}{ }_{b c} \equiv h_{b}{ }^{\mu} h_{c}{ }^{\nu}\left({ }^{0} D_{\mu} b^{a}{ }_{\nu}-{ }^{0} D_{\nu} b^{a}{ }_{\mu}\right)=0$	${}^0\mathcal{T}^a{}_{bc} \equiv h_b{}^\mu h_c{}^\nu({}^0D_\mu b^a{}_\nu - {}^0D_\nu b^a{}_\mu) = 0$. Thus, reduced
1510.06699_F00816	43	1.000	`{ }^{0} A^{\prime a b}{ }_{\mu}={ }^{0} A^{a b}{ }_{\mu}+b^{a}{ }_{\mu} \mathcal{P}^{b}-b^{b}{ }_{\mu} \mathcal{P}^{a}`	${ }^{0} A^{\prime a b}{ }_{\mu}={ }^{0} A^{a b}{ }_{\mu}+b^{a}{ }_{\mu} \mathcal{P}^{b}-b^{b}{ }_{\mu} \mathcal{P}^{a}$	${}^0A'^{ab}{}_\mu = {}^0A^{ab}{}_\mu + b^a{}_\mu\mathcal{P}^b - b^b{}_\mu\mathcal{P}^a$, which is independent
1510.06699_F00817	43	1.000	`{ }^{0} A^{a b}{ }_{\mu},{ }^{0} \mathcal{D}_{c} \varphi`	${ }^{0} A^{a b}{ }_{\mu},{ }^{0} \mathcal{D}_{c} \varphi$	${}^0A^{ab}{}_\mu$, ${}^0\mathcal{D}_c\varphi$ and ${}^0\mathcal{R}^{ab}{}_{cd}$ all transform independently of
1510.06699_F00818	43	0.999	`A^{a b}{ }_{\mu}, \mathcal{D}_{c} \varphi`	$A^{a b}{ }_{\mu}, \mathcal{D}_{c} \varphi$	$A^{ab}{}_\mu$, $\mathcal{D}_c\varphi$ and $\mathcal{R}^{ab}{}_{cd}$ do assuming the 'special' transfor-

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1510.06699_F00819	43	1.000	`\mathcal{D}_{c} \rightarrow{ }^{0} \mathcal{D}_{c}`	$\mathcal{D}_{c} \rightarrow{ }^{0} \mathcal{D}_{c}$	replacements $\mathcal{D}_c \rightarrow {}^0\mathcal{D}_c$ and $\mathcal{T}^a{}_{bc} \rightarrow {}^0\mathcal{T}^a{}_{bc} \equiv 0$.
1510.06699_F00820	43	1.000	`\mathcal{T}^{a}{ }_{b c} \rightarrow{ }^{0} \mathcal{T}^{a}{ }_{b c} \equiv 0`	$\mathcal{T}^{a}{ }_{b c} \rightarrow{ }^{0} \mathcal{T}^{a}{ }_{b c} \equiv 0$	replacements $\mathcal{D}_c \rightarrow {}^0\mathcal{D}_c$ and $\mathcal{T}^a{}_{bc} \rightarrow {}^0\mathcal{T}^a{}_{bc} \equiv 0$.
1510.06699_F00821	43	0.915	`L_{G}`	L_{G}	Following the discussion in Appendix B, if L_G contains
1510.06699_F00822	43	1.000	`2 \alpha_{3}+\alpha_{2}=0`	$2 \alpha_{3}+\alpha_{2}=0$	$2\alpha_3 + \alpha_2 = 0$ and $3\alpha_1 - \alpha_3 = 0$. Letting $\alpha = \alpha_3$ for con-
1510.06699_F00823	43	1.000	`3 \alpha_{1}-\alpha_{3}=0`	$3 \alpha_{1}-\alpha_{3}=0$	$2\alpha_3 + \alpha_2 = 0$ and $3\alpha_1 - \alpha_3 = 0$. Letting $\alpha = \alpha_3$ for con-
1510.06699_F00824	43	1.000	`\alpha=\alpha_{3}`	$\alpha=\alpha_{3}$	$2\alpha_3 + \alpha_2 = 0$ and $3\alpha_1 - \alpha_3 = 0$. Letting $\alpha = \alpha_3$ for con-
1510.06699_F00825	43	1.000	`{ }^{0} \mathcal{R}_{a b c d}{ }^{0} \mathcal{R}^{a b c d}`	${ }^{0} \mathcal{R}_{a b c d}{ }^{0} \mathcal{R}^{a b c d}$	proportional to ${}^0\mathcal{R}_{abcd}{}^0\mathcal{R}^{abcd}$ to obtain
1510.06699_F00826	43	0.899	`{ }^{\underline{101}}`	${ }^{\underline{101}}$	theory[101]. Indeed, as one might suspect, one can easily
1510.06699_F00827	43	0.999	`L_{\mathrm{W}}={ }^{0} \mathcal{C}_{a b c d}{ }^{0} \mathcal{C}^{a b c d}`	$L_{\mathrm{W}}={ }^{0} \mathcal{C}_{a b c d}{ }^{0} \mathcal{C}^{a b c d}$	grangian $L_{\mathrm{W}} = {}^0\mathcal{C}_{abcd}{}^0\mathcal{C}^{abcd}$, where ${}^0\mathcal{C}_{abcd}$ is the gauge
1510.06699_F00828	43	0.999	`{ }^{0} \mathcal{C}_{a b c d}`	${ }^{0} \mathcal{C}_{a b c d}$	grangian $L_{\mathrm{W}} = {}^0\mathcal{C}_{abcd}{}^0\mathcal{C}^{abcd}$, where ${}^0\mathcal{C}_{abcd}$ is the gauge
1510.06699_F00829	44	1.000	`{ }^{0} \mathcal{R}_{a b c d}`	${ }^{0} \mathcal{R}_{a b c d}$	This tensor represents the traceless part of ${}^0\mathcal{R}_{abcd}$; it has
1510.06699_F00830	44	0.988	`{ }^{0} \mathcal{C}_{a b c d}^{\prime}=e^{-2 \rho}{ }^{0} \mathcal{C}_{a b c d}`	${ }^{0} \mathcal{C}_{a b c d}^{\prime}=e^{-2 \rho}{ }^{0} \mathcal{C}_{a b c d}$	zero). It transforms covariantly as ${}^0\mathcal{C}'_{abcd} = e^{-2\rho}\,{}^0\mathcal{C}_{abcd}$
1510.06699_F00831	44	1.000	`L_{W}`	L_{W}	under a local change of scale, so that L_W also transforms
1510.06699_F00832	44	1.000	`w\left(L_{W}\right)=-4`	$w\left(L_{W}\right)=-4$	covariantly with a Weyl weight $w(L_W) = -4$, and hence
1510.06699_F00833	44	1.000	`{ }^{\underline{19}}`	${ }^{\underline{19}}$	nian and survives canonical quantisation[19]; it thus seems
1510.06699_F00834	44	1.000	`\mathcal{D}_{a} \rightarrow{ }^{0} \mathcal{D}_{a}`	$\mathcal{D}_{a} \rightarrow{ }^{0} \mathcal{D}_{a}$	grangian (256), but with the replacements $\mathcal{D}_a \rightarrow {}^0\mathcal{D}_a$,
1510.06699_F00835	44	0.554	`\mathcal{R} \rightarrow{ }^{0} \mathcal{R}`	$\mathcal{R} \rightarrow{ }^{0} \mathcal{R}$	$\mathcal{R} \rightarrow {}^0\mathcal{R}$ and without the term proportional to $L_{\mathcal{T}^2}$,
1510.06699_F00836	44	0.972	`{ }^{\underline{103}}`	${ }^{\underline{103}}$	namely[103]
1510.06699_F00837	44	1.000	`\frac{1}{2} \nu+`	$\frac{1}{2} \nu+$	resulting matter Lagrangian, obtained by setting $\frac{1}{2}\nu +$
1510.06699_F00838	44	1.000	`6 a=0`	$6 a=0$	$6a = 0$ in (268), coincides with that typically assumed in
1510.06699_F00839	44	0.657	`\phi^{20} \mathcal{R}`	$\phi^{20} \mathcal{R}$	term for ϕ and the $\phi^2\,{}^0\mathcal{R}$ term.
1510.06699_F00840	44	1.000	`{ }^{0} \mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{*}`	${ }^{0} \mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{*}$	placements ${}^0\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}^*_{abcd}$ or ${}^0\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}^\dagger_{abcd}$ (and
1510.06699_F00841	44	1.000	`{ }^{0} \mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{\dagger}`	${ }^{0} \mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}^{\dagger}$	placements ${}^0\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}^*_{abcd}$ or ${}^0\mathcal{R}_{abcd} \rightarrow {}^0\mathcal{R}^\dagger_{abcd}$ (and
1510.06699_F00842	44	1.000	`\frac{1}{2} \nu+6 a=`	$\frac{1}{2} \nu+6 a=$	Similarly, the matter Lagrangian (268) with $\frac{1}{2}\nu + 6a =$
1510.06699_F00843	44	0.996	`\mathcal{T}_{a} \equiv 0`	$\mathcal{T}_{a} \equiv 0$	$\mathcal{T}_a \equiv 0$ and either set of replacements $\mathcal{D}^\dagger_a \rightarrow {}^0\mathcal{D}^*_a$,
1510.06699_F00844	44	0.996	`\mathcal{D}_{a}^{\dagger} \rightarrow{ }^{0} \mathcal{D}_{a}^{*}`	$\mathcal{D}_{a}^{\dagger} \rightarrow{ }^{0} \mathcal{D}_{a}^{*}$	$\mathcal{T}_a \equiv 0$ and either set of replacements $\mathcal{D}^\dagger_a \rightarrow {}^0\mathcal{D}^*_a$,
1510.06699_F00845	44	1.000	`\mathcal{R}^{\dagger} \rightarrow{ }^{0} \mathcal{R}^{*} \mathcal{T}_{a b c}^{\dagger} \rightarrow{ }^{0} \mathcal{T}_{a b c}^{*} \equiv 0`	$\mathcal{R}^{\dagger} \rightarrow{ }^{0} \mathcal{R}^{*} \mathcal{T}_{a b c}^{\dagger} \rightarrow{ }^{0} \mathcal{T}_{a b c}^{*} \equiv 0$	$\mathcal{R}^\dagger \rightarrow {}^0\mathcal{R}^*$ $\mathcal{T}^\dagger_{abc} \rightarrow {}^0\mathcal{T}^*_{abc} \equiv 0$ or $\mathcal{D}^\dagger_a \rightarrow {}^0\mathcal{D}^\dagger_a$, $\mathcal{R}^\dagger \rightarrow {}^0\mathcal{R}^\dagger$
1510.06699_F00846	44	1.000	`\mathcal{D}_{a}^{\dagger} \rightarrow{ }^{0} \mathcal{D}_{a}^{\dagger}, \mathcal{R}^{\dagger} \rightarrow{ }^{0} \mathcal{R}^{\dagger}`	$\mathcal{D}_{a}^{\dagger} \rightarrow{ }^{0} \mathcal{D}_{a}^{\dagger}, \mathcal{R}^{\dagger} \rightarrow{ }^{0} \mathcal{R}^{\dagger}$	$\mathcal{R}^\dagger \rightarrow {}^0\mathcal{R}^*$ $\mathcal{T}^\dagger_{abc} \rightarrow {}^0\mathcal{T}^*_{abc} \equiv 0$ or $\mathcal{D}^\dagger_a \rightarrow {}^0\mathcal{D}^\dagger_a$, $\mathcal{R}^\dagger \rightarrow {}^0\mathcal{R}^\dagger$
1510.06699_F00847	44	0.996	`\mathcal{T}_{a b c}^{\dagger} \rightarrow{ }^{0} \mathcal{T}_{a b c}^{\dagger} \equiv 0`	$\mathcal{T}_{a b c}^{\dagger} \rightarrow{ }^{0} \mathcal{T}_{a b c}^{\dagger} \equiv 0$	$\mathcal{T}^\dagger_{abc} \rightarrow {}^0\mathcal{T}^\dagger_{abc} \equiv 0$. Thus, (268) with $\frac{1}{2}\nu + 6a = 0$ is equiv-
1510.06699_F00848	45	0.997	`w,-1`	$w,-1$	w, -1 and 0, respectively, under the 'extended' transfor-
1510.06699_F00849	45	1.000	`\mathcal{R}_{a b c d}^{\dagger}(h, A, \partial A, V, \partial V)`	$\mathcal{R}_{a b c d}^{\dagger}(h, A, \partial A, V, \partial V)$	field strength tensors $\mathcal{R}^\dagger_{abcd}(h, A, \partial A, V, \partial V)$,
1510.06699_F00850	45	1.000	`\mathcal{T}_{a b c}^{\dagger}(h, \partial h, A) \quad`	$\mathcal{T}_{a b c}^{\dagger}(h, \partial h, A) \quad$	$\mathcal{T}^\dagger_{abc}(h, \partial h, A)$ and $\mathcal{H}^\dagger_{ab}(h, \partial h, \partial^2 h, A, \partial A, \partial V)$,
1510.06699_F00851	45	1.000	`\quad \mathcal{H}_{a b}^{\dagger}\left(h, \partial h, \partial^{2} h, A, \partial A, \partial V\right)`	$\quad \mathcal{H}_{a b}^{\dagger}\left(h, \partial h, \partial^{2} h, A, \partial A, \partial V\right)$	$\mathcal{T}^\dagger_{abc}(h, \partial h, A)$ and $\mathcal{H}^\dagger_{ab}(h, \partial h, \partial^2 h, A, \partial A, \partial V)$,
1510.06699_F00852	45	1.000	`\mathcal{T}_{a}^{\dagger}`	$\mathcal{T}_{a}^{\dagger}$	• The trace $\mathcal{T}^\dagger_a$ of the eWGT translational gauge field

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1510.06699_F00853	45	0.971	`\tau_{a b}^{\dagger}`	$\tau_{a b}^{\dagger}$	versions $t^{\dagger}_{ab}$ and $\tau^{\dagger}_{ab}$ of free-gravitational and mat-
1510.06699_F00854	45	1.000	`A^{\dagger a b}{ }_{\mu} \equiv`	$A^{\dagger a b}{ }_{\mu} \equiv$	variables such that $A^{ab}{}_{\mu}$ is replaced by $A^{\dagger ab}{}_{\mu} \equiv$
1510.06699_F00855	45	0.932	`A^{a b}{ }_{\mu}+\left(\mathcal{V}^{a} b^{b}{ }_{\mu}-\mathcal{V}^{b} b^{a}{ }_{\mu}\right)`	$A^{a b}{ }_{\mu}+\left(\mathcal{V}^{a} b^{b}{ }_{\mu}-\mathcal{V}^{b} b^{a}{ }_{\mu}\right)$	$A^{ab}{}_{\mu} + (\mathcal{V}^a b^b{}_{\mu} - \mathcal{V}^b b^a{}_{\mu})$.
1510.06699_F00856	46	1.000	`t^{\dagger a}{ }_{a}`	$t^{\dagger a}{ }_{a}$	trace $t^{\dagger a}{}_a$ of the free-gravitational energy momen-
1510.06699_F00857	46	1.000	`\tau^{\dagger a}{ }_{a}`	$\tau^{\dagger a}{ }_{a}$	$\tau^{\dagger a}{}_a$ of the total matter energy-momentum tensor
1510.06699_F00858	46	0.948	`{ }^{0} \mathcal{R}_{a b c d}^{\dagger}\left(h, \partial h, \partial^{2} h, A, \partial A, V, \partial V\right)`	${ }^{0} \mathcal{R}_{a b c d}^{\dagger}\left(h, \partial h, \partial^{2} h, A, \partial A, V, \partial V\right)$	${}^0\mathcal{R}^{\dagger}_{abcd}(h, \partial h, \partial^2 h, A, \partial A, V, \partial V)$, depends on all the
1510.06699_F00859	47	1.000	`\not \partial \equiv \gamma^{\mu} \partial_{\mu}`	$\not \partial \equiv \gamma^{\mu} \partial_{\mu}$	metrical manner the dilation part of the Weyl trans-
1510.06699_F00860	47	0.938	`\bar{\psi} / t`	$\bar{\psi} / t$	vely) dilates the Minkowski spacetime such that the
1510.06699_F00861	47	0.938	`J^{\mu}=`	$J^{\mu}=$	vely) dilates the Minkowski spacetime such that the
1510.06699_F00862	47	1.000	`\bar{\psi} \gamma^{\mu} \psi`	$\bar{\psi} \gamma^{\mu} \psi$	distance between any two points is increased by
1510.06699_F00863	47	1.000	`\psi \bar{\psi}`	$\psi \bar{\psi}$	The spinor outer product $\psi\bar{\psi}$ may be rewritten using the
1510.06699_F00864	47	1.000	`\gamma^{5}=\gamma_{5}=i \gamma^{0} \gamma^{1} \gamma^{2} \gamma^{3}`	$\gamma^{5}=\gamma_{5}=i \gamma^{0} \gamma^{1} \gamma^{2} \gamma^{3}$	where $\gamma^5 = \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and $\sigma^{\nu\rho} = \frac{i}{2}[\gamma^\nu, \gamma^\rho]$ (and
1510.06699_F00865	47	1.000	`\sigma^{\nu \rho}=\frac{i}{2}\left[\gamma^{\nu}, \gamma^{\rho}\right]`	$\sigma^{\nu \rho}=\frac{i}{2}\left[\gamma^{\nu}, \gamma^{\rho}\right]$	where $\gamma^5 = \gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and $\sigma^{\nu\rho} = \frac{i}{2}[\gamma^\nu, \gamma^\rho]$ (and
1510.06699_F00866	47	1.000	`\rho \equiv \bar{\psi} \psi`	$\rho \equiv \bar{\psi} \psi$	It is convenient to define the real scalar $\rho \equiv \bar{\psi}\psi$ and
1510.06699_F00867	47	0.671	`\beta \equiv \bar{\psi} i \gamma^{5} \psi`	$\beta \equiv \bar{\psi} i \gamma^{5} \psi$	real pseudoscalar $\beta \equiv \bar{\psi} i\gamma^5\psi$, so that (A4) reads $\bar{\psi}\not{J} =$
1510.06699_F00868	47	0.671	`\bar{\psi} / \hbar=`	$\bar{\psi} / \hbar=$	real pseudoscalar $\beta \equiv \bar{\psi} i\gamma^5\psi$, so that (A4) reads $\bar{\psi}\not{J} =$
1510.06699_F00869	47	1.000	`\bar{\psi}\left(\rho+\beta i \gamma^{5}\right)`	$\bar{\psi}\left(\rho+\beta i \gamma^{5}\right)$	$\bar{\psi}(\rho+\beta i\gamma^5)$. Postmultiplying this expression by $(\rho-\beta i\gamma^5)$
1510.06699_F00870	47	1.000	`\left(\rho-\beta i \gamma^{5}\right)`	$\left(\rho-\beta i \gamma^{5}\right)$	$\bar{\psi}(\rho+\beta i\gamma^5)$. Postmultiplying this expression by $(\rho-\beta i\gamma^5)$
1510.06699_F00871	47	0.982	`\nexists \not \partial \psi`	$\nexists \not \partial \psi$	In particular, we note that the quantity $\not{J}\not{\partial}\psi$ appearing
1510.06699_F00872	47	1.000	`J^{\mu} \partial_{\mu} \psi`	$J^{\mu} \partial_{\mu} \psi$	The quantity $J^\mu\partial_\mu\psi$ is clearly interpreted as a derivative
1510.06699_F00873	47	1.000	`n^{\nu} \equiv-i \sigma^{\mu \nu} J_{\mu}`	$n^{\nu} \equiv-i \sigma^{\mu \nu} J_{\mu}$	the direction $n^\nu \equiv -i\sigma^{\mu\nu}J_\mu$, which is orthogonal to J^μ,
1510.06699_F00874	47	1.000	`J_{\nu} n^{\nu}=-i \sigma^{\mu \nu} J_{\mu} J_{\nu}=0`	$J_{\nu} n^{\nu}=-i \sigma^{\mu \nu} J_{\mu} J_{\nu}=0$	since $J_\nu n^\nu = -i\sigma^{\mu\nu}J_\mu J_\nu = 0$. Thus, $-i\sigma^{\mu\nu}J_\mu\partial_\nu\psi$ is in-
1510.06699_F00875	47	1.000	`-i \sigma^{\mu \nu} J_{\mu} \partial_{\nu} \psi`	$-i \sigma^{\mu \nu} J_{\mu} \partial_{\nu} \psi$	since $J_\nu n^\nu = -i\sigma^{\mu\nu}J_\mu J_\nu = 0$. Thus, $-i\sigma^{\mu\nu}J_\mu\partial_\nu\psi$ is in-
1510.06699_F00876	47	1.000	`x^{\mu}(\lambda)`	$x^{\mu}(\lambda)$	the particle is described by a worldline $x^\mu(\lambda)$, together
1510.06699_F00877	47	1.000	`R(\lambda)`	$R(\lambda)$	with a ‘rotor’ $R(\lambda)$, which is a Dirac spinor defined along
1510.06699_F00878	47	1.000	`\bar{R} R=1`	$\bar{R} R=1$	$\bar{R}R = 1$ and $\bar{R}i\gamma^5 R = 0$, and contains information about
1510.06699_F00879	47	1.000	`\bar{R} i \gamma^{5} R=0`	$\bar{R} i \gamma^{5} R=0$	$\bar{R}R = 1$ and $\bar{R}i\gamma^5 R = 0$, and contains information about
1510.06699_F00880	47	1.000	`R`	R	one replaces ψ by R in (A6), restricts derivatives to lie
1510.06699_F00881	47	1.000	`\dot{x}^{\mu} \equiv d x^{\mu} / d \lambda`	$\dot{x}^{\mu} \equiv d x^{\mu} / d \lambda$	locity $\dot{x}^\mu \equiv dx^\mu/d\lambda$ with $\bar{R}\gamma^\mu R$. The last identification
1510.06699_F00882	47	1.000	`\bar{R} \gamma^{\mu} R`	$\bar{R} \gamma^{\mu} R$	locity $\dot{x}^\mu \equiv dx^\mu/d\lambda$ with $\bar{R}\gamma^\mu R$. The last identification
1510.06699_F00883	47	1.000	`p_{\mu}(\lambda)`	$p_{\mu}(\lambda)$	multiplier $p_\mu(\lambda)$, which is identified with the momentum
1510.06699_F00884	47	1.000	`p_{\mu}`	p_{μ}	of the particle; note that, in general, p_μ and $\dot{x}^\mu$ are not
1510.06699_F00885	47	1.000	`\dot{x}^{\mu}`	$\dot{x}^{\mu}$	of the particle; note that, in general, p_μ and $\dot{x}^\mu$ are not
1510.06699_F00886	47	0.999	`m e(\lambda)`	$m e(\lambda)$	ten as $me(\lambda)$ for later convenience) to ensure reparam-
1510.06699_F00887	47	1.000	`x^{\mu}(\lambda), R(\lambda), p^{\mu}(\lambda)`	$x^{\mu}(\lambda), R(\lambda), p^{\mu}(\lambda)$	to the dynamical variables $x^\mu(\lambda)$, $R(\lambda)$, $p^\mu(\lambda)$ and $e(\lambda)$
1510.06699_F00888	47	1.000	`p^{\mu} / m`	p^{μ} / m	replacing $\bar{R}\gamma^\mu R$ by p^μ/m and then neglecting the particle
1510.06699_F00889	47	1.000	`e(\lambda) \rightarrow \frac{1}{2} e(\lambda)`	$e(\lambda) \rightarrow \frac{1}{2} e(\lambda)$	einbein $e(\lambda) \to \frac{1}{2}e(\lambda)$ for later convenience, this process
1510.06699_F00890	47	1.000	`x^{\mu}(\lambda), p^{\mu}(\lambda)`	$x^{\mu}(\lambda), p^{\mu}(\lambda)$	variables $x^\mu(\lambda)$, $p^\mu(\lambda)$ and $e(\lambda)$ leads to the classical
1510.06699_F00891	47	1.000	`\rho=0`	$\rho=0$	ticular, by setting $\rho = 0$ in the global Weyl transforma-
1510.06699_F00892	47	1.000	`\rho(x)=0`	$\rho(x)=0$	reduce to local Poincaré transformations if $\rho(x) = 0$. In
1510.06699_F00893	48	1.000	`D_{\mu} \varphi`	$D_{\mu} \varphi$	derivative $D_\mu\varphi$. Thus, all the results given in Sec-
1510.06699_F00894	48	0.981	`\rho(x)=0=`	$\rho(x)=0=$	tions II B and II C also hold in PGT if one sets $\rho(x) = 0 =$
1510.06699_F00895	48	1.000	`B_{\mu}=0`	$B_{\mu}=0$	valid in PGT if one sets $B_\mu = 0$ and removes all asterisks.
1510.06699_F00896	48	1.000	`H_{\mu \nu}=0`	$H_{\mu \nu}=0$	prescription of setting $B_\mu = 0$ (so that $H_{\mu\nu} = 0$) and re-

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1510.06699_F00897	48	0.999	`\mathcal{R} \equiv \mathcal{R}^{a b}{ }_{a b}, \Lambda`	$\mathcal{R} \equiv \mathcal{R}^{a b}{ }_{a b}, \Lambda$	where $\mathcal{R} \equiv \mathcal{R}^{ab}{}_{ab}$, Λ is a cosmological constant, a is
1510.06699_F00898	48	1.000	`a, \Lambda,\left\{\alpha_{i}\right\}`	$a, \Lambda,\left\{\alpha_{i}\right\}$	depends on the choice of the parameters a, Λ, $\{\alpha_i\}$ and
1510.06699_F00899	48	1.000	`\left\{\beta_{i}\right\}`	$\left\{\beta_{i}\right\}$	$\{\beta_i\}$ in L_{G}. As an illustration, in Appendix C we give
1510.06699_F00900	48	1.000	`\varphi, \partial \varphi, h`	$\varphi, \partial \varphi, h$	induce a dependence on φ, $\partial\varphi$, h and A, but one typically
1510.06699_F00901	48	0.546	`\mathcal{H}_{a b}, j^{a}, \zeta^{a}`	$\mathcal{H}_{a b}, j^{a}, \zeta^{a}$	$\delta\varphi \quad \partial\varphi \quad \partial(\mathcal{D}_a^*\varphi)\big)$
1510.06699_F00902	48	1.000	`\mu \phi \bar{\psi} \psi \rightarrow m \bar{\psi} \psi`	$\mu \phi \bar{\psi} \psi \rightarrow m \bar{\psi} \psi$	satisfy any further required invariances that were lost in
1510.06699_F00903	48	1.000	`\phi=1`	$\phi=1$	in Section II K.
1510.06699_F00904	48	0.992	`{ }^{0} \mathcal{T}^{a}{ }_{b c} \equiv 0`	${ }^{0} \mathcal{T}^{a}{ }_{b c} \equiv 0$	the contributions of the free gravitational sector to the
1510.06699_F00905	48	1.000	`{ }^{0} \mathcal{R}^{a b}{ }_{c d}\left(h, \partial h, \partial^{2} h\right)`	${ }^{0} \mathcal{R}^{a b}{ }_{c d}\left(h, \partial h, \partial^{2} h\right)$	gravitational field equations:
1510.06699_F00906	49	1.000	`{ }^{0} \mathcal{D}_{a} \mathcal{T}^{a}=h \partial_{\mu} \left(h^{-1} h_{a}{ }^{\mu} \mathcal{T}^{a}\right)`	${ }^{0} \mathcal{D}_{a} \mathcal{T}^{a}=h \partial_{\mu} \left(h^{-1} h_{a}{ }^{\mu} \mathcal{T}^{a}\right)$	terms of a total derivative as ${}^0\mathcal{D}_a\mathcal{T}^a = h\partial_\mu(h^{-1}h_a{}^\mu\mathcal{T}^a)$,
1510.06699_F00907	49	0.980	`{ }^{\underline{105}}`	${ }^{\underline{105}}$	whereas this is not possible in WGT[105].
1510.06699_F00908	49	0.935	`L_{0} \mathcal{R}^{2}`	$L_{0} \mathcal{R}^{2}$	where $L_{{}^0\mathcal{R}^2}$ is based on (2) with $\mathcal{R}^{ab}{}_{cd}$ replaced by
1510.06699_F00909	49	0.606	`L_{0^{\mathcal{R}^{2}}}`	$L_{0^{\mathcal{R}^{2}}}$	tinct terms in $L_{{}^0\mathcal{R}^2}$ is reduced to
1510.06699_F00910	49	0.999	`\mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}`	$\mathcal{R}_{a b c d} \rightarrow{ }^{0} \mathcal{R}_{a b c d}$	${}^0\mathcal{R}'^a{}_c = e^{-2\rho}\{{}^0\mathcal{R}^a{}_c - 2\,{}^0\mathcal{D}_c\mathcal{P}^a - \delta^a_c\,{}^0$
1510.06699_F00911	49	1.000	`\alpha_{3}=0`	$\alpha_{3}=0$	
1510.06699_F00912	49	0.999	`\alpha_{1}=\alpha_{2}=0`	$\alpha_{1}=\alpha_{2}=0$	h, then one must set $\alpha_1 = \alpha_2 = 0$ (at least in $D \leq 4$
1510.06699_F00913	49	1.000	`a=\frac{1}{2}`	$a=\frac{1}{2}$	spacetime dimensions), and on choosing $a = \frac{1}{2}$ the re-
1510.06699_F00914	49	0.999	`a=0=\Lambda, \alpha_{1}=-\frac{1}{3} \alpha_{2}`	$a=0=\Lambda, \alpha_{1}=-\frac{1}{3} \alpha_{2}$	is $a = 0 = \Lambda$, $\alpha_1 = -\frac{1}{3}\alpha_2$, which corresponds to the free
1510.06699_F00915	49	1.000	`U_{4}`	U_{4}	that the spacetime has, in general, a Riemann–Cartan U_4
1510.06699_F00916	49	1.000	`{ }^{0} R^{\rho}{ }_{\sigma \mu \nu}`	${ }^{0} R^{\rho}{ }_{\sigma \mu \nu}$	leads one to recognise ${}^0R^\rho{}_{\sigma\mu\nu}$ as the standard curvature
1510.06699_F00917	49	1.000	`R \equiv R^{\mu \nu}{ }_{\mu \nu}`	$R \equiv R^{\mu \nu}{ }_{\mu \nu}$	which relates the Ricci–Cartan scalar $R \equiv R^{\mu\nu}{}_{\mu\nu}$ in
1510.06699_F00918	49	0.999	`{ }^{0} R \equiv{ }^{0} R^{\mu \nu}{ }_{\mu \nu}`	${ }^{0} R \equiv{ }^{0} R^{\mu \nu}{ }_{\mu \nu}$	U_4 spacetime to the Ricci scalar ${}^0R \equiv {}^0R^{\mu\nu}{}_{\mu\nu}$ in V_4
1510.06699_F00919	49	1.000	`L_{\mathrm{G}}=-R /(2 \kappa)`	$L_{\mathrm{G}}=-R /(2 \kappa)$	taken to be $L_{\mathrm{G}} = -R/(2\kappa)$ and the theory is interpreted
1510.06699_F00920	49	1.000	`T_{\mu \nu \sigma}`	$T_{\mu \nu \sigma}$	that includes a tensor field $T_{\mu\nu\sigma}$ (obeying the appropri-
1510.06699_F00921	49	1.000	`T_{\mu \nu \lambda}`	$T_{\mu \nu \lambda}$	Lagrangian is given by the terms containing $T_{\mu\nu\lambda}$ in (B6)
1510.06699_F00922	49	1.000	`\mathcal{R}=\mathcal{R}^{a b}{ }_{a b}`	$\mathcal{R}=\mathcal{R}^{a b}{ }_{a b}$	$\mathcal{R} = \mathcal{R}^{ab}{}_{ab}$. Thus, in Kibble's original theory, one has
1510.06699_F00923	49	1.000	`\Lambda=\alpha_{i}=\beta_{i}=0`	$\Lambda=\alpha_{i}=\beta_{i}=0$	which corresponds to setting $a = \frac{1}{2}$ and $\Lambda = \alpha_i = \beta_i = 0$
1510.06699_F00924	49	0.983	`{ }^{\underline{106}}`	${ }^{\underline{106}}$	theory[106], which, when re-interpreted geometrically, is a
1510.06699_F00925	50	1.000	`\mathcal{T}_{a} \equiv \mathcal{T}^{c}{ }_{a c}`	$\mathcal{T}_{a} \equiv \mathcal{T}^{c}{ }_{a c}$	where $\mathcal{T}_a \equiv \mathcal{T}^c{}_{ac}$, as defined in (66). In particular, by
1510.06699_F00926	50	0.999	`\mathcal{A}^{a b}{ }_{c}=h_{c}{ }^{\mu} A^{a b}{ }_{\mu}`	$\mathcal{A}^{a b}{ }_{c}=h_{c}{ }^{\mu} A^{a b}{ }_{\mu}$	its fully Lorentz-index counterpart $\mathcal{A}^{ab}{}_c = h_c{}^\mu A^{ab}{}_\mu$. In
1510.06699_F00927	50	1.000	`\sigma_{a b}{ }^{c} \neq 0`	$\sigma_{a b}{ }^{c} \neq 0$	isks). In the general case, in which $\sigma_{ab}{}^c \neq 0$, this is not,
1510.06699_F00928	50	1.000	`\sigma_{a b c}`	$\sigma_{a b c}$	derivatives of the matter field(s), then σ_{abc} is indepen-
1510.06699_F00929	50	1.000	`\mathcal{T}^{c}{ }_{a b}`	$\mathcal{T}^{c}{ }_{a b}$	field strength $\mathcal{T}^c{}_{ab}$, and so the h-field itself must satisfy
1510.06699_F00930	50	0.997	`\mu \phi \equiv m`	$\mu \phi \equiv m$	Lagrangian density is given by (70) with $\mu\phi \equiv m$, sev-
1510.06699_F00931	50	1.000	`\mathcal{T}^{c}{ }_{a b}=2 \kappa h \sigma_{a b}{ }^{c}`	$\mathcal{T}^{c}{ }_{a b}=2 \kappa h \sigma_{a b}{ }^{c}$	$\mathcal{T}^c{}_{ab} = 2\kappa h\sigma_{ab}{}^c$. More interestingly, the condition (C9)
1510.06699_F00932	50	1.000	`R_{\mu \nu} \equiv R^{\lambda}{ }_{\mu \lambda \nu}`	$R_{\mu \nu} \equiv R^{\lambda}{ }_{\mu \lambda \nu}$	where $R_{\mu\nu} \equiv R^\lambda{}_{\mu\lambda\nu}$ and $R \equiv R^\mu{}_\mu$ have their usual forms,
1510.06699_F00933	50	1.000	`R \equiv R^{\mu}{ }_{\mu}`	$R \equiv R^{\mu}{ }_{\mu}$	where $R_{\mu\nu} \equiv R^\lambda{}_{\mu\lambda\nu}$ and $R \equiv R^\mu{}_\mu$ have their usual forms,
1510.06699_F00934	50	1.000	`\Gamma^{\lambda}{ }_{\mu \nu}`	$\Gamma^{\lambda}{ }_{\mu \nu}$	but one must remember that the affine connection $\Gamma^\lambda{}_{\mu\nu}$
1510.06699_F00935	51	1.000	`\tau_{\mu \nu}=`	$\tau_{\mu \nu}=$	is not, in general, symmetric in μ and ν. Also, $\tau_{\mu\nu} =$
1510.06699_F00936	51	1.000	`e^{a}{ }_{\mu} e^{b}{ }_{\nu} \tau_{a b}`	$e^{a}{ }_{\mu} e^{b}{ }_{\nu} \tau_{a b}$	$e^a{}_\mu e^b{}_\nu \tau_{ab}$ and $\sigma_{\mu\nu}{}^\lambda = e^a{}_\mu e^b{}_\nu e_c{}^\lambda \sigma_{ab}{}^c$, where the former
1510.06699_F00937	51	1.000	`\sigma_{\mu \nu}{ }^{\lambda}=e^{a}{ }_{\mu} e^{b}{ }_{\nu} e_{c}{ }^{\lambda} \sigma_{a b}{ }^{c}`	$\sigma_{\mu \nu}{ }^{\lambda}=e^{a}{ }_{\mu} e^{b}{ }_{\nu} e_{c}{ }^{\lambda} \sigma_{a b}{ }^{c}$	$e^a{}_\mu e^b{}_\nu \tau_{ab}$ and $\sigma_{\mu\nu}{}^\lambda = e^a{}_\mu e^b{}_\nu e_c{}^\lambda \sigma_{ab}{}^c$, where the former
1510.06699_F00938	51	1.000	`\sigma_{\mu \nu}{ }^{\lambda}`	$\sigma_{\mu \nu}{ }^{\lambda}$	angular-momentum $\sigma_{\mu\nu}{}^\lambda$ of the matter field vanishes,
1510.06699_F00939	51	1.000	`T^{\lambda}{ }_{\mu \nu}`	$T^{\lambda}{ }_{\mu \nu}$	then so too does the torsion $T^\lambda{}_{\mu\nu}$ and hence the affine
1510.06699_F00940	51	1.000	`R_{\mu \nu}`	$R_{\mu \nu}$	case, $R_{\mu\nu}$ reduces to the standard symmetric Ricci ten-

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1510.06699_F00941	51	1.000	`{ }^{0} R_{\mu \nu} \equiv R_{\mu \nu}\left(V_{4}\right)`	${ }^{0} R_{\mu \nu} \equiv R_{\mu \nu}\left(V_{4}\right)$	sor ${}^{0}R_{\mu\nu} \equiv R_{\mu\nu}(V_4)$ in a Riemann spacetime, expressed
1510.06699_F00942	51	0.979	`{ }^{\underline{108}}`	${ }^{\underline{108}}$	grangian suggested by Dirac has the simple form[108]
1510.06699_F00943	51	1.000	`-\frac{1}{4}`	$-\frac{1}{4}$	sionless constant (usually positive); the factor of $-\frac{1}{4}$ is
1510.06699_F00944	51	1.000	`\phi^{4}`	ϕ^{4}	ϕ^4, and a term proportional to $\phi^2\mathcal{R}$ that non-minimally
1510.06699_F00945	51	1.000	`L_{\varphi}(\varphi, \partial \varphi, h, A, B, \phi)`	$L_{\varphi}(\varphi, \partial \varphi, h, A, B, \phi)$	for the matter field φ are $L_\varphi(\varphi, \partial\varphi, h, A, B, \phi)$; note that
1510.06699_F00946	51	1.000	`\mathcal{L}_{T}=h^{-1} L_{T}`	$\mathcal{L}_{T}=h^{-1} L_{T}$	The total Lagrangian density is then $\mathcal{L}_T = h^{-1}L_T$,
1510.06699_F00947	51	1.000	`L_{T}`	L_{T}	where L_T has the form
1510.06699_F00948	51	0.993	`(a, \xi, \nu, \lambda)`	(a, ξ, ν, λ)	(a, ξ, ν, λ) are dimensionless (and usually positive). In
1510.06699_F00949	51	1.000	`\phi^{-2}`	ϕ^{-2}	plied ϕ^{-2} should replace Newton's gravitational constant
1510.06699_F00950	51	0.967	`h_{a}{ }^{\mu}, A^{a b}{ }_{\mu}, B_{\mu}`	$h_{a}{ }^{\mu}, A^{a b}{ }_{\mu}, B_{\mu}$	independent fields $h_a{}^\mu$, $A^{ab}{}_\mu$, B_μ and ϕ respectively[50],
1510.06699_F00951	51	1.000	`B_{\mu}, \phi`	B_{μ}, ϕ	with respect to B_μ, ϕ and the spacetime metric $g_{\mu\nu}$. Our
1510.06699_F00952	52	1.000	`\left(\tau_{\varphi}\right)^{a}{ }_{b}`	$\left(\tau_{\varphi}\right)^{a}{ }_{b}$	where $(\tau_\varphi)^a{}_b$ is the energy-momentum tensor of the mat-
1510.06699_F00953	52	1.000	`\left(\sigma_{\varphi}\right)_{a b}{ }^{c}`	$\left(\sigma_{\varphi}\right)_{a b}{ }^{c}$	where $(\sigma_\varphi)_{ab}{}^c$ is the spin-angular-momentum tensor of
1510.06699_F00954	52	1.000	`\mathcal{D}_{a}^{*} \rightarrow \mathcal{D}_{a}, \mathcal{T}^{* c}{ }_{a b} \rightarrow \mathcal{T}^{c}{ }_{a b}`	$\mathcal{D}_{a}^{*} \rightarrow \mathcal{D}_{a}, \mathcal{T}^{* c}{ }_{a b} \rightarrow \mathcal{T}^{c}{ }_{a b}$	if one makes the replacements $\mathcal{D}^*_a \rightarrow \mathcal{D}_a$, $\mathcal{T}^{*c}{}_{ab} \rightarrow \mathcal{T}^c{}_{ab}$
1510.06699_F00955	52	1.000	`\mathcal{T}_{a}^{*} \rightarrow \mathcal{T}_{a}`	$\mathcal{T}_{a}^{*} \rightarrow \mathcal{T}_{a}$	and $\mathcal{T}^*_a \rightarrow \mathcal{T}_a$ to PGT quantities, although the resulting
1510.06699_F00956	52	1.000	`\left(\zeta_{\varphi}\right)^{c}`	$\left(\zeta_{\varphi}\right)^{c}$	where $(\zeta_\varphi)^c$ is the dilation current of the φ matter field.
1510.06699_F00957	52	1.000	`\left.\mathcal{D}_{a}^{*} \phi\right|_{\phi=\phi_{0}}=-\phi_{0} \mathcal{B}_{a}`	$\left.\mathcal{D}_{a}^{*} \phi\right\vert_{\phi=\phi_{0}}=-\phi_{0} \mathcal{B}_{a}$	replacement $\mathcal{D}^*_a\phi\vert_{\phi=\phi_0} = -\phi_0\mathcal{B}_a$ throughout. It is also
1510.06699_F00958	52	1.000	`a=1 /\left(\kappa \phi_{0}^{2}\right)`	$a=1 /\left(\kappa \phi_{0}^{2}\right)$	convenient to set $a = 1/(\kappa\phi_0^2)$, where κ is Einstein's grav-
1510.06699_F00959	52	1.000	`\Lambda=\lambda \phi_{0}^{2} / a=\kappa \lambda`	$\Lambda=\lambda \phi_{0}^{2} / a=\kappa \lambda$	where $\Lambda = \lambda\phi_0^2/a = \kappa\lambda$ has the correct dimensions of
1510.06699_F00960	52	1.000	`m^{2}=\nu \phi_{0}^{2} / \xi`	$m^{2}=\nu \phi_{0}^{2} / \xi$	vector field (or Proca field) of mass-squared $m^2 = \nu\phi_0^2/\xi$.
1510.06699_F00961	53	1.000	`\mathcal{R}^{a b}{ }_{c d}, \mathcal{T}^{a}{ }_{b c}=\mathcal{T}^{* a}{ }_{b c}-\delta_{c}^{a} \mathcal{B}_{b}+\delta_{b}^{a} \mathcal{B}_{c}`	$\mathcal{R}^{a b}{ }_{c d}, \mathcal{T}^{a}{ }_{b c}=\mathcal{T}^{* a}{ }_{b c}-\delta_{c}^{a} \mathcal{B}_{b}+\delta_{b}^{a} \mathcal{B}_{c}$	tensors $\mathcal{R}^{ab}{}_{cd}$, $\mathcal{T}^a{}_{bc} = \mathcal{T}^{*a}{}_{bc} - \delta^a_c\mathcal{B}_b + \delta^a_b\mathcal{B}_c$ and $\mathcal{H}_{ab}$
1510.06699_F00962	53	1.000	`\left(\zeta_{\psi}\right)^{\mu}=0`	$\left(\zeta_{\psi}\right)^{\mu}=0$	does not appear in (D12), the dilation current $(\zeta_\psi)^\mu = 0$
1510.06699_F00963	53	1.000	`\mathcal{T}_{a}=0`	$\mathcal{T}_{a}=0$	theory, from (C10), we again have $\mathcal{T}_a = 0$. Thus, the
1510.06699_F00964	53	1.000	`\tau_{\psi}=h^{-1} \mu \phi \bar{\psi} \psi`	$\tau_{\psi}=h^{-1} \mu \phi \bar{\psi} \psi$	of the Dirac field is easily found to be $\tau_\psi = h^{-1}\mu\phi\bar{\psi}\psi$.
1510.06699_F00965	53	1.000	`m^{2} \equiv \nu \phi_{0}^{2} / \xi`	$m^{2} \equiv \nu \phi_{0}^{2} / \xi$	where we have defined $m^2 \equiv \nu\phi_0^2/\xi$ and $\sigma_{ab}{}^c$ is the spin-
1510.06699_F00966	53	1.000	`\mathcal{B}_{a}`	$\mathcal{B}_{a}$	field $\mathcal{B}_a$ has a non-minimal coupling to the gravitational
1510.06699_F00967	53	1.000	`D_{\mu}^{*}=D_{\mu}+w B_{\mu}`	$D_{\mu}^{*}=D_{\mu}+w B_{\mu}$	variant derivative (20), namely $D^*_\mu = D_\mu + wB_\mu$, the
1510.06699_F00968	54	1.000	`P_{\mu} \equiv \partial_{\mu} \rho`	$P_{\mu} \equiv \partial_{\mu} \rho$	where $P_\mu \equiv \partial_\mu\rho$. This transformation for V_μ implies
1510.06699_F00969	54	1.000	`(w=0)`	$(w=0)$	variant $(w = 0)$ under extended local dilations. One
1510.06699_F00970	54	1.000	`\mathcal{R}^{\dagger a b}{ }_{c d} \equiv h_{c}{ }^{\mu} h_{d}{ }^{\nu} R^{\dagger a b}{ }_{\mu \nu}`	$\mathcal{R}^{\dagger a b}{ }_{c d} \equiv h_{c}{ }^{\mu} h_{d}{ }^{\nu} R^{\dagger a b}{ }_{\mu \nu}$	can then define $\mathcal{R}^{\dagger ab}{}_{cd} \equiv h_c{}^\mu h_d{}^\nu R^{\dagger ab}{}_{\mu\nu}$ as before, which
1510.06699_F00971	54	0.997	`D_{\mu}^{\natural} \varphi`	$D_{\mu}^{\natural} \varphi$	ever, one quickly finds that $D^\natural_\mu\varphi$ does *not* transform co-
1510.06699_F00972	54	1.000	`D_{\mu}^{\natural}`	$D_{\mu}^{\natural}$	ing our reluctance thus far to call $D^\natural_\mu$ (or $\mathcal{D}^\natural_a$) a covariant
1510.06699_F00973	54	1.000	`\nu=\beta_{1}=\beta_{2}=0`	$\nu=\beta_{1}=\beta_{2}=0$	gravitational Lagrangian L_G and $\nu = \beta_1 = \beta_2 = 0$ in the
1510.06699_F00974	54	0.994	`\xi=\nu=\beta_{1}=\beta_{2}=0`	$\xi=\nu=\beta_{1}=\beta_{2}=0$	in Section IV A, but with $\xi = \nu = \beta_1 = \beta_2 = 0$. These
1510.06699_F00975	54	1.000	`\mathcal{D}_{a}^{\natural} \chi`	$\mathcal{D}_{a}^{\natural} \chi$	to $\mathcal{D}^\natural_a\chi$ to assemble the appropriate *covariant* derivative
1510.06699_F00976	54	1.000	`\mathcal{D}_{a}^{\dagger} \chi=h_{a}{ }^{\mu} D_{\mu}^{\dagger} \chi`	$\mathcal{D}_{a}^{\dagger} \chi=h_{a}{ }^{\mu} D_{\mu}^{\dagger} \chi$	$\mathcal{D}^\dagger_a\chi$. In each case, one finds that $\mathcal{D}^\dagger_a\chi = h_a{}^\mu D^\dagger_\mu\chi$, where
1510.06699_F00977	54	0.991	`D_{\mu}^{\dagger}`	$D_{\mu}^{\dagger}$	$D^\dagger_\mu$ has the form (143), but with the appropriate value of
1510.06699_F00978	54	0.997	`\left(U_{4}\right)`	$\left(U_{4}\right)$	respectively, of a Riemann–Cartan (U_4) spacetime.
1510.06699_F00979	54	1.000	`\delta \varphi`	$\delta \varphi$	given action, for example, then the 'total' variation $\delta\varphi$ and the

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1510.06699_F00980	54	■ 1.000	`\left[\mathcal{D}_{c}, \mathcal{D}_{d}\right] \varphi \equiv \frac{1}{2} \mathcal{R}^{a b}{ }_{c d} \Sigma_{c d} \varphi-\mathcal{T}^{a}{ }_{c d} \mathcal{D}_{a} \varphi, \mathcal{R}^{a}{ }_{b} \equiv \mathcal{R}^{a c}{ }_{b c}`	$\left[\mathcal{D}_{c}, \mathcal{D}_{d}\right] \varphi \equiv \frac{1}{2} \mathcal{R}^{a b}{ }_{c d} \Sigma_{c d} \varphi-\mathcal{T}^{a}{ }_{c d} \mathcal{D}_{a} \varphi, \mathcal{R}^{a}{ }_{b} \equiv \mathcal{R}^{a c}{ }_{b c}$	signature, $[\mathcal{D}_c, \mathcal{D}_d]\varphi \equiv \frac{1}{2}\mathcal{R}^{ab}{}_{cd}\Sigma_{cd}\varphi - \mathcal{T}^a{}_{cd}\mathcal{D}_a\varphi$, $\mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}$
1510.06699_F00981	54	■ 1.000	`\mathcal{T}_{a} \equiv \mathcal{T}^{b}{ }_{a b}`	$\mathcal{T}_{a} \equiv \mathcal{T}^{b}{ }_{a b}$	and $\mathcal{T}_a \equiv \mathcal{T}^b{}_{ab}$.
1510.06699_F00982	55	■ 1.000	`\widehat{\mathcal{D}}_{a}^{*}`	$\widehat{\mathcal{D}}_{a}^{*}$	(Weyl weight $w = 0$) versions of the quantities listed below are denoted with a caret, for example $\widehat{h}_a{}^\mu$ or $\widehat{\mathcal{D}}^*_a$ or $\widehat{\mathcal{R}}^{ab}{}_{cd}$; see
1510.06699_F00983	55	■ 1.000	`\widehat{\mathcal{R}}^{a b}{ }_{c d}`	$\widehat{\mathcal{R}}^{a b}{ }_{c d}$	(Weyl weight $w = 0$) versions of the quantities listed below are denoted with a caret, for example $\widehat{h}_a{}^\mu$ or $\widehat{\mathcal{D}}^*_a$ or $\widehat{\mathcal{R}}^{ab}{}_{cd}$; see
1510.06699_F00984 **[conf 0.000]**	55	■ 0.000	`A^{\dagger a b}{ }_{\mu} \equiv A^{a b}{ }_{\mu}+2 \eta^{c[a} b^{b]}{ }_{\mu} h_{c}{ }^{\nu} V_{\nu}`	$A^{\dagger a b}{ }_{\mu} \equiv A^{a b}{ }_{\mu}+2 \eta^{c[a} b^{b]}{ }_{\mu} h_{c}{ }^{\nu} V_{\nu}$	*Gravitational gauge field variables* $h_a{}^\mu$ translational gauge field (h-field) h determinant of $h_a{}^\mu$ $b^a{}_\mu$ inverse h-field $A^{ab}{}_\mu$ rotational gauge field (A-field) B_μ WGT dilational gauge field (B-field) V_μ eWGT dilational gauge field (V-field) $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ eWGT extended rotational gauge field ($A^\dagger$-field) *Derivative operators* $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ PGT (generalised) covariant derivative[a] $\partial^*_\mu \equiv \partial_\mu + wB_\mu$ WGT 'augmented' partial derivative $\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ WGT (generalised) covariant derivative[a] $\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}\mathcal{T}_\mu)$ eWGT 'augmented' partial derivative $\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ eWGT (generalised) covariant derivative[a] $\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ eWGT (generalised) semi-covariant derivative[a] *Gauge field strengths* $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$ PGT and WGT rotational gauge field strength[a,b] $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$ eWGT rotational gauge field strength[a,b] $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$ PGT translational gauge field strength[b] $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$ WGT translational gauge field strength[b] $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$ eWGT translational gauge field strength[b] $\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$ eWGT translational gauge 'semi' field strength[a,b] $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$ WGT dilational gauge field strength $\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu + \frac{1}{3}\mathcal{T}_{\nu]})$ eWGT dilational gauge field strength *Reduced A-fields and related quantities* $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$ PGT Ricci rotation coefficients $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$ WGT Ricci rotation coefficients $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^\dagger_{[\mu}b^a{}_{\nu]}$ eWGT Ricci rotation coefficients ${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$ WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$ eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) *Currents and related quantities* L Lagrangian $\mathcal{L} \equiv h^{-1}L$ Lagrangian density $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$ gravitational sector energy-momentum tensor[c] $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$ gravitational sector spin-angular-momentum tensor[c] $j^a \equiv j^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_G/\delta B_\mu$ gravitational sector dilation current[c] $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$ gravitational sector eWGT covariant energy-momentum tensor[c] $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ gravitational sector eWGT covariant dilation current[c] $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$ matter sector energy-momentum tensor[c] $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$ matter sector spin-angular-momentum tensor[c] $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$ matter sector dilation current[c] $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$ gravitational sector eWGT covariant energy-momentum tensor[c] $\zeta^{\dagger a} \equiv \zeta^a - 2\sigma^{ab}{}_b$ gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00985 **[conf 0.000]**	55	■ 0.000	`\mathcal{D}_{a} \equiv h_{a}{ }^{\mu} D_{\mu} \equiv h_{a}{ }^{\mu}\left(\partial_{\mu}+\frac{1}{2} A^{a b}{ }_{\mu} \Sigma_{a b}\right)`	$\mathcal{D}_{a} \equiv h_{a}{ }^{\mu} D_{\mu} \equiv h_{a}{ }^{\mu}\left(\partial_{\mu}+\frac{1}{2} A^{a b}{ }_{\mu} \Sigma_{a b}\right)$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{*ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^*-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu\delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b = \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00986 **[conf 0.000]**	55	■0.000	`\partial_{\mu}^{*} \equiv \partial_{\mu}+w B_{\mu}`	$\partial_{\mu}^{*} \equiv \partial_{\mu}+w B_{\mu}$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^+_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^+_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^+_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_{\nu]} + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{eb}{}^a\mathcal{V}^e - s^{ae}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} = j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b = \tau^a{}_b + 2(\sigma_{eb}{}^a\mathcal{V}^e - \sigma^{ae}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00987 **[conf 0.000]**	55	■ 0.000	`\mathcal{D}_{a}^{*} \equiv h_{a}{ }^{\mu} D_{\mu}^{*} \equiv h_{a}{ }^{\mu}\left(\partial_{\mu}^{*}+\frac{1}{2} A^{a b}{ }_{\mu} \Sigma_{a b}\right)`	$\mathcal{D}_{a}^{*} \equiv h_{a}{ }^{\mu} D_{\mu}^{*} \equiv h_{a}{ }^{\mu}\left(\partial_{\mu}^{*}+\frac{1}{2} A^{a b}{ }_{\mu} \Sigma_{a b}\right)$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}\mathcal{T}_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_\nu + \frac{1}{3}\mathcal{T}_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00988 **[conf 0.000]**	55	■ 0.000	`\partial_{\mu}^{\dagger} \equiv \partial_{\mu}-w\left(V_{\mu}+\frac{1}{3} T_{\mu}\right)`	$\partial_{\mu}^{\dagger} \equiv \partial_{\mu}-w\left(V_{\mu}+\frac{1}{3} T_{\mu}\right)$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a} b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\dagger$-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial_\mu^* \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}_a^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu(\partial_\mu^* + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu(\partial_\mu^\dagger + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}_a^\natural \equiv h_a{}^\mu D_\mu^\natural \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu} A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a],[b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu} A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a],[b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a],[b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_{\nu]} + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a \equiv j^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00989 **[conf 0.000]**	55	■ 0.000	`\mathcal{D}_{a}^{\dagger} \equiv h_{a}{ }^{\mu} D_{\mu}^{\dagger} \equiv h_{a}{ }^{\mu}\left(\partial_{\mu}^{\dagger}+\frac{1}{2} A^{\dagger a b}{ }_{\mu} \Sigma_{a b}\right)`	$\mathcal{D}_{a}^{\dagger} \equiv h_{a}{ }^{\mu} D_{\mu}^{\dagger} \equiv h_{a}{ }^{\mu}\left(\partial_{\mu}^{\dagger}+\frac{1}{2} A^{\dagger a b}{ }_{\mu} \Sigma_{a b}\right)$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial_\mu^* \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}_a^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu(\partial_\mu^* + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial_\mu^\dagger \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu(\partial_\mu^\dagger + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}_a^\natural \equiv h_a{}^\mu D_\mu^\natural \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu} A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu} A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^+_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} = j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00990 **[conf 0.000]**	55	■ 0.000	`\mathcal{D}_{a}^{\natural} \equiv h_{a}{ }^{\mu} D_{\mu}^{\natural} \equiv h_{a}{ }^{\mu}\left(\partial_{\mu}+\frac{1}{2} A^{\dagger a b}{ }_{\mu} \Sigma_{a b}\right)`	$\mathcal{D}_{a}^{\natural} \equiv h_{a}{ }^{\mu} D_{\mu}^{\natural} \equiv h_{a}{ }^{\mu}\left(\partial_{\mu}+\frac{1}{2} A^{\dagger a b}{ }_{\mu} \Sigma_{a b}\right)$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a \equiv j^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00991 **[conf 0.000]**	55	■ 0.000	`{ }^{\mathrm{a}}`	a	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\dagger$-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu} A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu} A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_{\nu]} + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{eb}{}^a \mathcal{V}^e - s^{ae}{}_c \mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{eb}{}^a \mathcal{V}^e - \sigma^{ae}{}_c \mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_ F00992 **[conf 0.000]**	55	■ 0.000	\mathcal{R}^{a b}{ }_{c d}=\mathcal{R}^{* a b}{ }_{c d}=2 h_{a}{ }^{\mu} h_{b}{ }^{\nu}\left(\partial_{[\mu} A^{a b}{ }_{\nu]}+A^{a}{ }_{c[\mu} A^{c b}{ }_{\nu]}\right) \mathcal{R}^{\dagger a b}{ }_{c d}=2 h_{a}{ }^{\mu} h_{b}{ }^{\nu}\left(\partial_{[\mu} A^{\dagger a b}{ }_{\nu]}+A^{\dagger a}{ }_{c[\mu} A^{\dagger c b}{ }_{\nu]}\right) \mathcal{T}^{a}{ }_{b c}=2 h_{b}{ }^{\mu} h_{c}{ }^{\nu} D_{[\mu} b^{a}{ }_{\nu]} \mathcal{T}^{* a}{ }_{b c}=2 h_{b}{ }^{\mu} h_{c}{ }^{\nu} D_{[\mu}^{*} b^{a}{ }_{\nu]} \mathcal{T}^{\dagger a}{ }_{b c}=2 h_{b}{ }^{\mu} h_{c}{ }^{\nu} D_{[\mu}^{\dagger} b^{a}{ }_{\nu]} \mathcal{T}^{\natural a}{ }_{b c}=2 h_{b}{ }^{\mu} h_{c}{ }^{\nu} D_{[\mu}^{\natural} b^{a}{ }_{\nu]} \mathcal{H}_{a b}=2 h_{a}{ }^{\mu} h_{b}{ }^{\nu} \partial_{[\mu} B_{\nu]} \mathcal{H}_{a b}^{\dagger}=2 h_{a}{ }^{\mu} h_{b}{ }^{\nu} \partial_{[\mu}\left(V_{\nu]}+\frac{1}{3} T_{\nu]}\right)	$\mathcal{R}^{a b}{ }_{c d}=\mathcal{R}^{* a b}{ }_{c d}=2 h_{a}{ }^{\mu} h_{b}{ }^{\nu}\left(\partial_{[\mu} A^{a b}{ }_{\nu]}+A^{a}{ }_{c[\mu} A^{c b}{ }_{\nu]}\right) \mathcal{R}^{\dagger a b}{ }_{c d}=2 h_{a}{ }^{\mu} h_{b}{ }^{\nu}\left(\partial_{[\mu} A^{\dagger a b}{ }_{\nu]}+A^{\dagger a}{ }_{c[\mu} A^{\dagger c b}{ }_{\nu]}\right) \mathcal{T}^{a}{ }_{b c}=2 h_{b}{ }^{\mu} h_{c}{ }^{\nu} D_{[\mu} b^{a}{ }_{\nu]} \mathcal{T}^{* a}{ }_{b c}=2 h_{b}{ }^{\mu} h_{c}{ }^{\nu} D_{[\mu}^{*} b^{a}{ }_{\nu]} \mathcal{T}^{\dagger a}{ }_{b c}=2 h_{b}{ }^{\mu} h_{c}{ }^{\nu} D_{[\mu}^{\dagger} b^{a}{ }_{\nu]} \mathcal{T}^{\natural a}{ }_{b c}=2 h_{b}{ }^{\mu} h_{c}{ }^{\nu} D_{[\mu}^{\natural} b^{a}{ }_{\nu]} \mathcal{H}_{a b}=2 h_{a}{ }^{\mu} h_{b}{ }^{\nu} \partial_{[\mu} B_{\nu]} \mathcal{H}_{a b}^{\dagger}=2 h_{a}{ }^{\mu} h_{b}{ }^{\nu} \partial_{[\mu}\left(V_{\nu]}+\frac{1}{3} T_{\nu]}\right)$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a} b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^+_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^+_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu} A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a],[b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu} A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a],[b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a],[b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^+_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_{\nu]} + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{eb}{}^a \mathcal{V}^e - s^{ae}{}_c \mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} = j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{eb}{}^a \mathcal{V}^e - \sigma^{ae}{}_c \mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00993 **[conf 0.000]**	55	■0.000	`{ }^{\mathrm{a}}{ }^{\mathrm{b}}`	ab	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\dagger$-field)
Derivative operators	
$\mathcal{D}_a = h_a{}^\mu D_\mu = h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_{\nu]} + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2\sigma^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00994 **[conf 0.000]**	55	■0.000	`{ }^{\text {b }}`	b	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\dagger$-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}\mathcal{T}_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_{\nu]} + \frac{1}{3}\mathcal{T}_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a \equiv j^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00995 **[conf 0.000]**	55	■ 0.000	`{ }^{\mathrm{b}}`	b	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\sim ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\sim$-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_{\nu]} + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu\delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b = \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2\sigma^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00996 **[conf 0.000]**	55	■ 0.000	c^{a}{ }_{b c} \equiv 2 h_{b}{ }^{\mu} h_{c}{ }^{\nu} \partial_{[\mu} b^{a}{ }_{\nu]}	$c^{a}{ }_{b c} \equiv 2 h_{b}{ }^{\mu} h_{c}{ }^{\nu} \partial_{[\mu} b^{a}{ }_{\nu]}$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a} b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^+_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}\mathcal{T}_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^+_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu} A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a],[b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu} A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a],[b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a],[b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^+_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_\nu + \frac{1}{3}\mathcal{T}_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{eb}{}^a \mathcal{V}^e - s^{ae}{}_e \mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{eb}{}^a \mathcal{V}^e - \sigma^{ae}{}_e \mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00997 **[conf 0.000]**	55	■ 0.000	`c^{* a}{ }_{b c} \equiv 2 h_{b}{ }^{\mu} h_{c}{ }^{\nu} \partial_{[\mu}^{*} b^{a}{ }_{\nu]}`	$c^{*a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial^{*}_{[\mu} b^a{}_{\nu]}$	

Gravitational gauge field variables	
$h_a{}^{\mu}$	translational gauge field (h-field)
h	determinant of $h_a{}^{\mu}$
$b^a{}_{\mu}$	inverse h-field
$A^{ab}{}_{\mu}$	rotational gauge field (A-field)
B_{μ}	WGT dilational gauge field (B-field)
V_{μ}	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_{\mu} \equiv A^{ab}{}_{\mu} + 2\eta^{c[a} b^{b]}{}_{\mu} h_c{}^{\nu} V_{\nu}$	eWGT extended rotational gauge field ($A^{\dagger}$-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^{\mu} D_{\mu} \equiv h_a{}^{\mu}(\partial_{\mu} + \frac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^{*}_{\mu} \equiv \partial_{\mu} + wB_{\mu}$	WGT 'augmented' partial derivative
$\mathcal{D}^{*}_a \equiv h_a{}^{\mu} D^{*}_{\mu} \equiv h_a{}^{\mu}(\partial^{*}_{\mu} + \frac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^{\dagger}_{\mu} \equiv \partial_{\mu} - w(V_{\mu} + \frac{1}{3}T_{\mu})$	eWGT 'augmented' partial derivative
$\mathcal{D}^{\dagger}_a \equiv h_a{}^{\mu} D^{\dagger}_{\mu} \equiv h_a{}^{\mu}(\partial^{\dagger}_{\mu} + \frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^{\natural}_a \equiv h_a{}^{\mu} D^{\natural}_{\mu} \equiv h_a{}^{\mu}(\partial_{\mu} + \frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu}(\partial_{[\mu} A^{ab}{}_{\nu]} + A^{a}{}_{e[\mu} A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu}(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu} A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^{a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D^{*}_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D^{\dagger}_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D^{\natural}_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^{\dagger}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu}(V_{\nu} + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial^{*}_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial^{\dagger}_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^{0}A^{ab}{}_{\mu} \equiv \frac{1}{2}b^c{}_{\mu}(c^{ab}{}_c + c^{b}{}_c{}^{a} - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^{0}A^{*ab}{}_{\mu} \equiv \frac{1}{2}b^c{}_{\mu}(c^{*ab}{}_c + c^{*b}{}_c{}^{a} - c_c^{*ab})$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^{0}A^{\dagger ab}{}_{\mu} \equiv \frac{1}{2}b^c{}_{\mu}(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^{a} - c_c^{\dagger ab})$	eWGT reduced $A^{\dagger}$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu}\delta\mathcal{L}_G/\delta h_a{}^{\mu}$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^{c} \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu}\delta\mathcal{L}_G/\delta A^{ab}{}_{\mu}$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^{\mu} b^a{}_{\mu} = b^a{}_{\mu}\delta\mathcal{L}_G/\delta B_{\mu}$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{eb}{}^{a}\mathcal{V}^e - s^{ae}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu}\delta\mathcal{L}_M/\delta h_a{}^{\mu}$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^{c} \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu}\delta\mathcal{L}_M/\delta A^{ab}{}_{\mu}$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu}\delta\mathcal{L}_M/\delta B_{\mu}$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{eb}{}^{a}\mathcal{V}^e - \sigma^{ae}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2\sigma^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00998 **[conf 0.000]**	55	■0.000	`c^{\dagger a}{ }_{b c} \equiv 2 h_{b}{ }^{\mu} h_{c}{ }^{\nu} \partial_{[\mu}^{\dagger} b^{a}{ }_{\nu]}`	$c^{\dagger a}{ }_{b c} \equiv 2 h_{b}{ }^{\mu} h_{c}{ }^{\nu} \partial_{[\mu}^{\dagger} b^{a}{ }_{\nu]}$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\dagger$-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}\mathcal{T}_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu} A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a],[b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu} A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a],[b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a],[b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_\nu + \frac{1}{3}\mathcal{T}_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a \equiv j^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{eb}{}^a\mathcal{V}^e - s^{ae}{}_e\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{eb}{}^a\mathcal{V}^e - \sigma^{ae}{}_e\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F00999 **[conf 0.000]**	55	■ 0.000	`{ }^{0} A^{a b}{ }_{\mu} \equiv \frac{1}{2} b^{c}{ }_{\mu}\left(c^{a b}{ }_{c}+c^{b}{ }_{c}{ }^{a}-c_{c}{ }^{a b}\right)`	${ }^{0} A^{a b}{ }_{\mu} \equiv \frac{1}{2} b^{c}{ }_{\mu}\left(c^{a b}{ }_{c}+c^{b}{ }_{c}{ }^{a}-c_{c}{ }^{a b}\right)$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\dagger$-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu} A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu} A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} = j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01000 **[conf 0.000]**	55	■ 0.000	`{ }^{0} A^{* a b}{ }_{\mu} \equiv \frac{1}{2} b^{c}{ }_{\mu}\left(c^{* a b}{ }_{c}+c^{* b}{ }_{c}{ }^{a}-c_{c}^{* a b}\right)`	${ }^{0} A^{* a b}{ }_{\mu} \equiv \frac{1}{2} b^{c}{ }_{\mu}\left(c^{* a b}{ }_{c}+c^{* b}{ }_{c}{ }^{a}-c_{c}^{* a b}\right)$	(see table below)

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^+_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^+_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^+_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu\delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} = j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01001 **[conf 0.000]**	55	■ 0.000	`{ }^{0} A^{\dagger a b}{ }_{\mu} \equiv \frac{1}{2} b^{c}{ }_{\mu}\left(c^{\dagger a b}{ }_{c}+c^{\dagger b}{ }_{c}{ }^{a}-c_{c}^{\dagger a b}\right)`	${ }^{0} A^{\dagger a b}{ }_{\mu} \equiv \frac{1}{2} b^{c}{ }_{\mu}\left(c^{\dagger a b}{ }_{c}+c^{\dagger b}{ }_{c}{ }^{a}-c_{c}^{\dagger a b}\right)$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\dagger$-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu\delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{eb}{}^a\mathcal{V}^e - s^{ae}{}_e\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{eb}{}^a\mathcal{V}^e - \sigma^{ae}{}_e\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01002 **[conf 0.000]**	55	■ 0.000	`\mathcal{T}^{a}{ }_{b c} \equiv 0`	$\mathcal{T}^{a}{ }_{b c} \equiv 0$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^+_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^+_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^+_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu\delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{eb}{}^a\mathcal{V}^e - s^{ae}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{eb}{}^a\mathcal{V}^e - \sigma^{ae}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01003 **[conf 0.000]**	55	■ 0.000	`\mathcal{T}^{* a}{ }_{b c} \equiv 0`	$\mathcal{T}^{*a}{}_{bc} \equiv 0$	(see below)

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_{\nu]} + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu\delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} = j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01004 **[conf 0.000]**	55	■ 0.000	`\mathcal{T}^{\dagger a}{ }_{b c} \equiv 0`	$\mathcal{T}^{\dagger a}{ }_{b c} \equiv 0$	*Gravitational gauge field variables* $h_a{}^\mu$ translational gauge field (h-field) h determinant of $h_a{}^\mu$ $b^a{}_\mu$ inverse h-field $A^{ab}{}_\mu$ rotational gauge field (A-field) B_μ WGT dilational gauge field (B-field) V_μ eWGT dilational gauge field (V-field) $A^{*ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ eWGT extended rotational gauge field (A^*-field) *Derivative operators* $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ PGT (generalised) covariant derivative[a] $\partial^*_\mu \equiv \partial_\mu + wB_\mu$ WGT 'augmented' partial derivative $\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ WGT (generalised) covariant derivative[a] $\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$ eWGT 'augmented' partial derivative $\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ eWGT (generalised) covariant derivative[a] $\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ eWGT (generalised) semi-covariant derivative[a] *Gauge field strengths* $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$ PGT and WGT rotational gauge field strength[a,b] $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$ eWGT rotational gauge field strength[a,b] $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$ PGT translational gauge field strength[b] $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$ WGT translational gauge field strength[b] $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$ eWGT translational gauge field strength[b] $\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$ eWGT translational gauge 'semi' field strength[a,b] $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$ WGT dilational gauge field strength $\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_{\nu} + \frac{1}{3}T_{\nu]})$ eWGT dilational gauge field strength *Reduced A-fields and related quantities* $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$ PGT Ricci rotation coefficients $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$ WGT Ricci rotation coefficients $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^\dagger_{[\mu}b^a{}_{\nu]}$ eWGT Ricci rotation coefficients ${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$ WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$ eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) *Currents and related quantities* L Lagrangian $\mathcal{L} \equiv h^{-1}L$ Lagrangian density $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$ gravitational sector energy-momentum tensor[c] $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$ gravitational sector spin-angular-momentum tensor[c] $j^a = j^\mu b^a{}_\mu = b^a{}_\mu\delta\mathcal{L}_G/\delta B_\mu$ gravitational sector dilation current[c] $t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$ gravitational sector eWGT covariant energy-momentum tensor[c] $j^{\dagger a} = j^a - 2s^{ab}{}_b$ gravitational sector eWGT covariant dilation current[c] $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$ matter sector energy-momentum tensor[c] $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$ matter sector spin-angular-momentum tensor[c] $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$ matter sector dilation current[c] $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$ gravitational sector eWGT covariant energy-momentum tensor[c] $\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$ gravitational sector eWGT covariant dilation current[c]

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Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a],[b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a],[b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a],[b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu\delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01006 **[conf 0.000]**	55	■0.000	\mathcal{L} \equiv h^{-1} L	$\mathcal{L} \equiv h^{-1}L$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\dagger$-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_{\nu]} + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01007 **[conf 0.000]**	55	■ 0.000	`t^{a}{ }_{b} \equiv t^{a}{ }_{\mu} h_{b}{ }^{\mu} \equiv h_{b}{ }^{\mu} \delta \mathcal{L}_{\mathrm{G}} / \delta h_{a}{ }^{\mu}`	$t^{a}{ }_{b} \equiv t^{a}{ }_{\mu} h_{b}{ }^{\mu} \equiv h_{b}{ }^{\mu} \delta \mathcal{L}_{\mathrm{G}} / \delta h_{a}{ }^{\mu}$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^+_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^+_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^+_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_{\nu]} + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a \equiv j^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{eb}{}^a\mathcal{V}^e - s^{ae}{}_e\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{eb}{}^a\mathcal{V}^e - \sigma^{ae}{}_e\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01008 **[conf 0.000]**	55	■ 0.000	`s_{a b}{ }^{c} \equiv s_{a b}{ }^{\mu} b^{c}{ }_{\mu} \equiv b^{c}{ }_{\mu} \delta \mathcal{L}_{\mathrm{G}} / \delta A^{a b}{ }_{\mu}`	$s_{ab}{}^{c} \equiv s_{ab}{}^{\mu} b^{c}{}_{\mu} \equiv b^{c}{}_{\mu} \delta \mathcal{L}_{\mathrm{G}} / \delta A^{ab}{}_{\mu}$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a} b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\dagger$-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu} A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu} A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{eb}{}^a \mathcal{V}^e - s^{ae}{}_e \mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} = j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{eb}{}^a \mathcal{V}^e - \sigma^{ae}{}_e \mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01009 **[conf 0.000]**	55	■ 0.000	`j^{a} \equiv j^{\mu} b^{a}{ }_{\mu} \equiv b^{a}{ }_{\mu} \delta \mathcal{L}_{\mathrm{G}} / \delta B_{\mu}`	$j^{a} \equiv j^{\mu} b^{a}{ }_{\mu} \equiv b^{a}{ }_{\mu} \delta \mathcal{L}_{\mathrm{G}} / \delta B_{\mu}$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a} b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\dagger$-field)
Derivative operators	
$\mathcal{D}_a = h_a{}^\mu D_\mu = h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu} A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu} A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_{\rm G}/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_{\rm G}/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_{\rm G}/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_{\rm M}/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_{\rm M}/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_{\rm M}/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01010 **[conf 0.000]**	55	■ 0.000	`t^{\dagger a}{ }_{b} \equiv t^{a}{ }_{b}+2\left(s_{c b}{ }^{a} \mathcal{V}^{c}-s^{a c}{ }_{c} \mathcal{V}_{b}\right)`	$t^{\dagger a}{ }_{b} \equiv t^{a}{ }_{b}+2\left(s_{c b}{ }^{a} \mathcal{V}^{c}-s^{a c}{ }_{c} \mathcal{V}_{b}\right)$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a = h_a{}^\mu D_\mu = h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^+_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^+_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_c{}^\mu h_d{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu} A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_c{}^\mu h_d{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu} A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^+_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2\sigma^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01011 **[conf 0.000]**	55	■ 0.000	`j^{\dagger a} \equiv j^{a}-2 s^{a b}{ }_{b}`	$j^{\dagger a} \equiv j^{a}-2 s^{a b}{ }_{b}$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^+_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^+_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a],[b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a],[b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a],[b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^+_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a \equiv j^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01012 **[conf 0.000]**	55	■ 0.000	`\tau^{a}{ }_{b} \equiv \tau^{a}{ }_{\mu} h_{b}{ }^{\mu} \equiv h_{b}{ }^{\mu} \delta \mathcal{L}_{\mathrm{M}} / \delta h_{a}{ }^{\mu}`	$\tau^{a}{ }_{b} \equiv \tau^{a}{ }_{\mu} h_{b}{ }^{\mu} \equiv h_{b}{ }^{\mu} \delta \mathcal{L}_{\mathrm{M}} / \delta h_{a}{ }^{\mu}$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^+_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^+_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^+_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_{\nu]} + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2\sigma^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01013 **[conf 0.000]**	55	■ 0.000	`\sigma_{a b}{ }^{c} \equiv \sigma_{a b}{ }^{\mu} b^{c}{ }_{\mu} \equiv b^{c}{ }_{\mu} \delta \mathcal{L}_{ \mathrm{M}} / \delta A^{a b}{ }_{\mu}`	$\sigma_{ab}{}^{c} \equiv \sigma_{ab}{}^{\mu} b^{c}{}_{\mu} \equiv b^{c}{}_{\mu} \delta \mathcal{L}_{\mathrm{M}} / \delta A^{ab}{}_{\mu}$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\dagger$-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu} A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu} A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} = j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b = \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01014 **[conf 0.000]**	55	■0.000	\zeta^{a} \equiv \zeta^{\mu} b^{a}{ }_{\mu} \equiv b^{a}{ }_{\mu} \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu}	$\zeta^{a} \equiv \zeta^{\mu} b^{a}{ }_{\mu} \equiv b^{a}{ }_{\mu} \delta \mathcal{L}_{\mathrm{M}} / \delta B_{\mu}$	(see table below)

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01015 **[conf 0.000]**	55	■ 0.000	\tau^{\dagger a}{ }_{b} \equiv \tau^{a}{ }_{b}+2\left(\sigma_{c b}{ }^{a} \mathcal{V}^{c}-\sigma^{a c}{ }_{c} \mathcal{V}_{b}\right)	$\tau^{\dagger a}{}_{b} \equiv \tau^{a}{}_{b}+2\left(\sigma_{cb}{}^{a} \mathcal{V}^{c}-\sigma^{ac}{}_{c} \mathcal{V}_{b}\right)$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{+ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field (A^+-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^+_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^+_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^+_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu\delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} = j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2\sigma^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01016 **[conf 0.000]**	55	■ 0.000	`\zeta^{\dagger a} \equiv \zeta^{a}-2 s^{a b}{ }_{b}`	$\zeta^{\dagger a} \equiv \zeta^{a}-2 s^{ab}{}_{b}$	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\dagger$-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^\dagger_{[\mu}b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a = j^\mu b^a{}_\mu = b^a{}_\mu\delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b = t^a{}_b + 2(s_{eb}{}^a\mathcal{V}^e - s^{ae}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{eb}{}^a\mathcal{V}^e - \sigma^{ae}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_ F01017 **[conf 0.000]**	55	■0.000	`{ }^{\mathrm{c}}`	c	

Gravitational gauge field variables	
$h_a{}^\mu$	translational gauge field (h-field)
h	determinant of $h_a{}^\mu$
$b^a{}_\mu$	inverse h-field
$A^{ab}{}_\mu$	rotational gauge field (A-field)
B_μ	WGT dilational gauge field (B-field)
V_μ	eWGT dilational gauge field (V-field)
$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a} b^{b]}{}_\mu h_c{}^\nu V_\nu$	eWGT extended rotational gauge field ($A^\dagger$-field)
Derivative operators	
$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	PGT (generalised) covariant derivative[a]
$\partial^*_\mu \equiv \partial_\mu + wB_\mu$	WGT 'augmented' partial derivative
$\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu \Sigma_{ab})$	WGT (generalised) covariant derivative[a]
$\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$	eWGT 'augmented' partial derivative
$\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) covariant derivative[a]
$\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu \Sigma_{ab})$	eWGT (generalised) semi-covariant derivative[a]
Gauge field strengths	
$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{e[\mu} A^{eb}{}_{\nu]})$	PGT and WGT rotational gauge field strength[a,b]
$\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu} A^{\dagger eb}{}_{\nu]})$	eWGT rotational gauge field strength[a,b]
$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$	PGT translational gauge field strength[b]
$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$	WGT translational gauge field strength[b]
$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge field strength[b]
$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu} b^a{}_{\nu]}$	eWGT translational gauge 'semi' field strength[a,b]
$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$	WGT dilational gauge field strength
$\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_\nu + \frac{1}{3}T_{\nu]})$	eWGT dilational gauge field strength
Reduced A-fields and related quantities	
$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	PGT Ricci rotation coefficients
$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$	WGT Ricci rotation coefficients
$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^\dagger_{[\mu} b^a{}_{\nu]}$	eWGT Ricci rotation coefficients
${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$	PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$)
${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$	WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$)
${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$	eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$)
Currents and related quantities	
L	Lagrangian
$\mathcal{L} \equiv h^{-1}L$	Lagrangian density
$t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_G/\delta h_a{}^\mu$	gravitational sector energy-momentum tensor[c]
$s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_G/\delta A^{ab}{}_\mu$	gravitational sector spin-angular-momentum tensor[c]
$j^a \equiv j^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_G/\delta B_\mu$	gravitational sector dilation current[c]
$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]
$\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta\mathcal{L}_M/\delta h_a{}^\mu$	matter sector energy-momentum tensor[c]
$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu \delta\mathcal{L}_M/\delta A^{ab}{}_\mu$	matter sector spin-angular-momentum tensor[c]
$\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu \delta\mathcal{L}_M/\delta B_\mu$	matter sector dilation current[c]
$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$	gravitational sector eWGT covariant energy-momentum tensor[c]
$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$	gravitational sector eWGT covariant dilation current[c]

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1510.06699_F01018 **[conf 0.000]**	55	■0.000	`{ }^{\text {c }}`	c	*Gravitational gauge field variables* $h_a{}^\mu$ translational gauge field (h-field) h determinant of $h_a{}^\mu$ $b^a{}_\mu$ inverse h-field $A^{ab}{}_\mu$ rotational gauge field (A-field) B_μ WGT dilational gauge field (B-field) V_μ eWGT dilational gauge field (V-field) $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ eWGT extended rotational gauge field ($A^\dagger$-field) *Derivative operators* $\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ PGT (generalised) covariant derivative[a] $\partial^*_\mu \equiv \partial_\mu + wB_\mu$ WGT 'augmented' partial derivative $\mathcal{D}^*_a \equiv h_a{}^\mu D^*_\mu \equiv h_a{}^\mu(\partial^*_\mu + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab})$ WGT (generalised) covariant derivative[a] $\partial^\dagger_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)$ eWGT 'augmented' partial derivative $\mathcal{D}^\dagger_a \equiv h_a{}^\mu D^\dagger_\mu \equiv h_a{}^\mu(\partial^\dagger_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ eWGT (generalised) covariant derivative[a] $\mathcal{D}^\natural_a \equiv h_a{}^\mu D^\natural_\mu \equiv h_a{}^\mu(\partial_\mu + \frac{1}{2}A^{\dagger ab}{}_\mu\Sigma_{ab})$ eWGT (generalised) semi-covariant derivative[a] *Gauge field strengths* $\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{ab}{}_{\nu]} + A^a{}_{e[\mu}A^{eb}{}_{\nu]})$ PGT and WGT rotational gauge field strength[a,b] $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu(\partial_{[\mu}A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{e[\mu}A^{\dagger eb}{}_{\nu]})$ eWGT rotational gauge field strength[a,b] $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]}$ PGT translational gauge field strength[b] $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]}$ WGT translational gauge field strength[b] $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\dagger_{[\mu}b^a{}_{\nu]}$ eWGT translational gauge field strength[b] $\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^\natural_{[\mu}b^a{}_{\nu]}$ eWGT translational gauge 'semi' field strength[a,b] $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}B_{\nu]}$ WGT dilational gauge field strength $\mathcal{H}^\dagger_{ab} = 2h_a{}^\mu h_b{}^\nu\partial_{[\mu}(V_{\nu]} + \frac{1}{3}T_{\nu]})$ eWGT dilational gauge field strength *Reduced A-fields and related quantities* $c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial_{[\mu}b^a{}_{\nu]}$ PGT Ricci rotation coefficients $c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^*_{[\mu}b^a{}_{\nu]}$ WGT Ricci rotation coefficients $c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu\partial^\dagger_{[\mu}b^a{}_{\nu]}$ eWGT Ricci rotation coefficients ${}^0A^{ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ PGT reduced A-field (when $\mathcal{T}^a{}_{bc} \equiv 0$) ${}^0A^{*ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{*ab}{}_c + c^{*b}{}_c{}^a - c^{*ab}_c)$ WGT reduced A-field (when $\mathcal{T}^{*a}{}_{bc} \equiv 0$) ${}^0A^{\dagger ab}{}_\mu \equiv \frac{1}{2}b^c{}_\mu(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c^{\dagger ab}_c)$ eWGT reduced $A^\dagger$-field (when $\mathcal{T}^{\dagger a}{}_{bc} \equiv 0$) *Currents and related quantities* L Lagrangian $\mathcal{L} \equiv h^{-1}L$ Lagrangian density $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_G/\delta h_a{}^\mu$ gravitational sector energy-momentum tensor[c] $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_G/\delta A^{ab}{}_\mu$ gravitational sector spin-angular-momentum tensor[c] $j^a = j^\mu b^a{}_\mu = b^a{}_\mu\delta\mathcal{L}_G/\delta B_\mu$ gravitational sector dilation current[c] $t^{\dagger a}{}_b = t^a{}_b + 2(s_{cb}{}^a\mathcal{V}^c - s^{ac}{}_c\mathcal{V}_b)$ gravitational sector eWGT covariant energy-momentum tensor[c] $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ gravitational sector eWGT covariant dilation current[c] $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu\delta\mathcal{L}_M/\delta h_a{}^\mu$ matter sector energy-momentum tensor[c] $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_\mu \equiv b^c{}_\mu\delta\mathcal{L}_M/\delta A^{ab}{}_\mu$ matter sector spin-angular-momentum tensor[c] $\zeta^a \equiv \zeta^\mu b^a{}_\mu \equiv b^a{}_\mu\delta\mathcal{L}_M/\delta B_\mu$ matter sector dilation current[c] $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a\mathcal{V}^c - \sigma^{ac}{}_c\mathcal{V}_b)$ gravitational sector eWGT covariant energy-momentum tensor[c] $\zeta^{\dagger a} = \zeta^a - 2s^{ab}{}_b$ gravitational sector eWGT covariant dilation current[c]
1510.06699_F01019	55	■0.999	`\mathcal{R}^{a}{ }_{b} \equiv \mathcal{R}^{a c}{ }_{b c}, \mathcal{R} \equiv \mathcal{R}^{a}{ }_{a}`	$\mathcal{R}^{a}{}_{b} \equiv \mathcal{R}^{ac}{}_{bc}, \mathcal{R} \equiv \mathcal{R}^{a}{}_{a}$	[b] Contractions are defined by $\mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}$, $\mathcal{R} \equiv \mathcal{R}^a{}_a$ and $\mathcal{T}_a = \mathcal{T}^b{}_{ab}$, and similarly for other versions.
1510.06699_F01020	55	■0.999	`\mathcal{T}_{a}=\mathcal{T}^{b}{ }_{a b}`	$\mathcal{T}_{a}=\mathcal{T}^{b}{}_{ab}$	[b] Contractions are defined by $\mathcal{R}^a{}_b \equiv \mathcal{R}^{ac}{}_{bc}$, $\mathcal{R} \equiv \mathcal{R}^a{}_a$ and $\mathcal{T}_a = \mathcal{T}^b{}_{ab}$, and similarly for other versions.
1510.06699_F01021	56	■1.000	`S L(2, C)`	$SL(2, C)$	Lorentz transformations by Λ, rather than $SL(2,C)$ elements.
1510.06699_F01022	56	■1.000	`h^{\mu}{ }_{a} J_{\mu} \neq \mathcal{J}_{a}`	$h^{\mu}{}_{a} J_{\mu} \neq \mathcal{J}_{a}$	$\mathcal{J}$, so that, for example, $h^\mu{}_a J_\mu \neq \mathcal{J}_a$. We will not, however,
1510.06699_F01023	56	■1.000	`{ }^{0} \mathcal{A}_{a b c}`	${}^{0}\mathcal{A}_{abc}$	${}^0\mathcal{A}_{abc}$ and $\mathcal{K}_{abc}$ in PGT (see Appendix B), one has ${}^0\mathcal{A}^{*ab}{}_c =$
1510.06699_F01024	56	■1.000	`\mathcal{K}_{a b c}`	$\mathcal{K}_{abc}$	${}^0\mathcal{A}_{abc}$ and $\mathcal{K}_{abc}$ in PGT (see Appendix B), one has ${}^0\mathcal{A}^{*ab}{}_c =$
1510.06699_F01025	56	■1.000	`{ }^{0} \mathcal{A}^{* a b}{ }_{c}=`	${}^{0}\mathcal{A}^{*ab}{}_{c} =$	${}^0\mathcal{A}_{abc}$ and $\mathcal{K}_{abc}$ in PGT (see Appendix B), one has ${}^0\mathcal{A}^{*ab}{}_c =$
1510.06699_F01026	56	■1.000	`{ }^{0} \mathcal{A}^{a b}{ }_{c}+\delta_{c}^{a} \mathcal{B}^{b}-\delta_{c}^{b} \mathcal{B}^{a}`	${}^{0}\mathcal{A}^{ab}{}_{c}+\delta_{c}^{a}\mathcal{B}^{b}-\delta_{c}^{b}\mathcal{B}^{a}$	${}^0\mathcal{A}^{ab}{}_c+\delta^a_c\mathcal{B}^b-\delta^b_c\mathcal{B}^a$ and $\mathcal{K}^{*ab}{}_c = \mathcal{K}^{ab}{}_c-\delta^a_c\mathcal{B}^b+\delta^b_c\mathcal{B}^a$, so that the
1510.06699_F01027	56	■1.000	`\mathcal{K}^{* a b}{ }_{c}=\mathcal{K}^{a b}{ }_{c}-\delta_{c}^{a} \mathcal{B}^{b}+\delta_{c}^{b} \mathcal{B}^{a}`	$\mathcal{K}^{*ab}{}_{c}=\mathcal{K}^{ab}{}_{c}-\delta_{c}^{a}\mathcal{B}^{b}+\delta_{c}^{b}\mathcal{B}^{a}$	${}^0\mathcal{A}^{ab}{}_c+\delta^a_c\mathcal{B}^b-\delta^b_c\mathcal{B}^a$ and $\mathcal{K}^{*ab}{}_c = \mathcal{K}^{ab}{}_c-\delta^a_c\mathcal{B}^b+\delta^b_c\mathcal{B}^a$, so that the
1510.06699_F01028	56	■1.000	`t^{a}{ }_{\mu} \equiv h \delta \mathcal{L}_{\mathrm{G}} / \delta h_{a}{ }^{\mu}, s_{a b}{ }^{\mu} \equiv`	$t^{a}{}_{\mu} \equiv h\delta\mathcal{L}_{\mathrm{G}}/\delta h_{a}{}^{\mu}, s_{ab}{}^{\mu} \equiv$	[72]One could alternatively define $t^a{}_\mu \equiv h\,\delta\mathcal{L}_{\mathrm{G}}/\delta h_a{}^\mu$, $s_{ab}{}^\mu \equiv$
1510.06699_F01029	56	■1.000	`h \delta \mathcal{L}_{\mathrm{G}} / \delta A^{a b}{ }_{\mu}`	$h\delta\mathcal{L}_{\mathrm{G}}/\delta A^{ab}{}_{\mu}$	$h\,\delta\mathcal{L}_{\mathrm{G}}/\delta A^{ab}{}_\mu$, and $j^\mu \equiv h\,\delta\mathcal{L}_{\mathrm{G}}/\delta B_\mu$, and similarly for the mat-
1510.06699_F01030	56	■1.000	`j^{\mu} \equiv h \delta \mathcal{L}_{\mathrm{G}} / \delta B_{\mu}`	$j^{\mu} \equiv h\delta\mathcal{L}_{\mathrm{G}}/\delta B_{\mu}$	$h\,\delta\mathcal{L}_{\mathrm{G}}/\delta A^{ab}{}_\mu$, and $j^\mu \equiv h\,\delta\mathcal{L}_{\mathrm{G}}/\delta B_\mu$, and similarly for the mat-
1510.06699_F01031	56	■1.000	`\dot{R} \equiv`	$\dot{R} \equiv$	point particle one must, in addition, make the replacement $\dot{R} \equiv$
1510.06699_F01032	56	■1.000	`\dot{x}^{\mu} \partial_{\mu} R \rightarrow \dot{v}^{a} \mathcal{D}_{a}^{*} R`	$\dot{x}^{\mu}\partial_{\mu}R \rightarrow \dot{v}^{a}\mathcal{D}_{a}^{*}R$	$\dot{x}^\mu\partial_\mu R \rightarrow \dot{v}^a\mathcal{D}^*_a R$.
1510.06699_F01033	56	■0.994	`{ }^{77,78}`	^{77,78}	theory[77,78]. In the standard interpretation, the scalar field in
1510.06699_F01034	56	■0.999	`\bar{\psi} \gamma^{a}{ }^{0} \stackrel{\leftrightarrow}{\mathcal{D}}_{a}^{*} \psi=\bar{\psi} \gamma^{a}{ }^{0} \stackrel{\leftrightarrow}{\mathcal{D}}_{a} \psi`	$\bar{\psi}\gamma^{a0}\stackrel{\leftrightarrow}{\mathcal{D}}_{a}^{*}\psi=\bar{\psi}\gamma^{a0}\stackrel{\leftrightarrow}{\mathcal{D}}_{a}\psi$	[81]It is easily shown that $\bar{\psi}\gamma^a\,{}^0\stackrel{\leftrightarrow}{\mathcal{D}}{}^*_a\psi = \bar{\psi}\gamma^a\,{}^0\stackrel{\leftrightarrow}{\mathcal{D}}_a\psi$ (which does not
1510.06699_F01035	56	■1.000	`\theta(x)`	$\theta(x)$	a field $\theta(x)$. Provided it has Weyl weight $w = 0$, one would ob-

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1510.06699_F01036	57	1.000	`D_{\mu}^{\dagger} \varphi=`	$D_{\mu}^{\dagger} \varphi=$	ticular, we note that the eWGT covariant derivative $D^{\dagger}_{\mu}\varphi =$
1510.06699_F01037	57	1.000	`\left[D_{\mu}+\frac{1}{2}\left(\mathcal{V}^{a} b^{b}{ }_{\mu}-\mathcal{V}^{b} b^{a}{ }_{\mu}\right) \Sigma_{a b}-w V_{\mu}-\frac{1}{3} w T_{\mu}\right] \varphi`	$\left[D_{\mu}+\frac{1}{2}\left(\mathcal{V}^{a} b^{b}{ }_{\mu}-\mathcal{V}^{b} b^{a}{ }_{\mu}\right) \Sigma_{a b}-w V_{\mu}-\frac{1}{3} w T_{\mu}\right] \varphi$	$[D_\mu + \frac{1}{2}(\mathcal{V}^a b^b{}_\mu - \mathcal{V}^b b^a{}_\mu)\Sigma_{ab} - wV_\mu - \frac{1}{3}wT_\mu]\varphi$ represents a far
1510.06699_F01038	57	1.000	`D_{\mu}^{*} \varphi=\left(D_{\mu}+w B_{\mu}\right) \varphi`	$D_{\mu}^{*} \varphi=\left(D_{\mu}+w B_{\mu}\right) \varphi$	derivative $D^{*}_{\mu}\varphi = (D_\mu + wB_\mu)\varphi$.
1510.06699_F01039	57	0.995	`{ }^{0} \mathcal{A}^{\dagger a b}{ }_{c}={ }^{0} \mathcal{A}^{a b}{ }_{c}-\delta_{c}^{a}\left( \mathcal{V}^{b}+\frac{1}{3} \mathcal{T}^{b}\right)+ \delta_{c}^{b}\left(\mathcal{V}^{a}+\right.`	${ }^{0} \mathcal{A}^{\dagger a b}{ }_{c}={ }^{0} \mathcal{A}^{a b}{ }_{c}-\delta_{c}^{a}\left(\mathcal{V}^{b}+\frac{1}{3} \mathcal{T}^{b}\right)+ \delta_{c}^{b}\left(\mathcal{V}^{a}+\right.$	pendix B), one has ${}^0\mathcal{A}^{\dagger ab}{}_c = {}^0\mathcal{A}^{ab}{}_c - \delta^a_c(\mathcal{V}^b + \frac{1}{3}\mathcal{T}^b) + \delta^b_c(\mathcal{V}^a +$
1510.06699_F01040	57	0.600	`\frac{1}{3} \mathcal{T}^{a}`	$\frac{1}{3} \mathcal{T}^{a}$	$\frac{1}{3}\mathcal{T}^a)$ and $\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3}\delta^a_c\mathcal{T}^b - \frac{1}{3}\delta^b_c\mathcal{T}^a$, so confirming that
1510.06699_F01041	57	0.600	`\mathcal{K}^{\dagger a b}{ }_{c}=\mathcal{K}^{a b}{ }_{c}+\frac{1}{3} \delta_{c}^{a} \mathcal{T}^{b}-\frac{1}{3} \delta_{c}^{b} \mathcal{T}^{a}`	$\mathcal{K}^{\dagger a b}{ }_{c}=\mathcal{K}^{a b}{ }_{c}+\frac{1}{3} \delta_{c}^{a} \mathcal{T}^{b}-\frac{1}{3} \delta_{c}^{b} \mathcal{T}^{a}$	$\frac{1}{3}\mathcal{T}^a)$ and $\mathcal{K}^{\dagger ab}{}_c = \mathcal{K}^{ab}{}_c + \frac{1}{3}\delta^a_c\mathcal{T}^b - \frac{1}{3}\delta^b_c\mathcal{T}^a$, so confirming that
1510.06699_F01042	57	0.965	`\bar{\psi} \gamma^{a}{ }^{0} \mathcal{D}_{a}^{\dagger} \psi=\bar{\psi} \gamma^{a}{ }^{0} \stackrel{\leftrightarrow}{\mathcal{D}}_{a} \psi`	$\bar{\psi} \gamma^{a}{ }^{0} \mathcal{D}_{a}^{\dagger} \psi=\bar{\psi} \gamma^{a}{ }^{0} \stackrel{\leftrightarrow}{\mathcal{D}}_{a} \psi$	[94]It is easily shown that $\bar\psi\gamma^a\,{}^0\overleftrightarrow{\mathcal{D}}^\dagger_a\psi = \bar\psi\gamma^a\,{}^0\overleftrightarrow{\mathcal{D}}_a\psi$ (which does not
1510.06699_F01043	57	1.000	`\nabla_{\sigma} g_{\mu \nu}=`	$\nabla_{\sigma} g_{\mu \nu}=$	that given in (124) for WGT can both be written as $\nabla_\sigma g_{\mu\nu} =$
1510.06699_F01044	57	1.000	`-\frac{1}{2}\left(\Gamma^{\lambda}{ }_{\lambda \sigma}-{ }^{0} \Gamma^{\lambda}{ }_{\lambda \sigma}\right) g_{\mu \nu}`	$-\frac{1}{2}\left(\Gamma^{\lambda}{ }_{\lambda \sigma}-{ }^{0} \Gamma^{\lambda}{ }_{\lambda \sigma}\right) g_{\mu \nu}$	$-\frac{1}{2}(\Gamma^\lambda{}_{\lambda\sigma} - {}^0\Gamma^\lambda{}_{\lambda\sigma})g_{\mu\nu}$, where ${}^0\Gamma^\lambda{}_{\mu\nu}$ is the metric connection;
1510.06699_F01045	57	1.000	`L_{\mathcal{R}^{2}}=`	$L_{\mathcal{R}^{2}}=$	which its symmetries are more manifest, namely $L_{\mathcal{R}^2} =$
1510.06699_F01046	57	1.000	`\alpha_{1} \mathcal{R}^{\dagger 2}+\alpha_{2}^{\prime} \mathcal{R}_{(a b)}^{\dagger} \mathcal{R}^{\dagger(a b)}+\alpha_{3}^{\prime} \mathcal{R}_{[a b]}^{\dagger} \mathcal{R}^{\dagger[a b]}+\alpha_{4}^{\prime} \mathcal{R}_{a(b c) d}^{\dagger} \mathcal{R}^{\dagger a(b c) d}+`	$\alpha_{1} \mathcal{R}^{\dagger 2}+\alpha_{2}^{\prime} \mathcal{R}_{(a b)}^{\dagger} \mathcal{R}^{\dagger(a b)}+\alpha_{3}^{\prime} \mathcal{R}_{[a b]}^{\dagger} \mathcal{R}^{\dagger[a b]}+\alpha_{4}^{\prime} \mathcal{R}_{a(b c) d}^{\dagger} \mathcal{R}^{\dagger a(b c) d}+$	$\alpha_1\mathcal{R}^{\dagger 2} + \alpha_2'\mathcal{R}^\dagger_{(ab)}\mathcal{R}^{\dagger(ab)} + \alpha_3'\mathcal{R}^\dagger_{[ab]}\mathcal{R}^{\dagger[ab]} + \alpha_4'\mathcal{R}^\dagger_{a(bc)d}\mathcal{R}^{\dagger a(bc)d} +$
1510.06699_F01047	57	1.000	`\alpha_{5}^{\prime} \mathcal{R}_{a[b c] d}^{\dagger} \mathcal{R}^{\dagger a[b c] d}+\alpha_{6} \mathcal{R}_{a b c d}^{\dagger} \mathcal{R}^{\dagger c d a b}`	$\alpha_{5}^{\prime} \mathcal{R}_{a[b c] d}^{\dagger} \mathcal{R}^{\dagger a[b c] d}+\alpha_{6} \mathcal{R}_{a b c d}^{\dagger} \mathcal{R}^{\dagger c d a b}$	$\alpha_5'\mathcal{R}^\dagger_{a[bc]d}\mathcal{R}^{\dagger a[bc]d} + \alpha_6\mathcal{R}^\dagger_{abcd}\mathcal{R}^{\dagger cdab}$, where the new coefficients
1510.06699_F01048	57	0.994	`\alpha_{2}^{\prime}=\alpha_{2}+\alpha_{3}, \alpha_{3}^{\prime}=\alpha_{2}-\alpha_{3}, \alpha_{4}^{\prime}=\alpha_{4}+\alpha_{5}`	$\alpha_{2}^{\prime}=\alpha_{2}+\alpha_{3}, \alpha_{3}^{\prime}=\alpha_{2}-\alpha_{3}, \alpha_{4}^{\prime}=\alpha_{4}+\alpha_{5}$	are $\alpha_2' = \alpha_2+\alpha_3$, $\alpha_3' = \alpha_2-\alpha_3$, $\alpha_4' = \alpha_4+\alpha_5$ and $\alpha_5' = \alpha_4-\alpha_5$.
1510.06699_F01049	57	0.994	`\alpha_{5}^{\prime}=\alpha_{4}-\alpha_{5}`	$\alpha_{5}^{\prime}=\alpha_{4}-\alpha_{5}$	are $\alpha_2' = \alpha_2+\alpha_3$, $\alpha_3' = \alpha_2-\alpha_3$, $\alpha_4' = \alpha_4+\alpha_5$ and $\alpha_5' = \alpha_4-\alpha_5$.
1510.06699_F01050	57	1.000	`\alpha_{2}^{\prime}=\alpha_{3}^{\prime}`	$\alpha_{2}^{\prime}=\alpha_{3}^{\prime}$	allows one to set α_1 or α_6 to zero as before, or set $\alpha_2' = \alpha_3'$.
1510.06699_F01051	57	0.999	`L_{\mathcal{T}^{2}}=\beta_{1}^{\prime} \mathcal{T}_{(a b) c}^{\dagger} \mathcal{T}^{\dagger(a b) c}+\beta_{2}^{\prime} \mathcal{T}_{[a b] c}^{\dagger} \mathcal{T}^{\dagger[a b] c}`	$L_{\mathcal{T}^{2}}=\beta_{1}^{\prime} \mathcal{T}_{(a b) c}^{\dagger} \mathcal{T}^{\dagger(a b) c}+\beta_{2}^{\prime} \mathcal{T}_{[a b] c}^{\dagger} \mathcal{T}^{\dagger[a b] c}$	symmetric form $L_{\mathcal{T}^2} = \beta_1'\mathcal{T}^\dagger_{(ab)c}\mathcal{T}^{\dagger(ab)c} + \beta_2'\mathcal{T}^\dagger_{[ab]c}\mathcal{T}^{\dagger[ab]c}$, where
1510.06699_F01052	57	0.998	`\beta_{1}^{\prime}=\beta_{1}+\beta_{2}`	$\beta_{1}^{\prime}=\beta_{1}+\beta_{2}$	$\beta_1' = \beta_1+\beta_2$ and $\beta_2' = \beta_1-\beta_2$.
1510.06699_F01053	57	0.998	`\beta_{2}^{\prime}=\beta_{1}-\beta_{2}`	$\beta_{2}^{\prime}=\beta_{1}-\beta_{2}$	$\beta_1' = \beta_1+\beta_2$ and $\beta_2' = \beta_1-\beta_2$.
1510.06699_F01054	57	1.000	`S_{\mathrm{T}}=S_{\mathrm{G}}+S_{\mathrm{M}}`	$S_{\mathrm{T}}=S_{\mathrm{G}}+S_{\mathrm{M}}$	lations, but the total action $S_{\mathrm{T}} = S_{\mathrm{G}} + S_{\mathrm{M}}$ is so. We will not
1510.06699_F01055	57	0.999	`{ }^{0} \mathcal{R}^{* a b}{ }_{c d}={ }^{0} \mathcal{R}^{a b}{ }_{c d}-2 \delta_{d}^{[b}\left({ }^{0} \mathcal{D}_{c}-\mathcal{B}_{c}\right) \mathcal{B}^{a]}+`	${ }^{0} \mathcal{R}^{* a b}{ }_{c d}={ }^{0} \mathcal{R}^{a b}{ }_{c d}-2 \delta_{d}^{[b}\left({ }^{0} \mathcal{D}_{c}-\mathcal{B}_{c}\right) \mathcal{B}^{a]}+$	quantity ${}^0\mathcal{R}^{*ab}{}_{cd}$ by ${}^0\mathcal{R}^{*ab}{}_{cd} = {}^0\mathcal{R}^{ab}{}_{cd} - 2\delta^{[b}_d({}^0\mathcal{D}_c - \mathcal{B}_c)\mathcal{B}^{a]} +$
1510.06699_F01056	57	1.000	`2 \delta_{c}^{[b}\left({ }^{0} \mathcal{D}_{d}-\mathcal{B}_{d}\right) \mathcal{B}^{a]}-2 \mathcal{B}^{e} \mathcal{B}_{e} \delta_{c}^{[a} \delta_{d}^{b]}`	$2 \delta_{c}^{[b}\left({ }^{0} \mathcal{D}_{d}-\mathcal{B}_{d}\right) \mathcal{B}^{a]}-2 \mathcal{B}^{e} \mathcal{B}_{e} \delta_{c}^{[a} \delta_{d}^{b]}$	$2\delta^{[b}_c({}^0\mathcal{D}_d - \mathcal{B}_d)\mathcal{B}^{a]} - 2\mathcal{B}^e\mathcal{B}_e\delta^{[a}_c\delta^{b]}_d$, from which one finds that their
1510.06699_F01057	57	1.000	`{ }^{0} \mathcal{R}^{* a}{ }_{c}={ }^{0} \mathcal{R}^{a}{ }_{c}-2\left({ }^{0} \mathcal{D}_{c}-\mathcal{B}_{c}\right) \mathcal{B}^{a}-`	${ }^{0} \mathcal{R}^{* a}{ }_{c}={ }^{0} \mathcal{R}^{a}{ }_{c}-2\left({ }^{0} \mathcal{D}_{c}-\mathcal{B}_{c}\right) \mathcal{B}^{a}-$	contractions are related by ${}^0\mathcal{R}^{*a}{}_c = {}^0\mathcal{R}^a{}_c - 2({}^0\mathcal{D}_c - \mathcal{B}_c)\mathcal{B}^a -$
1510.06699_F01058	57	1.000	`\left({ }^{0} \mathcal{D}_{e}+2 \mathcal{B}_{e}\right) \mathcal{B}^{e} \delta_{c}^{a}`	$\left({ }^{0} \mathcal{D}_{e}+2 \mathcal{B}_{e}\right) \mathcal{B}^{e} \delta_{c}^{a}$	$({}^0\mathcal{D}_e + 2\mathcal{B}_e)\mathcal{B}^e\delta^a_c$ and ${}^0\mathcal{R}^* = {}^0\mathcal{R} - 6({}^0\mathcal{D}_e + \mathcal{B}_e)\mathcal{B}^e$. Similarly,

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1510.06699_F01059	57	■1.000	`{ }^{0} \mathcal{R}^{*}={ }^{0} \mathcal{R}-6 \left({ }^{0} \mathcal{D}_{e}+\mathcal{B}_{e}\right) \mathcal{B}^{e}`	${ }^{0} \mathcal{R}^{*}={ }^{0} \mathcal{R}-6 \left({ }^{0} \mathcal{D}_{e}+\mathcal{B}_{e}\right) \mathcal{B}^{e}$	$({}^0\mathcal{D}_e + 2\mathcal{B}_e)\mathcal{B}^e\delta^a_c$ and ${}^0\mathcal{R}^* = {}^0\mathcal{R} - 6({}^0\mathcal{D}_e + \mathcal{B}_e)\mathcal{B}^e$. Similarly,
1510.06699_F01060	57	■0.996	`{ }^{0} \mathcal{R}^{* a b}{ }_{c d} \rightarrow{ }^{0} \mathcal{R}^{\dagger a b}{ }_{c d}`	${ }^{0} \mathcal{R}^{* a b}{ }_{c d} \rightarrow{ }^{0} \mathcal{R}^{\dagger a b}{ }_{c d}$	placements ${}^0\mathcal{R}^{*ab}{}_{cd} \to {}^0\mathcal{R}^{\dagger ab}{}_{cd}$ and $\mathcal{B}_a \to -(\mathcal{V}_a + \frac{1}{3}\mathcal{T}_a)$.
1510.06699_F01061	57	■0.996	`\mathcal{B}_{a} \rightarrow-\left(\mathcal{V}_{a}+ \frac{1}{3} \mathcal{T}_{a}\right)`	$\mathcal{B}_{a} \rightarrow-\left(\mathcal{V}_{a}+ \frac{1}{3} \mathcal{T}_{a}\right)$	placements ${}^0\mathcal{R}^{*ab}{}_{cd} \to {}^0\mathcal{R}^{\dagger ab}{}_{cd}$ and $\mathcal{B}_a \to -(\mathcal{V}_a + \frac{1}{3}\mathcal{T}_a)$.
1510.06699_F01062	57	■1.000	`{ }^{0} \mathcal{R}^{*}`	${ }^{0} \mathcal{R}^{*}$	'reduced' versions ${}^0\mathcal{D}^*_a$ and ${}^0\mathcal{R}^*$, respectively, as described in